PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM

A Research Proposal

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DEDICATION

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CHAPTER I

INTRODUCTION

Research papers, thesis manuscripts, and capstone project papers are among the primary forms of academic documents required in higher education institutions. These scholarly works serve as evidence of students' knowledge, critical thinking, and research competencies. To ensure academic quality and integrity, higher education institutions require academic documents to comply with established institutional standards, including document structure, formatting, citation style, grammar, and other academic writing conventions (Gupta et al., 2022).

Despite the availability of institutional writing manuals and manuscript templates, many students continue to experience difficulties in preparing academic documents that meet these requirements. They often encounter challenges in applying formatting guidelines, organizing document sections, maintaining citation accuracy, and identifying grammatical and writing errors. As a result, students frequently undergo multiple revision cycles, experience delays in manuscript approval, and spend considerable time manually reviewing their documents before submission (Sitompul & Anditasari, 2022).

The compliance-checking process is largely performed manually by students, advisers, and research coordinators. Although existing writing applications can assist with grammar and spelling correction, they provide limited support for verifying compliance with institutional formatting requirements, document structure, and academic writing standards. Consequently, many compliance-related issues remain undetected until the academic document undergoes formal evaluation, increasing the workload of both students and evaluators while prolonging the review process (Bulqiyah et al., 2021).

Maintaining proper document structure, formatting consistency, and clear academic writing is essential because these elements contribute to the quality, credibility, and professionalism of scholarly work. However, manually reviewing these requirements can be time-consuming and prone to oversight, particularly for lengthy academic documents. These challenges highlight the need for a more efficient approach to ensuring compliance before submission (Rezeki, 2018).

In response to these challenges, there is a growing need for an intelligent system

that assists users in evaluating the compliance of academic documents with institutional standards. An automated compliance-checking system can help identify formatting inconsistencies, structural deficiencies, citation issues, grammatical errors, and other writing-related concerns while providing organized feedback that enables users to revise their documents more efficiently before submission.

To address this need, this study proposes PaperPilot: An AI-Powered Academic Document Analysis and Compliance Checking System. The system is designed to analyze academic documents and evaluate their compliance with predefined institutional standards. Specifically, PaperPilot assesses document formatting, structural organization, citation compliance, and writing quality while providing structured feedback to guide users in correcting identified issues. Through intelligent document analysis, the system aims to support students in producing academic documents that meet institutional requirements before formal evaluation.

The implementation of PaperPilot is expected to reduce compliance-related errors, minimize repetitive manual checking, improve adherence to institutional manuscript standards, and streamline the document preparation process. By assisting students in identifying and resolving compliance issues at an early stage, the system aims to improve the quality of academic documents while supporting advisers and research coordinators in conducting more efficient manuscript evaluations.

Rationale of the Study

Many academic documents require strict compliance with institutional formatting guidelines and documentation requirements. However, many students encounter challenges in identifying formatting errors within their academic documents. These difficulties often result in repeated revisions, increased workload, and delays in the approval process (Perdomo & Morales, 2022).

Although existing writing tools can assist with grammar correction and language enhancement, they generally provide limited support for academic compliance checking. Students frequently experience difficulties in applying the prescribed academic standards, which may affect the quality and credibility of their work (Rezeki, 2018). As a result, many still rely heavily on manual review and adviser feedback to verify document compliance.

The development of PaperPilot is driven by the need for a dedicated system that assists students in evaluating the compliance of their academic documents before

submission. The system automatically analyzes formatting requirements and identifies potential compliance issues based on predefined academic rules. By providing immediate feedback and recommendations, it enables users to correct errors efficiently and improve the quality of their academic documents.

This study is significant because it offers a practical solution for improving academic document preparation and compliance verification. Through automated checking and guided assessment, PaperPilot aims to reduce common errors, support adherence to academic standards, and assist students in producing accurate, organized, and submission-ready academic documents.

Objectives of the Study

This study aims to design and develop PaperPilot, an AI-powered academic document analysis and compliance checking system available on both web and mobile platforms that assists students in organizing, formatting, and checking their academic documents. It seeks to provide a more efficient and reliable way for students to meet academic standards through automated tools and guided assistance delivered seamlessly via web and mobile applications, ultimately improving the overall quality and organization of these academic documents while reducing common errors.

Specifically, the study pursues the following objectives:

1. to gather primary data from researchers and research coordinators regarding their experiences and challenges in preparing academic documents that comply with institution-specific formatting standards.
2. to develop an AI-powered academic document analysis and compliance checking system that evaluates academic documents based on user-specific formatting standards.
3. to integrate Explainable Artificial Intelligence (XAI) that provides analysis results and recommendations.

Scope and Limitations of the Study

This study defines the scope and limitations of PaperPilot, which focuses on the design and development of an AI-powered academic document analysis and compliance checking system. The system helps users review the formatting of academic documents based on institution-specific formatting standards. PaperPilot provides both web-based and mobile-accessible platforms where users can register, log in, manage their accounts, upload academic documents, analyze their files, receive feedback, and monitor the

compliance status of revised versions. The system includes an AI-powered document analysis feature that automatically checks the formatting including citation formats APA, MLA, and IEEE.

In addition, PaperPilot includes an Explainable Artificial Intelligence (XAI) feature that provides clear explanations of the analysis results. This helps users understand the detected issues and the recommended corrections, making the feedback more transparent and easier to apply. The system analyzes only the documents uploaded by users and does not generate or evaluate the quality of the research content, methodology, findings, or originality of the study. It is intended to assist the researchers in reviewing the technical compliance of their academic documents before submission. The initial deployment and evaluation of the system will be conducted at the University of Cebu - Banilad Campus.

Despite its capabilities, PaperPilot has several limitations. The system can only analyze document formatting, citation format, and other technical compliance requirements based on the supported institution-specific formatting standards. Its analysis depends on the quality and completeness of the uploaded document, and incorrect or incomplete files may affect the accuracy of the results. The system does not evaluate the quality of the research content, the validity of the methodology, the accuracy of data analysis, the significance of findings, or the originality of the study. In addition, the Explainable Artificial Intelligence (XAI) feature provides explanations only of the compliance issues detected by the system and should be used as guidance rather than as a substitute for the evaluation and judgment of the research advisers, research coordinators. In addition, PaperPilot requires a stable internet connection to function properly, as both the web and mobile applications rely on cloud-based processing, storage, and real-time communication with the server. Users will not be able to access the system, upload manuscripts, or receive compliance analysis results without an active internet connection.

Significance of the Study

The study brings multiple benefits to various stakeholders involved in or affected by the development and implementation of PaperPilot.

Researchers. This study primarily benefits researchers and students who prepare academic documents. PaperPilot provides an AI-powered system that helps users check the structure, formatting, and compliance of their academic documents based on APA,

MLA, and IEEE writing styles. It also provides clear explanations of the analysis results through Explainable Artificial Intelligence (XAI), helping users understand detected issues and apply the recommended corrections. This reduces the time and effort required for manual checking and helps users prepare well-formatted academic documents before submission.

Thesis and Capstone Advisers. This study benefits thesis and capstone advisers by reducing the time spent checking the technical requirements of academic documents, including structure, formatting, and references. Since documents are analyzed before adviser review, advisers can spend more time evaluating the quality of the research, methodology, analysis, and findings, making the review process more efficient.

Research Coordinators. This study benefits research coordinators by providing an automated system for analyzing the technical compliance of academic documents with institutional formatting standards. The system supports the initial review process by identifying formatting and structural issues, helping reduce manual checking and promoting consistency in academic document preparation across the institution.

Flow of the Study

The diagram below presents the flow of the study for the development of PaperPilot: An AI-Powered Academic Document Analysis and Compliance Checking System. It illustrates the major Input, Process, and Output (IPO) components involved in the research and system development. The study begins by gathering and analyzing information about the common challenges faced by students and researchers in preparing academic documents, particularly in terms of document structure, formatting, references, and compliance with academic writing standards. Based on these identified needs, the study focuses on developing an AI-powered system that supports academic document formatting, automatically analyzes document structure, formatting, and references, and provides clear explanations of the analysis results through Explainable Artificial Intelligence (XAI). The system is designed to help users identify technical compliance issues and improve their academic documents before submission.

INPUT **PROCESS** **OUTPUT**

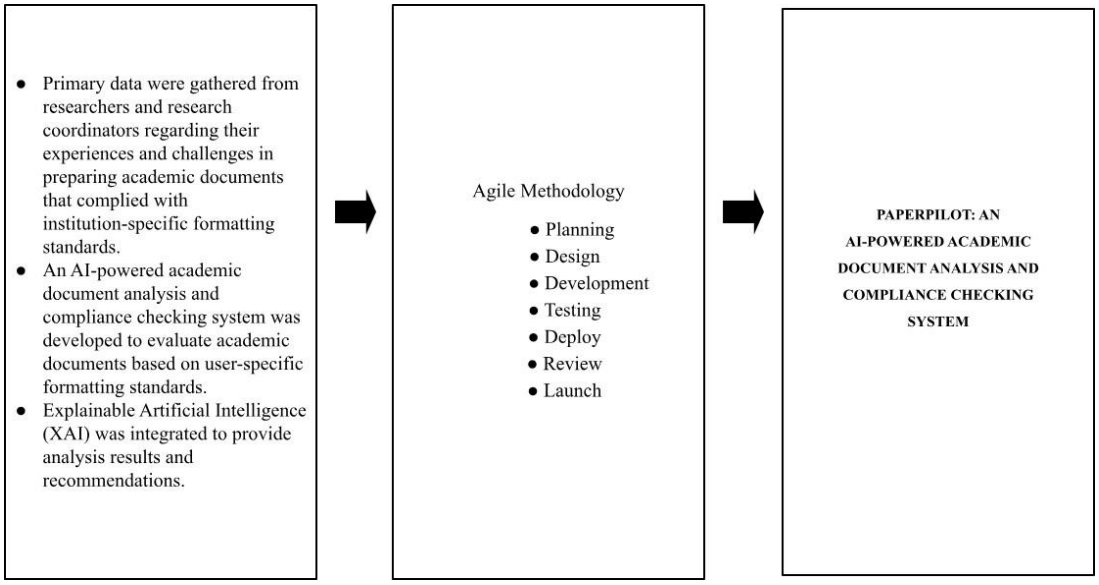


Figure 1: Flow of the Study

Figure 1 illustrates the development flow of PaperPilot: An AI-Powered Academic Document Analysis and Compliance Checking System. The process begins with collecting and analyzing academic document formatting guidelines from selected higher education institutions to identify common formatting standards and institutional requirements for system implementation. Based on these findings, the study developed an AI-powered research analyzer system that supports the formatting of academic documents based on different academic writing styles, a document analysis feature that checks the structure, formatting, and references of academic documents and provides feedback for improvement, and integrated Explainable Artificial Intelligence (XAI) that explains the analysis results and helps users understand the detected issues and suggested corrections.

The development of the system follows the Agile methodology, which consists of the planning, design, development, testing, deploy, review, and launch phases. This approach supports continuous improvement throughout the development process and allows the system to be refined based on user feedback.

The final output of the study is PaperPilot: An AI-Powered Academic Document Analysis and Compliance Checking System, a platform designed to assist users in formatting, analyzing, and improving the technical compliance of academic documents before submission.

Definition of Terms

PaperPilot. Refers to the proposed AI-powered academic document analysis and compliance-checking system designed to help researchers verify that academic documents conform to institution-specific formatting standards.

Academic Documents. Refers to formal scholarly documents prepared by students or researchers as part of the academic requirements or research activities. These include research papers, thesis, manuscripts, dissertations, and other similar scholarly works that follow institutional formatting standards.

Researchers. In this study, researchers is used as a generic term to describe all users of PaperPilot, including students and other individuals who prepare academic documents and use the system to verify compliance with institution-specific formatting standards.

Institutional Formatting Standards. Refers to the formatting guidelines established by a university or academic institution, including requirements for font style, font size, margins, spacing, headings, pagination, and document organization.

Document Compliance Checking. Refers to the automated process of verifying whether an academic document satisfies the formatting and structural requirements specified by an academic institution.

Compliance Report. Refers to the system-generated report that identifies formatting inconsistencies, non-compliant elements, and recommended corrections based on institutional standards.

Formatting. Refers to the presentation and arrangement of document elements according to institutional requirements, including font style, font size, margins, spacing, paragraph alignment, headings, indentation, and page layout.

Heading. Refers to the titles and subheadings used to organize sections of an academic document according to institutional formatting guidelines.

Margins. Refers to the blank spaces surrounding the text on each page that must comply with the measurement requirements specified by the institution.

Line Spacing. Refers to the vertical distance between lines of text, which must conform to institutional formatting requirements.

Font Style. Refers to the required typeface used throughout the academic document, such as Times New Roman or Arial, as prescribed by the institution.

Font Size. Refers to the required size of text used in the academic document to maintain consistency and readability.

Pagination. Refers to the numbering and placement of page numbers throughout the academic document in accordance with institutional guidelines.

Citation. Refers to the acknowledgment of sources within an academic document following the citation style required by the institution.

APA (American Psychological Association). Refers to a citation and formatting style commonly used in the social sciences that provides standards for references, in-text citations, headings, and document formatting.

MLA (Modern Language Association). Refers to a citation and formatting style primarily used in the humanities that specifies guidelines for in-text citations, references, and document formatting.

IEEE (Institute of Electrical and Electronics Engineers). Refers to a numbered citation and reference style commonly used in engineering, computer science, and technical research.

Researcher. Refers to a student or individual who prepares and submits an academic document for compliance checking using PaperPilot.

Word Processing Document. Refers to an electronic document, such as a Microsoft Word (.docx) file, that contains formatted text and is supported by the system for compliance checking.

Portable Document Format (PDF). Refers to a file format used to preserve the layout and appearance of electronic documents across different platforms and devices.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter discusses the related literature and studies following the researchers' careful and in-depth research. This introduces the framework of the study that comprises the main focus of the research described. This aligns with PaperPilot's purpose of assisting students in producing well-structured and compliant academic documents.

Theoretical Background

To understand this research, it is important to establish a theoretical foundation that explains how technology supports the preparation of academic documents. In recent years, digital tools have become increasingly important in helping students and researchers perform technical writing tasks such as document formatting, reference management, document organization, and compliance with academic writing standards. These technologies reduce the time and effort required for manual checking while helping users prepare well-structured and technically compliant academic documents. This study is anchored on the Digital Academic Writing Support Framework (Zhang & Hyland, 2022), which emphasizes the role of digital technologies in improving academic writing and document preparation. Supporting this framework are the Concepts of Structured Writing Revision Systems (Mekheimer, 2025), Academic Writing Assistance and Organization Tools (Kim et al., 2024), the Language-Based Document Evaluation Framework (Jurafsky & Martin, 2025), which supports Automated Document Analysis, and the Technology Acceptance Model (TAM) (Sánchez et al., 2023), which explains the factors that influence users' acceptance and adoption of technology. Together, these theories provide the foundation for developing PaperPilot as an AI-powered system that supports the formatting, analysis, and technical compliance of academic documents while providing understandable analysis results through Explainable Artificial Intelligence (XAI).

The development of PaperPilot: An AI-Powered Academic Document Analysis and Compliance Checking System is anchored on the Academic Writing Support Systems Framework, as discussed by Zhang and Hyland (2022). This framework explains that students and researchers benefit from digital systems that support the preparation of academic documents by improving technical writing tasks and promoting compliance with academic writing standards.

In this context, PaperPilot is based on this framework because it provides AI-powered support for formatting academic documents, analyzing document structure, formatting, and identifying technical compliance issues based on supported academic writing styles, helping users understand detected issues and apply the recommended corrections. Through these features, PaperPilot assists users in preparing academic documents that are well organized, properly formatted, and compliant with academic writing standards.

Supporting this foundation is the concept of structured writing revision systems, as reflected in the study of Mekheimer (2025). This concept emphasizes that students improve their academic documents through systematic review and structured revision processes that help them identify errors, and enhance overall writing quality. Mekheimer found that guided revision systems significantly improve accuracy, consistency, and the quality of academic outputs. This supports PaperPilot as a system that assists students in improving their Academic documents through organized compliance checking and structured feedback on writing errors.

Further strengthening this framework is the concept of academic writing assistance and organization tools, as discussed by Kim et al. (2024). Their study highlights that students benefit from structured writing support systems that help them organize ideas, improve clarity, and enhance the overall structure of long-form academic requirements such as research papers, thesis, and capstone projects. These tools reduce writing difficulties and improve academic productivity by helping students manage the complexity of formal academic writing. This aligns with PaperPilot’s purpose of assisting students in producing well-structured and compliant manuscripts.

In addition, the system is supported by the language-based document evaluation framework, as described by Jurafsky and Martin (2025). This framework explains that computational language systems can analyze written text to detect grammar issues, structural inconsistencies, and formatting errors. Such systems are commonly used in academic writing tools that evaluate document correctness, coherence, and adherence to writing conventions. In PaperPilot, this framework supports the system's ability to analyze academic documents for compliance with academic writing standards.

Lastly, the Technology Acceptance Model (TAM), as supported by Sánchez et al. (2023), explains user adoption of academic support systems based on perceived usefulness and ease of use. The model suggests that students are more likely to adopt systems that simplify academic tasks such as writing, formatting, and document

preparation. This implies that document compliance checking systems like PaperPilot will be widely accepted when they provide clear benefits in improving efficiency and reducing difficulties in academic document preparation. When integrated, these frameworks establish a comprehensive theoretical foundation for PaperPilot. The Academic Writing Support Systems Framework provides the core concept of structured writing assistance, Mekheimer (2025) emphasizes systematic revision and improvement processes, Kim et al. (2024) highlights structured writing organization tools, language-based document evaluation explains technical document checking, and TAM explains user adoption behavior. Together, these frameworks justify the development of a system that ensures accuracy, structure, and compliance in academic documents.

Review of Related Literature

This section presents related literature on artificial intelligence (AI) applications in academic document review, automated compliance and formatting checking, citation formatting, explainable artificial intelligence (XAI), and AI-generated feedback in educational contexts. The reviewed works provide the technological and conceptual grounding for the development of PaperPilot as an AI-powered academic document analysis and compliance checking system.

The use of artificial intelligence to screen and evaluate scholarly manuscripts before they undergo formal review has become an active area of research and development. Lin et al. (2023) examined the concept of automated scholarly paper review and found that computerized systems are already capable of performing format examination, plagiarism detection, and structural screening of manuscripts, tasks that traditionally required extensive manual effort from editors and reviewers. Their review identified format examination as one of the most mature functions of automated review systems, noting that document parsing tools can compare a submitted manuscript against a standard template to detect deviations in formatting. This finding directly supports the technical feasibility of PaperPilot's document analysis feature, which likewise checks an academic document's formatting against institutional standards.

Similarly, a review published by Proof-Reading-Service.com (2025) described how AI-powered manuscript screening tools use natural language processing and large academic databases to verify compliance with formatting guidelines, detect inconsistencies, and assess the overall readiness of a manuscript for submission. The review emphasized that such systems are most effective when used to automate repetitive, rule-based technical checks, while more nuanced judgments, such as

evaluating the soundness of a study's methodology, remain the responsibility of human reviewers. This distinction reinforces PaperPilot's defined scope, which focuses on the technical compliance of academic documents rather than the evaluation of research content, methodology, or originality.

Beyond document formatting, artificial intelligence has increasingly been applied to automate technical compliance checking in academic writing. Theobald (2023) explained that AI is now widely used in scholarly publishing to assist with manuscript screening, formatting validation, and other repetitive editorial tasks before peer review. The study emphasized that AI is particularly effective in evaluating technical aspects of academic documents, allowing authors to identify formatting inconsistencies and improve compliance with publication requirements while leaving content evaluation to human reviewers. These findings support the development of PaperPilot's document analysis feature, which automatically checks document structure, formatting, and citation style compliance and provides users with feedback to improve their academic documents before submission.

A central feature of PaperPilot is the integration of Explainable Artificial Intelligence (XAI) to help users understand why a document was flagged for a particular issue. Vilone and Longo (2020) conducted a systematic review of XAI methods and observed that the growing use of machine learning, particularly deep learning, has produced highly accurate but poorly interpretable models, prompting the development of techniques that make automated decisions more transparent and understandable to end users. Their classification of XAI methods provides a conceptual basis for designing a system in which users are not simply given a list of errors, but are also given an explanation of how each issue was identified.

The application of XAI specifically within educational settings has also been explored. Choi et al. (2025) conducted a systematic literature review of XAI techniques used to interpret student performance predictions in computer science and STEM education. Their review found that explanation methods such as SHapley Additive exPlanations (SHAP) were the most frequently used technique for helping students and educators understand the reasoning behind an automated system's output, and that such explanations supported better-informed interventions and decision-making. These findings support PaperPilot's intent to use XAI not merely as a technical feature, but as a way of making automated compliance feedback more transparent, trustworthy, and actionable for student researchers and their advisers.

Since PaperPilot is ultimately designed to give users feedback on their academic documents, it is also grounded in literature concerning the effectiveness of AI-generated feedback on writing quality. Song and Song (2023) conducted a mixed-methods study on the use of ChatGPT-based feedback among English as a Foreign Language (EFL) students and found that the group that received AI-assisted feedback demonstrated significant improvement in organization, coherence, grammar, and vocabulary compared to a control group that received only traditional instruction. Their findings suggest that automated feedback, when properly designed, can meaningfully improve specific, measurable aspects of a learner's written output.

A more recent quantitative study by Alnemrat et al. (2025) compared AI-generated feedback with teacher-generated feedback on the argumentative writing of EFL learners at different proficiency levels. The results showed that both feedback types produced significant improvement in writing performance, indicating that AI-generated feedback can serve as a reliable complement to, though not necessarily a full replacement for, human evaluation. This supports PaperPilot's design philosophy of positioning automated compliance feedback as a preparatory tool that assists students before their academic documents are reviewed by advisers and panel members, rather than as a substitute for expert academic judgment.

Taken together, the reviewed literature shows that the individual components proposed for PaperPilot automated format and structure checking, citation formatting, Explainable Artificial Intelligence (XAI), and AI-generated feedback are each independently supported by existing research and, in several cases, already implemented in isolated tools. However, the reviewed studies also reveal a gap: most existing systems focus primarily on journal-level manuscript screening and technical compliance checking (Lin et al., 2023; Proof-Reading-Service.com, 2025; Theobald, 2023), while XAI applications in education have largely been applied to student performance prediction rather than document compliance (Choi et al., 2025). PaperPilot addresses this gap by integrating document structure analysis, formatting validation, multi-style citation formatting, AI-generated feedback, and explainable AI into a single system designed specifically to assist undergraduate and graduate students in preparing academic documents that comply with institutional writing standards.

Review of Related Studies

Recent developments in academic writing-support technology explore various aspects of citation management and document formatting, ranging from AI-assisted

reference generators to integrated paper-formatting systems that connect academic writers directly with citation databases. These innovations aim to address growing concerns about citation accuracy, formatting compliance, and administrative burden through technological solutions. The academic writing-support landscape has evolved significantly, with companies developing specialized tools that leverage automation, cloud syncing, and browser-based technology to streamline reference collection, improve citation accuracy, and support academic document preparation.

PERRLA, founded in 1998 by Gary McGuire, employs a human-built reference engine to format academic documents in APA, MLA, and Turabian styles while automatically setting up margins, headers, page numbers, and title pages. The system integrates across Microsoft Word, PERRLA Online, and Google Docs, making it adaptable to different writing environments. Despite these advantages, PERRLA requires a paid subscription beyond its trial period and focuses solely on generating citations and a paper template rather than analyzing an already-completed document's structure. The platform operates through desktop, web, and browser-extension applications with nightly customer support, monetizing through per-student and per-institution subscription plans (PERRLA, n.d.).

Zotero, created in 2006 by researchers at the Roy Rosenzweig Center for History and New Media at George Mason University, takes a different approach by using a free, open-source browser extension to automatically detect and collect bibliographic metadata from academic databases and websites. Its focus on collaborative organization distinguishes it from purely citation-generation tools, letting researchers sort references into collections, attach PDFs, and share libraries with research groups. However, this open architecture also means Zotero depends on user-installed citation style files and browser connectors to function consistently across platforms. Zotero's non-profit funding model, supported by institutional grants rather than paid subscriptions, forms the core of its sustainability strategy, supporting its desktop, web, and mobile applications with community-forum support (Zotero, n.d.).

The academic reference-management landscape includes several specialized solutions addressing different aspects of the citation cycle. Mendeley, launched in 2007 by Victor Henning, Jan Reichelt, and Paul Föckler and later acquired by Elsevier in 2013, automatically extracts metadata from PDFs and combines reference management with academic networking features, though its cloud storage tiers place limits on how much free users can keep synced. EazyPaper, founded in 2004 by Michael Hu, imports bibliographic data directly from library databases and online bookstores into Microsoft Word, but functions strictly as a paid Word add-in with no independent web or mobile version (Mendeley, n.d.; EazyPaper, n.d.).

Cite This For Me, launched in October 2010, offers a free web-based platform that connects users directly with citation lookup services while automatically generating references from a DOI, ISBN, or URL. The platform allows immediate reference creation without requiring an account, creating an accessible entry point for citation

building. However, Cite This For Me's free tier displays advertisements and restricts users to a limited style selection unless they subscribe to premium access, and it offers no way to verify whether an entire document's structure or formatting complies with institutional requirements. The platform operates through a web application and browser extension with self-service help guides, employing a freemium model that monetizes through premium subscriptions and advertising (Cite This For Me, n.d.).

Collectively, these technological solutions highlight ongoing advancements and a persistent gap in academic writing-support systems. Automated metadata extraction, cloud-based syncing, open citation-style libraries, and browser-based reference collection all contribute to reducing the manual effort of citation formatting. However, across all five systems, the underlying function remains generative producing citations, references, or a formatted paper template. None of them re-analyze a completed academic document to detect formatting deviations against institution-specific standards, nor do they return a compliance score or an explainable account of why an issue was flagged. Future developments in this field should focus on extending citation and formatting tools into the diagnostic stage of academic writing, enabling systems to independently verify a finished document's compliance and explain detected issues to the user, ultimately reducing reliance on manual adviser review and promoting consistent formatting compliance across academic institutions.

Comparative Matrix

This section presents a comparative matrix utilized to organize and evaluate relevant applications alongside the Paperpilot platform. It enables the researchers to systematically compare the features, limitations, platform specifics, support options, and monetization strategies of each application. This aids in the decision-making and analysis by providing a structured summary of similarities, differences, and unique aspects among the applications included in the matrix.

Table

1

COMPARATIVE

MATRIX

Related Studies	Features	Limitations	Platform	Support	Monetization Scheme

Name: PERRLA Academic Writing Software URL: Owner: PERRLA, LLC	1. APA, MLA, and Turabian formatting 2. Automatically formats margins, headers, page numbers, and title pages	1. Does not analyze completed documents for formatting compliance 2. No institution-specific compliance checking	1. Desktop Application, 2. Web Application	1. Academic paper formatting and citation management	Subscription

Table 1.2

COMPARATIVE MATRIX CONT'D

Related Studies	**Features**	**Limitations**	**Platform**	**Support**	**Monetization Scheme**
Name: Mendeley Reference Manager URL: https://www.mendeley.com Owner: Elsevier	1. PDF organization 2. Citation generation 3. Research management	1. Metadata errors may affect accuracy 2. No citation validation system	Web-based, Desktop application	Academic reference management system	Freemium
Name: Zotero Reference Manager URL: https://www.zotero.org Owner: Corporation for Digital Scholarship	1. Reference organization	1. Depends on user-input metadata	Web-based, Desktop application	Reference management tool	Free
Name: EazyPaper URL: https://www.eazypaper.com Owner: Michael Hu	1. Citation generation 2.APA/MLA formatting support	1. Microsoft Word only 2. No document structure or formatting analysis	Microsoft Word Add-in	Citation generation for Microsoft Word	Freemium

Paperpilot **Year:** 2026 **Owner:** PaperPiloto	1.AI-powered academic document analysis 2.Institution-specific formatting compliance checking 3.Compliance report generation 4.Explainable AI (XAI) feedback and recommendations	1. Supports only supported institutional formatting templates	1. Web Application, 2.Mobile Application	1. Academic document compliance checking based on institutional standards	1. Freemium

Tables 1.1 to 1.2 present the evaluation of various academic writing, citation management, and paper-formatting systems based on their features, limitations, platform support, and monetization schemes. The reviewed systems include PERRLA, Zotero, Mendeley, EazyPaper, and Cite This For Me, all of which provide tools that assist users in preparing academic documents through citation generation, reference management, and formatting support.

PERRLA automates paper formatting by applying APA, MLA, and Turabian styles while generating citations and formatting essential document elements such as margins, headers, page numbers, and title pages. Similarly, EazyPaper assists users in formatting papers and generating citations through Microsoft Word integration. However, both systems focus on creating properly formatted documents rather than evaluating whether completed academic documents comply with institution-specific formatting requirements.

Reference management tools such as Zotero and Mendeley help researchers organize references, manage bibliographic information, and generate citations. These systems improve reference organization and research management but do not verify formatting compliance or adherence to institutional academic document standards. Likewise, Cite This For Me simplifies citation generation using DOI, ISBN, and URL lookups but is limited to creating references and does not evaluate the formatting of an entire academic document.

Overall, the existing systems perform specialized functions such as citation generation, reference organization, and paper formatting. However, none of them are specifically designed to evaluate completed academic documents based on institution-specific formatting standards. They do not verify formatting compliance or generate detailed compliance reports that explain detected formatting issues.

To address this gap, the proposed system, PaperPilot, is designed as an AI-powered academic document compliance-checking system that evaluates completed academic documents according to institution-specific formatting standards. The system verifies formatting elements such as font style, font size, margins, line spacing, headings, page numbering, and citation format. It also generates comprehensive compliance reports with Explainable Artificial Intelligence (XAI)-based feedback, enabling researchers to understand detected formatting issues and apply the necessary corrections before submitting their academic documents for adviser evaluation.

Table 1.3
COMPARATIVE MATRIX

Features	Zotero	PERRLA	Eazypaper	Mendeley	PaperPilot
APA/MLA/IEEE Formatting Support	✔	✔	✔	✔	✔
Institution-Specific Formatting Compliance Checking	✗	✗	✗	✗	✔
Compliance Report Generation	✔	✗	✗	✔	✔
Explainable AI (XAI) Feedback	✗	✗	✗	✗	✔

Table 1.3 presents a feature-based comparison of Zotero, PERRLA, EazyPaper, Mendeley, and the proposed system, PaperPilot. The comparison shows that all systems support APA/MLA/IEEE formatting and document management. However, only PaperPilot provides institution-specific formatting compliance checking and integrates Explainable Artificial Intelligence (XAI) to explain detected formatting issues and recommended corrections. While Zotero and Mendeley offer limited report generation, they do not verify compliance with university manuscript standards. By combining formatting support, document management, compliance checking, compliance report generation, and XAI-based feedback, PaperPilot provides a more comprehensive

solution for evaluating academic documents based on institutional formatting requirements.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the methodologies used in the design and development of PaperPilot, an AI-powered academic document analysis and compliance checking system designed to help students and researchers prepare academic documents that comply with academic writing and formatting standards. The planning phase employed the Business Model Canvas, Validation Board, Program Workflow, Gantt Chart, and Functional Decomposition Diagram to define the project's objectives, scope, and development process. The design phase included the creation of Use Case Diagrams, User Interface (UI) designs, Storyboards, an Entity-Relationship Diagram (ERD), and a Data Dictionary to illustrate the system's functionality, user interactions, and database structure. During the development phase, the technology stack, system requirements, and program specifications were established, while a modular approach was used to develop features for academic document formatting support, automated document analysis, compliance checking, and Explainable Artificial Intelligence (XAI) to provide clear explanations of the analysis results. These methodologies ensure that PaperPilot is functional, reliable, user-friendly, and capable of assisting users in preparing academic documents that comply with academic writing standards.

Software Engineering Methodology

The Agile software development life cycle model has been chosen as the preferred approach for developing PaperPilot, an intelligent academic document formatting and compliance system for researchers. Agile is a flexible and iterative approach that promotes collaboration, continuous improvement, and the delivery of functional features in incremental stages. It is well-suited for this project as it allows the system to adapt to evolving researcher needs and feedback, ensuring that features such as formatting support and compliance checking are continuously refined and improved throughout the development process.



Figure 2: **Agile Methodology**

The development team will utilize the iterative nature of the Agile methodology to effectively monitor project progress and ensure timely completion of each development phase. This approach is appropriate for Paperpilot because the system consists of multiple modules, including academic document upload, institution-specific formatting verification, compliance analysis, and compliance report generation. The flexibility of Agile enables each sprint to deliver a functional system increment that can be evaluated and refined using feedback from students, thesis advisers, and academic departments. This iterative process minimizes the need for extensive revisions during later stages of development and ensures that the system remains aligned with institutional formatting standards and user requirements.

Requirements Phase. This phase focuses on identifying the common challenges encountered by students and researchers when preparing academic documents. The research team reviews academic writing standards and identifies common issues related to document formatting. Based on these findings, the system requirements are defined, including academic document formatting support, automated document analysis, compliance checking, and Explainable Artificial Intelligence (XAI) to provide clear explanations of the analysis results.

Design Phase. During this phase, the identified requirements are translated into a structured system architecture and a user-friendly interface. The design illustrates how users upload academic documents, select an academic writing style, and receive analysis results and compliance feedback. Wireframes and prototypes are created for document upload, document analysis, issue visualization, XAI explanations, and report generation. The database and system architecture are also designed to manage user accounts, support academic writing styles, uploaded documents, analysis results, and compliance reports.

Development Phase. This phase involves implementing the system based on the approved design. Core modules are developed to support the formatting of academic documents, analyze document formatting and citation format, and detect compliance issues based on APA, MLA, and IEEE writing styles. The system also integrates Explainable Artificial Intelligence (XAI) to provide users with clear explanations of detected issues and recommended corrections. Development is carried out iteratively to allow continuous refinement and feature enhancement.

Testing Phase. The system undergoes unit, integration, and usability testing to ensure accurate document analysis and reliable performance. The academic documents

are tested using the supported academic writing styles to verify the accuracy of document analysis, compliance checking, and XAI-generated explanations. Identified issu es are resolved before deployment.

Deployment Phase. The completed prototype is deployed in a controlled environment and made available to selected students, advisers, and academic personnel for evaluation. Users assess the system's functionality, ease of use, accuracy of document analysis, quality of compliance feedback, and clarity of the Explainable AI (XAI) explanations.

Review Phase. The review phase evaluates user feedback and system performance after deployment. Feedback from students, advisers, and academic departments is analyzed to improve the accuracy of document analysis, enhance usability, refine XAI explanations, and update the supported academic writing styles when necessary. This continuous evaluation ensures that PaperPilot remains an effective tool for supporting the formatting, analysis, and technical compliance of academic documents while assisting users in understanding the system's analysis results.

Table 2.1
SPRINT PLAN

Agile Phase	Sprint 1 Goals: Establish project foundation and gather requirements Duration: 2 Weeks	Sprint 2 Goals: Complete literature synthesis and finalize related studies. Duration: 2 Weeks	Sprint 3 Goals: Finalize system analysis and initial design. Duration: 2 Weeks	Sprint 4 Goals: Complete Development Environment and begin initial coding Duration: 2 Weeks
Requirements	Define the PaperPilot objectives, problem background, rationale, scope and limitations, significance of the study, flow of the study, and definition of terms, including 10 local and 10 international journals.	Gather and synthesize 10 local and 10 international journals, identify supporting theories and frameworks, and evaluate existing academic document checking systems.	Finalize the functional and non-functional requirements for PaperPilot modules and Chapter III system analysis and design artifacts.	Prepare the development environment, tools, and initial app structure.

Design	Design the structure and organization of Chapter I based on the defined research objectives and requirements.	Design the research framework for Chapter II through the organization of the theoretical background, related literature, related studies, comparative analysis, and Business Model Canvas.	Design the Program Workflow, Functional Decomposition Diagram, Use Case Diagram/DFD, Validation Board, Gantt Chart, User Interface, Storyboard, Entity Relationship Diagram, Data Dictionary, and Network Model.	Design the repository architecture, frontend structure, backend architecture, Firebase structure, API structure, database structure, and reusable system components.
Development	Develop the PaperPilot research documentation structure for Chapter I content.	Finalize the Chapter II research content, including the Theoretical Background, Review of Related Literature, Review of Related Studies, Comparative Matrix, and Business Model Canvas.	Finalize Chapter III contents	Initialize Web, React Native, Node.js/Express, FastAPI, Firebase, GitHub branches, Android Studio, VS Code, environment variables, shared components, and project board. Begin initial coding by creating the basic web and mobile application structure, reusable components, and initial authentication framework.
Testing	Validate the completeness, consistency, and alignment of Chapter I with the PaperPilot research objectives and problem statement.	Validate the Comparative Matrix against the evaluated existing systems.	Validate diagram consistency and completeness against defined requirements.	Verify web/mobile builds, backend APIs, Git workflow, and Firebase integration.

Deployment	Organize collected research materials and literature for review.	Compile Chapter II and its supporting research materials for review.	Compile documents for review. And prepare for Oral Defense.	Deploy the initial project repository for team-wide access.
Review	Review Sprint 1 outputs (Chapter I draft).	Review Sprint 2 outputs (Chapter II).	Review Sprint 3 outputs (Chapter III).	Review Sprint 4 progress and finalize proposal documentation.

Table 2.2
SPRINT PLAN

Agile Phase	Sprint 5 Goal: Development Foundation Finalization. Duration: 2 Weeks	Sprint 6 Goal: Complete backend services setup. Duration: 2 Weeks	Sprint 7 Goal: Build the core Account Management authentication features. Duration: 2 Weeks	Sprint 8 Goal: Begin Manuscript Management Module. Duration: 2 Weeks
Requirements	Finalize the requirements for the application structure, shared components, and authentication framework.	Define the data fields, authentication rules, access permissions, and API requirements needed to connect PaperPilot user accounts with Firebase and backend services.	Define the remaining account-management requirements and establish the access requirements for manuscript storage, retrieval, and user-specific document ownership.	Define the manuscript processing requirements, including supported file types, format-mechanics selection, upload validation, storage handling, and document-processing inputs.
Design	Finalize the frontend component hierarchy, navigation structure, repository organization, Firebase configuration,	Design the user authentication flow, Users collection/schema, access-control rules, and communication between the frontend, Firebase, and	Design the account-management workflow and manuscript access structure, including profile management,	Design the manuscript upload workflow, file-validation process, format-mechanics selection interface, document storage structure, and initial processing pipeline.

	backend structure, and API communication flow.	backend services.	security controls, user-specific storage, and manuscript ownership.	
Development	Finalize the web and mobile application structures, reusable components, authentication framework.	Develop the user authentication and data layer by implementing Register, Login, Forgot Password, Firebase Authentication, Users collection, session handling	Develop the remaining Account Management functions, including Edit Profile, Profile Picture Upload, Change Password, Deactivate Account, and Logout, then connect them to the established user data layer.	Develop the initial Manuscript Management functions, including Format Mechanics selection, Manuscript Upload, File-Type Validation, manuscript storage, and user-specific manuscript access.
Testing	Verify the finalized frontend application structure.	Test registration, login, password recovery, authentication states, user data creation, access permissions, and Firebase/backend communication.	Test profile management, profile picture handling, password changes, account deactivation, logout, and authorized access to user-specific data.	Test manuscript uploading, file-type validation, format-mechanics selection, storage, retrieval, and user-to-manuscript access controls.
Deployment	Deploy the finalized development foundation.	Deploy the functional authentication and user-data layer to the development environment for integration with account-management functions.	Deploy the completed Account Management Module with secured user access and manuscript ownership support.	Deploy the initial Manuscript Management Module and validated upload workflow to the development/staging environment for the succeeding compliance-checking

				implementation.
Review	Review Sprint 5 progress.	Review Sprint 6 progress and confirm readiness for module development.	Review Sprint 7 outputs (Account Management progress).	Review Sprint 8 outputs (Account Management completion, Manuscript Management progress).

Table 2.3
SPRINT PLAN

Agile Phase	Sprint 9 Goals: Complete Manuscript Management and begin AI Analysis Structure. Duration: 2 Weeks	Sprint 10 Goals: Finalize the AI Analysis Structure Module. Duration: 2 Weeks	Sprint 11 Goals: Build the Overall Results Module. Duration: 2 Weeks	Sprint 12 Goals: Final System Validation, Documentation & Defense Preparation. Duration: 2 Weeks
Requirements	Define requirements for manuscript versioning, re-upload workflows, and formatting validation rules including font, spacing, and indentation consistency.	Define requirements for advanced AI document structure validation including margins, page breaks, table layouts, pagination rules, and citation format detection.	Define requirements for compliance scoring logic, automated report generation, and version-to-version comparison of manuscript outputs.	Finalize the system interface, compliance-checking workflow, user guides, presentation materials, system diagrams, and defense demonstration flow based on the completed system.
Design	Design the extended compliance-checking logic for formatting validation and version control structure for manuscripts	Design citation detection architecture supporting APA, MLA, IEEE, and Chicago styles with rule-based and pattern-based validation.	Design the compliance scoring model, results breakdown structure, and report layout for user interpretation and export.	Finalize the PaperPilot system by incorporating approved corrections from previous testing, completing remaining documentation,

				and preparing the final system build for deployment.
Development	Implement Manage Versions, Upload & Analyze, and Re-Upload Manuscript features; develop font, spacing, and indentation validation functions.	Implement margin validation, page-break detection, table structure analysis, pagination verification, and citation format checking engine.	Develop compliance scoring, score breakdown visualization, issue/warning detection system, and PDF report generation with version comparison.	Finalize the Account Management, Manuscript Management, AI Analysis Structure, and Overall Results modules.
Testing	Test manuscript versioning accuracy, re-upload, and formatting validation reliability across multiple document samples.	Test structural format validation accuracy across diverse academic manuscript formats.	Test scoring consistency, report generation accuracy, and version comparison correctness.	Conduct peer testing of the completed PaperPilot system and evaluate usability, functionality, and overall system performance. Test sample manuscripts using the compliance-checking functions to verify formatting, structure, citation, and identified compliance issues. Resolve and retest identified defects.
Deployment	Deploy Manuscript Management enhancements and initial AI Analysis	Deploy completed AI Analysis module.	Deploy completed Results and Reporting module for system-wide	Prepare the final production-ready build, complete system

	components to staging for validation.		evaluation.	documentation, organize the project repository, and prepare the final deployment package.
Review	Review Sprint 9 outputs.	Review Sprint 10 outputs (AI Analysis Structure completion).	Review Sprint 11 outputs (Overall Results completion).	Review the complete PaperPilot system and research manuscript with the development team. Finalize oral defense documents, presentation slides, system demonstration flow, user guide, testing results, and supporting defense materials. Conduct a final demonstration rehearsal and confirm readiness for oral defense and deployment.

Table 2.3 presents the complete Agile Sprint Plan for PaperPilot, consisting of twelve 12 two-week sprints. Each sprint follows the Requirements, Design, Development, Testing, Deployment, and Review phases to ensure continuous development, testing, and improvement of the system.

Sprints 1 to 3 focused on research preparation, requirements gathering, system analysis, and design. The researchers developed the research chapters, reviewed related literature and studies, finalized the methodology and system requirements, and prepared the major design artifacts such as the BMC, FDD, Use Case Diagram, ERD, Data Dictionary, UI Design, and Network Model. These outputs were reviewed and finalized before development.

Sprint 4 focused on preparing the development environment. The researchers finalized the technology stack, repository and database structure, backend architecture, APIs, development tools, and project configurations. Initial coding also began by creating the basic web and mobile application structures, reusable components, and authentication framework.

Sprints 5 to 6 focused on the application foundation and account/manuscript modules. The researchers developed the web and mobile interfaces, authentication, user profiles, manuscript upload, file validation, version history, Firebase Storage, and related APIs. These features were tested and deployed for review.

Sprints 7 to 9 focused on the AI document analysis and compliance features. The system was developed to extract document elements from DOCX and PDF files and check formatting requirements such as fonts, spacing, margins, headings, tables, and page numbers. Support for APA, MLA, and IEEE standards, institution-specific rules, XAI explanations, warnings, and recommendations was also implemented and tested.

Sprints 10 to 12 focused on results, integration, optimization, and final deployment. The researchers developed the compliance dashboard, score breakdown, reports, and version comparison features. The web, mobile, backend, database, and AI components were integrated and tested through regression testing and User Acceptance Testing (UAT). The final sprint included bug fixing, documentation, deployment, mock defense, and final preparation.

Planning/Conception-Initiation Phase

During the Planning/Conception-Initiation phase, the project's objectives, scope, and system requirements are clearly defined. This phase includes the development of a Validation Board that identifies the common challenges encountered by students and researchers when preparing academic documents, such as issues with formatting and compliance with academic writing standards, while presenting PaperPilot as an AI-powered solution to address these challenges. A Program Workflow is created to illustrate how users interact with the system, from uploading academic documents and selecting an academic writing style to receiving document analysis results, compliance feedback, and Explainable Artificial Intelligence (XAI) explanations. A Business Model Canvas is also developed to evaluate the system's feasibility, usability, and value in supporting the preparation of academic documents. In addition, a Gantt Chart is prepared to define the project timeline, development activities,

and target completion dates.

Lastly, a Functional Decomposition Diagram is constructed to divide the system into its major functional components, including academic document formatting support, citation format checking, and compliance checking. This provides a clear understanding of the system's processes and supports an organized and efficient development process.

Business Model Canvas

The following table presents the core building blocks that form the framework for the design and deployment of PaperPilot. This business model serves as a strategic guide to ensure the system delivers value to student researchers and academic institutions by streamlining the academic document formatting compliance checking process. PaperPilot's goal is to address inefficiencies in manual compliance review, reduce the time researchers spend correcting formatting errors, and support institutional formatting standards through AI-assisted formatting analysis and automated formatting compliance validation.

Table 3

BUSINESS MODEL CANVAS

Key Partners	**Key Activities**	**Value Propositions**	**Customer Relationships**	**Customer Segments**

• Research Coordinator • Thesis and Capstone Advisers • Resear-chers	• Development and maintenance of the PaperPilot platform • Academic document formatting checking • Testing and improving document analysis accuracy • Collecting user feedback for system enhancement	• Automatically checks compliance with institutional standards • Detects formatting errors such as font style, font size, margins, spacing, headings, and pagination • Verifies adherence to institution-required formats	• Self-service web platform • User-friendly compliance reports • Technical support and • help documentatio n • Continuous improvement through user feedback • Collaboration with academic Institutions for customized standards	• Researchers

	Key Resources		**Channels**
	• Programming tools and frameworks • Database system for storing documents • Institution Academic document s formattin g guideline. • Rule-based checking system		• Web-based and mobile application • Direct google chrome access • Google play listing

Cost Structures	**Revenue Streams**
• System development costs • Hosting and server expenses • Maintenance and updates • Software tools and licenses (if needed) • Testing and deployment costs	• Freemium -Free tier: 3 scans Per month -Paid tier : 50 scans

Table 3 presents the Business Model Canvas of PaperPilot, an academic document compliance checking system designed to assist researchers, students, and academic institutions in verifying academic documents based on university formatting standards. Key partners include researchers, thesis and capstone advisers, and research coordinators that support system implementation. The system's key activities focus on developing and maintaining the PaperPilot platform, designing and updating university-specific formatting and compliance rules, automating academic document formatting verification, generating compliance reports, improving document analysis accuracy, and continuously enhancing the system through user feedback. PaperPilot's value proposition lies in automatically evaluating academic documents against institutional requirements by detecting formatting issues involving font style, font size, margins, spacing, headings, pagination, and required citation formats, enabling researchers to identify and correct formatting errors before submission. Customer relationships are maintained through a self-service web and mobile platform, institution-based user access, user-friendly compliance reports, technical support, and continuous collaboration to accommodate institution-specific standards. The system primarily serves undergraduate and graduate researchers, thesis and capstone advisers, and research coordinators. Key resources include software developers, programming tools and frameworks, a database management system, academic formatting guidelines, and a rule-based compliance checking engine. PaperPilot is delivered through web and mobile applications, direct browser access. Operating costs include system development, maintenance, software licensing, testing, and deployment, while revenue may be generated through university-wide deployment contracts, customized compliance rule configuration services, system integration and implementation, and technical support and maintenance agreements.

Program Workflow

This section presents the program workflow of the system, illustrating the sequence of processes performed as the researcher interacts with its features. It provides a visual representation of the system's operations, showing how each function is executed, the decisions made during processing, and the transitions between different modules. Through these workflow diagrams, readers can better understand the logical flow of the system and how its components work together to accomplish each task efficiently.

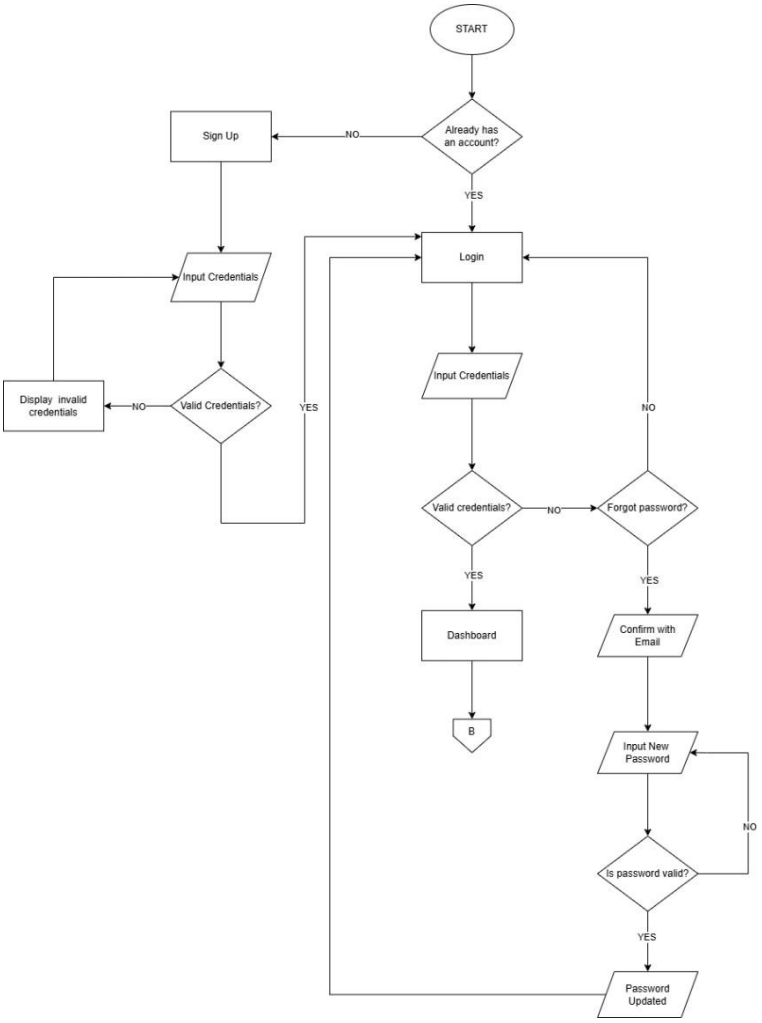


Figure 3: **Program Workflow (Researcher)**

Figure 3 presents the flowchart for the Account Management module of PaperPilot. The process begins when the researcher accesses the system and determines whether they already have an account. If the researcher does not have an existing account, they proceed to the Sign Up process, where they enter the required registration credentials. The system validates the submitted information, and if the credentials are invalid, an error message is displayed, prompting the researcher to re-enter the required information. Once the registration credentials are successfully validated, the researcher is redirected to the Login page. If the researcher already has an account, they proceed directly to the login process by entering their login credentials. The system then verifies the submitted credentials, and if they are valid, the researcher is granted access to the Dashboard, where they can access the system's features. If the login credentials are invalid, the researcher may choose the Forgot Password option to recover their account. The password recovery process requires the researcher to verify their identity through

email confirmation before entering a new password. The system validates the new password, and if it does not meet the required criteria, the researcher is prompted to enter another password. Once a valid password is provided, the system updates the account credentials and redirects the researcher back to the Login page to access the system. This workflow ensures secure account registration, authentication, and password recovery while maintaining controlled access to the manuscript compliance checking system.

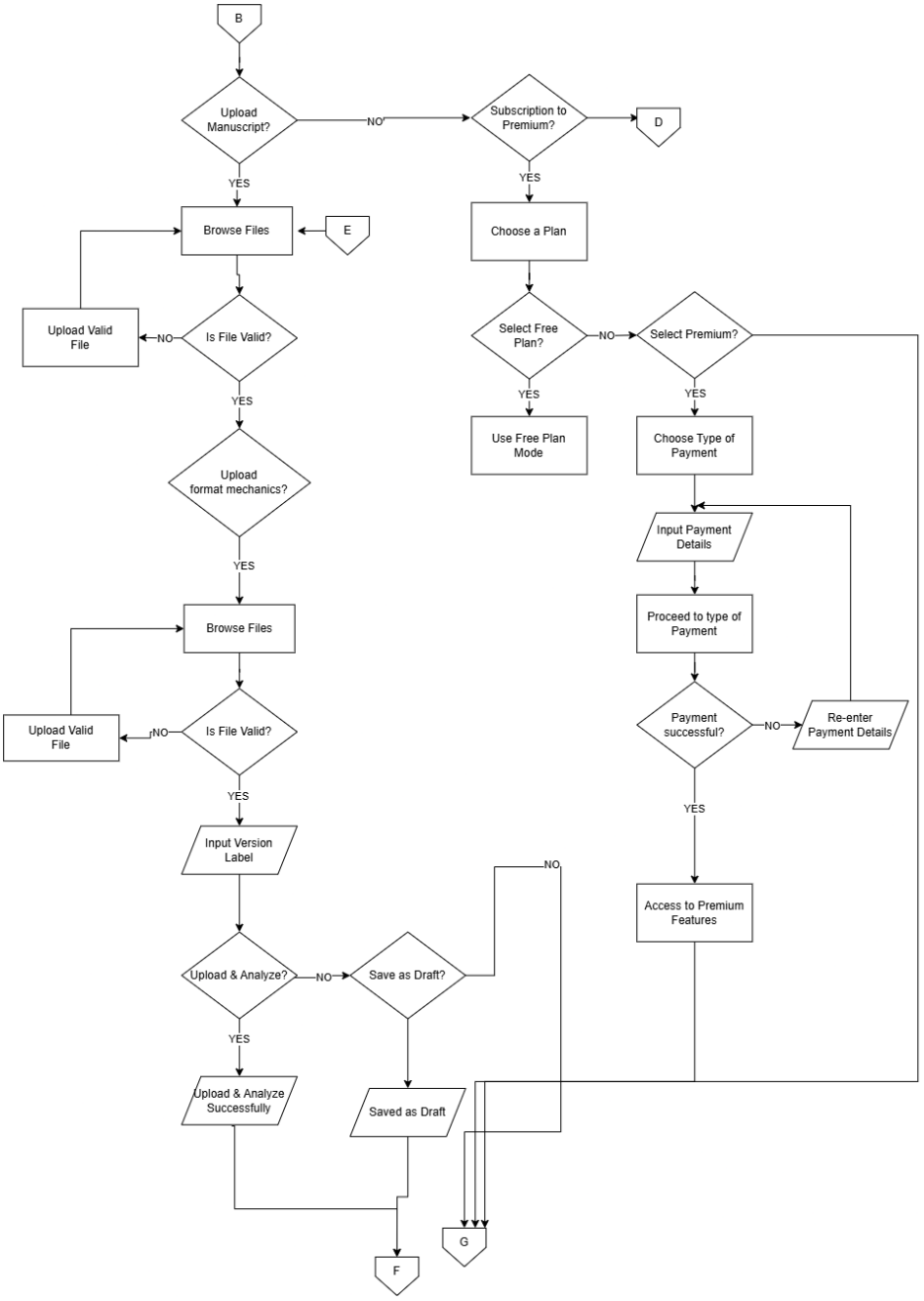


Figure 3.1: **Program Workflow (Researcher)**

Figure 3.1 presents the flowchart for the Manuscript Upload and Subscription Management process. The workflow begins when the researcher decides whether to upload a manuscript. If the researcher chooses to upload a file, the system prompts them to browse and select a manuscript, which is then validated to ensure it meets the supported file requirements. Once the file is successfully validated, the researcher provides the required manuscript information, including the institution, course or department, and version label. After completing the necessary details, the researcher may either proceed with the Upload and Analyze process, which uploads the manuscript and starts the compliance analysis, or choose to save the manuscript as a draft for future editing and submission. If the researcher does not upload a manuscript, the system provides the option to subscribe to a plan. The researcher may choose to continue using the Free Plan or subscribe to the Premium Plan. For premium subscriptions, the researcher selects a payment method, enters the required payment details, and completes the payment process. If the payment is unsuccessful, the system prompts the researcher to re-enter the payment information until the transaction is successfully completed. Once the payment is successful, the researcher gains access to premium features. This workflow ensures that researchers can efficiently manage manuscript uploads while providing flexible subscription options that accommodate both free and premium users.

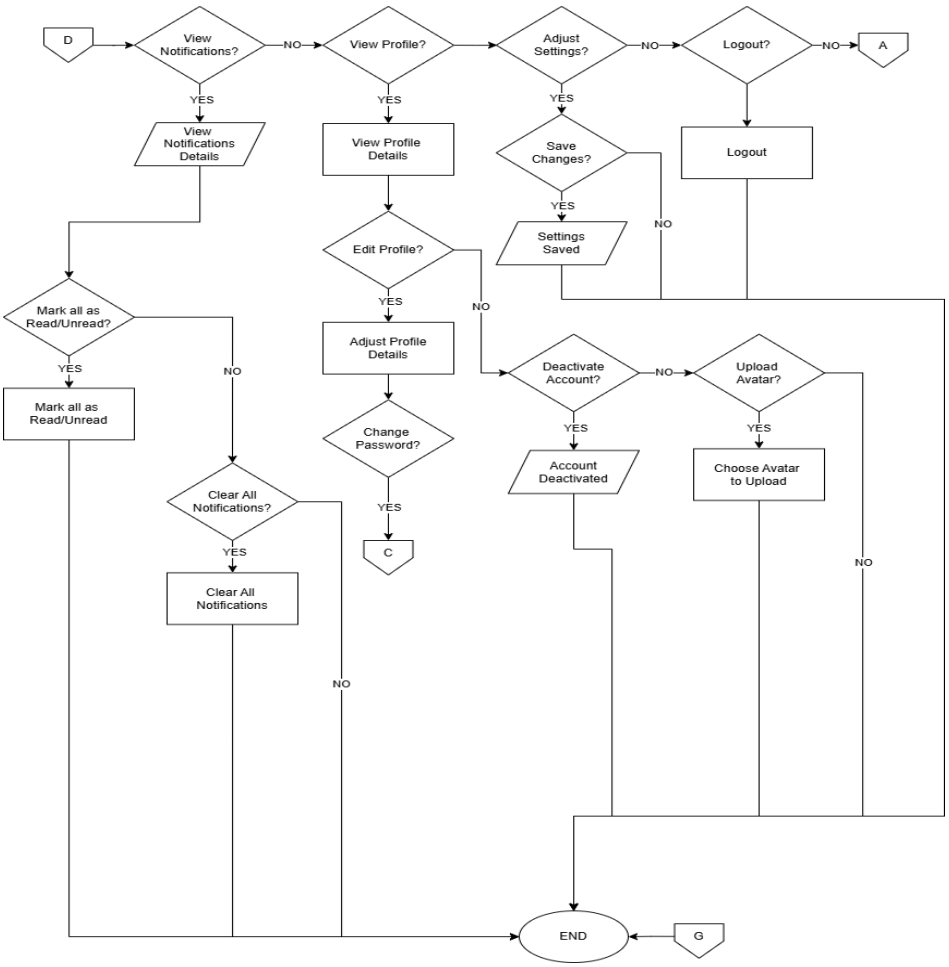


Figure 3.2: **Program Workflow (Researcher)**

Figure 3.2 presents the flowchart for the User Account Management process. The workflow begins when the researcher chooses to access account-related functions from the dashboard. The researcher may view notifications, access profile information, modify account settings, or log out of the system. When viewing notifications, the system displays the notification details and allows the researcher to either mark notifications as read or unread or clear all notifications. If the researcher accesses their profile, the system displays the profile details and provides the option to edit profile information, including changing the account password. The flowchart also illustrates additional account management functions available to the researcher. After updating profile information, the researcher may save the changes, deactivate the account, or upload a new avatar. If the researcher chooses to adjust the system settings, the changes are saved before returning to the account management process. When uploading an avatar, the researcher selects an image to update the profile picture, while selecting account deactivation permanently disables the account. The researcher may also log out

of the system at any time, which securely ends the current session. This workflow ensures that researchers can efficiently manage their account information, personalize their profiles, and maintain the security of their accounts.

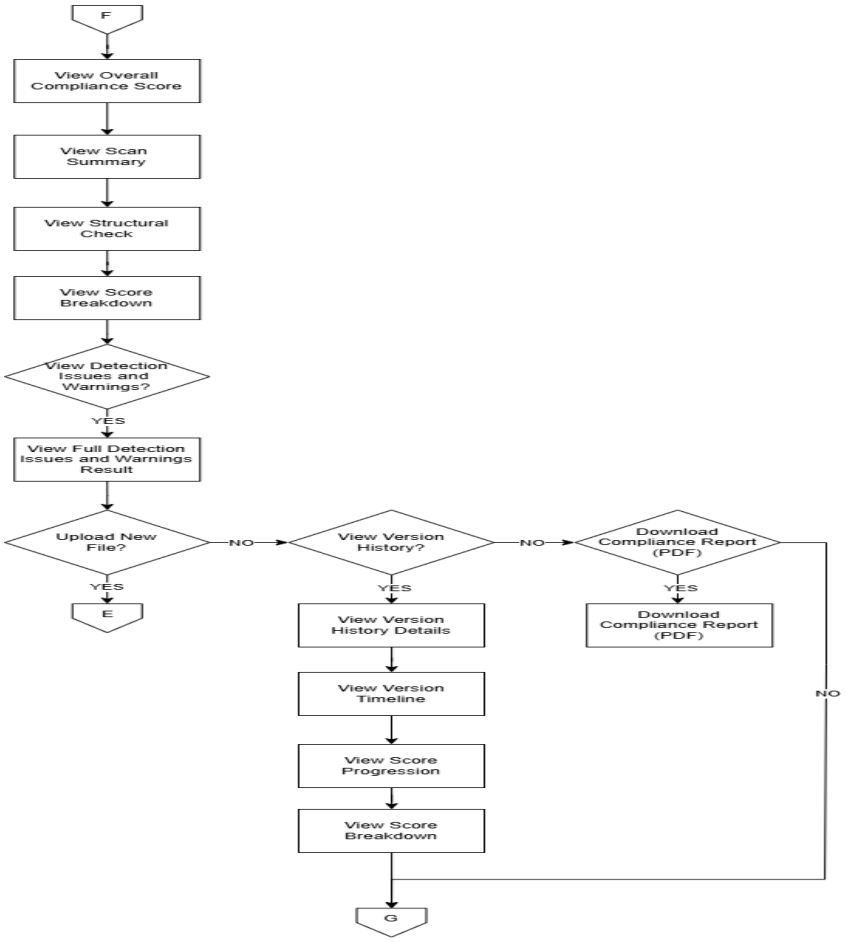


Figure 3.3: **Program Workflow (Researcher) CONT'D**

Figure 3.3 presents the flowchart for the Analysis Results process. The workflow begins after the manuscript has been successfully analyzed by the system. The researcher first views the Overall Compliance Score, which provides an overall assessment of the manuscript's compliance with the required formatting and structural standards. The system then displays the Scan Summary, followed by the Structural Check and the Score Breakdown, allowing the researcher to review the results of each evaluated component. If formatting or structural issues are detected, the researcher may choose to view the Full Detection Issues and Warnings Result, where all identified errors, warnings, and recommendations are presented to guide the revision of the manuscript. After reviewing the analysis results, the researcher is provided with several options for the next step. The researcher may upload a new manuscript to perform another compliance analysis, access the Version History to review previous

submissions, or download the Compliance Report (PDF) for documentation or sharing purposes. When Version History is selected, the system displays the version details, version timeline, score progression, and score breakdown, enabling the researcher to compare the compliance results of different manuscript versions. This workflow supports continuous manuscript improvement by allowing researchers to monitor revisions, track compliance progress, and generate reports for future reference.

Validation Board

This section presents the validation board which is a strategic tool used in the development process to systematically test and validate assumptions about a product or service. It helps teams identify the riskiest assumptions, design experiments to test these assumptions, and gather data to inform decision-making.

Table 4

VALIDATION BOARD

	1	2	3
Users	**Researchers**	**Thesis/Capstone Advisers**	**Research Coordinators**
Problem	Researcher struggle or overlook the formatting errors in their thesis and capstone manuscripts resulting in repeated revisions and delayed submissions.	Advisers spend significant time checking technical Academic document requirements instead of focusing on the research content, methodology, and findings.	Research Coordinators challenges in ensuring that all thesis and capstone manuscripts consistently comply with institutional standards before approval.
Solution	Provide an intelligent compliance-checking system that evaluates formatting, and document structure for correction.	Generate compliance reports that summarize manuscript issues and required revisions before adviser evaluation.	Implement the institution-specific compliance standards that help ensure consistent manuscript quality and adherence to academic

			requirements.
Riskiest Assumption	Researchers will trust the system's Recommendations and apply the suggested corrections before submission.	Advisers will consider the generated Compliance reports accurate and useful during manuscript evaluation.	Research Coordinators will support the use of PaperPilot as a preliminary manuscript compliance-checking tool.
Method & Success Criteria	Researchers successfully identify and correct manuscript compliance issues before adviser submission.	Reduction in adviser time spent checking formatting, and document structure.	Manuscripts meet institutional compliance requirements before final review.
Result & Destination	Researchers submit higher-quality thesis and capstone manuscripts with fewer compliance-related errors and revisions.	Advisers can dedicate more time to evaluating research content, methodology, analysis, and findings, improving the overall quality of academic guidance.	Students will achieve more consistent manuscript quality, streamlined approval processes, and improved compliance with institutional academic standards.
Learning	Researchers feedback helps improve the accuracy and usefulness of compliance recommendations.	Adviser feedback helps refine report generation, issue detection, and recommendation accuracy.	Feedback helps improve compliance rules, customization features, and alignment with university manuscript standards.

Table 4 presents the Validation Board for PaperPilot, which identifies the primary users, their existing challenges, proposed solutions, assumptions, validation

methods, expected outcomes, and learning objectives. The validation focuses on three key user groups: researchers, thesis and capstone advisers, and research coordinators. Researchers often experience difficulties in preparing academic documents, particularly in maintaining proper formatting, citation format, and compliance with academic writing standards. As a result, they may encounter repeated revisions and delays before submission. Advisers spend significant time reviewing technical requirements instead of focusing on the quality of the research, while research coordinators face challenges in efficiently reviewing the technical compliance of academic documents. To address these challenges, PaperPilot provides an AI-powered academic document analysis and compliance checking system that supports document formatting, analyzes document formatting and citation format, and provides Explainable Artificial Intelligence (XAI) to clearly explain the analysis results and recommended corrections. The validation assumes that users will apply the system's recommendations to improve their academic documents, advisers will find the analysis results useful during the review process, and research coordinators will use the system to support the initial technical review of academic documents.

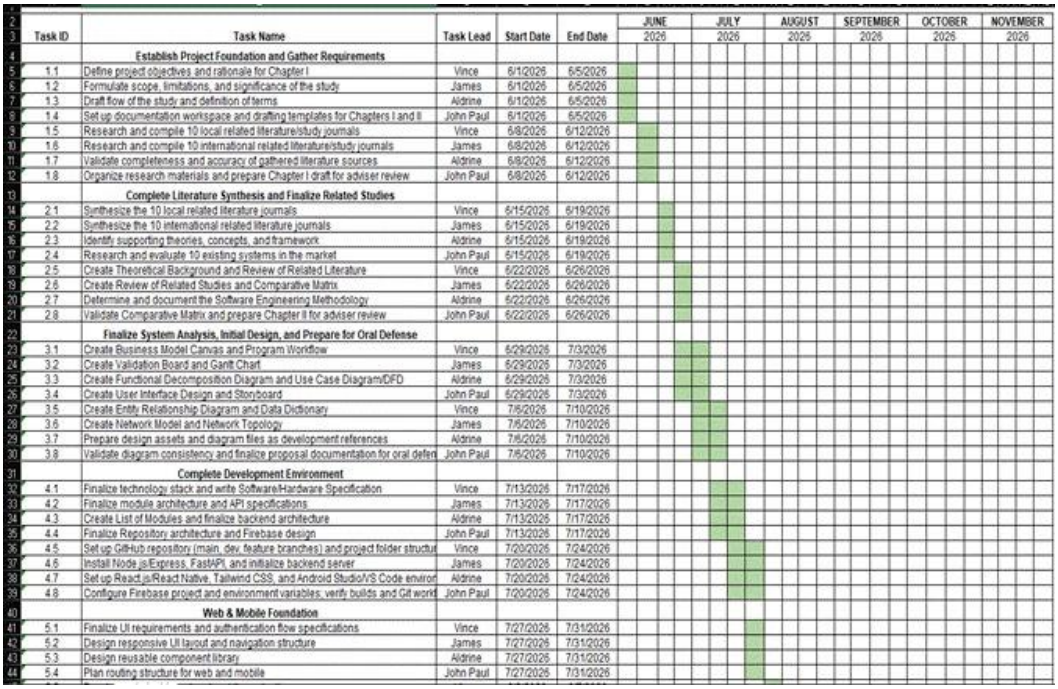
Success is measured by the users' ability to identify and correct technical compliance issues before submission, the reduction of time spent by advisers on manual technical checking, and the improved efficiency of research coordinators in reviewing academic documents. The expected outcomes include better-prepared academic documents, a more efficient review process, increased compliance with academic writing standards, and continuous system improvement through feedback collected from researchers, advisers, and research coordinators.

Gantt Chart

This section presents the researchers' Gantt chart for the project. It visually shows the schedule of tasks and the time needed to finish each one. The chart uses horizontal bars to represent each activity, showing when it starts and ends during the development process. Tasks are arranged in order, and some tasks depend on others to be done first. This helps the team understand the right flow of work. With the Gantt chart, the researchers can easily see what needs to be done, track how the project is moving, and manage time wisely. It also helps in making sure that all parts of the system are completed on time and that everyone stays on track throughout the project.

Table 5.1

GANTT CHART

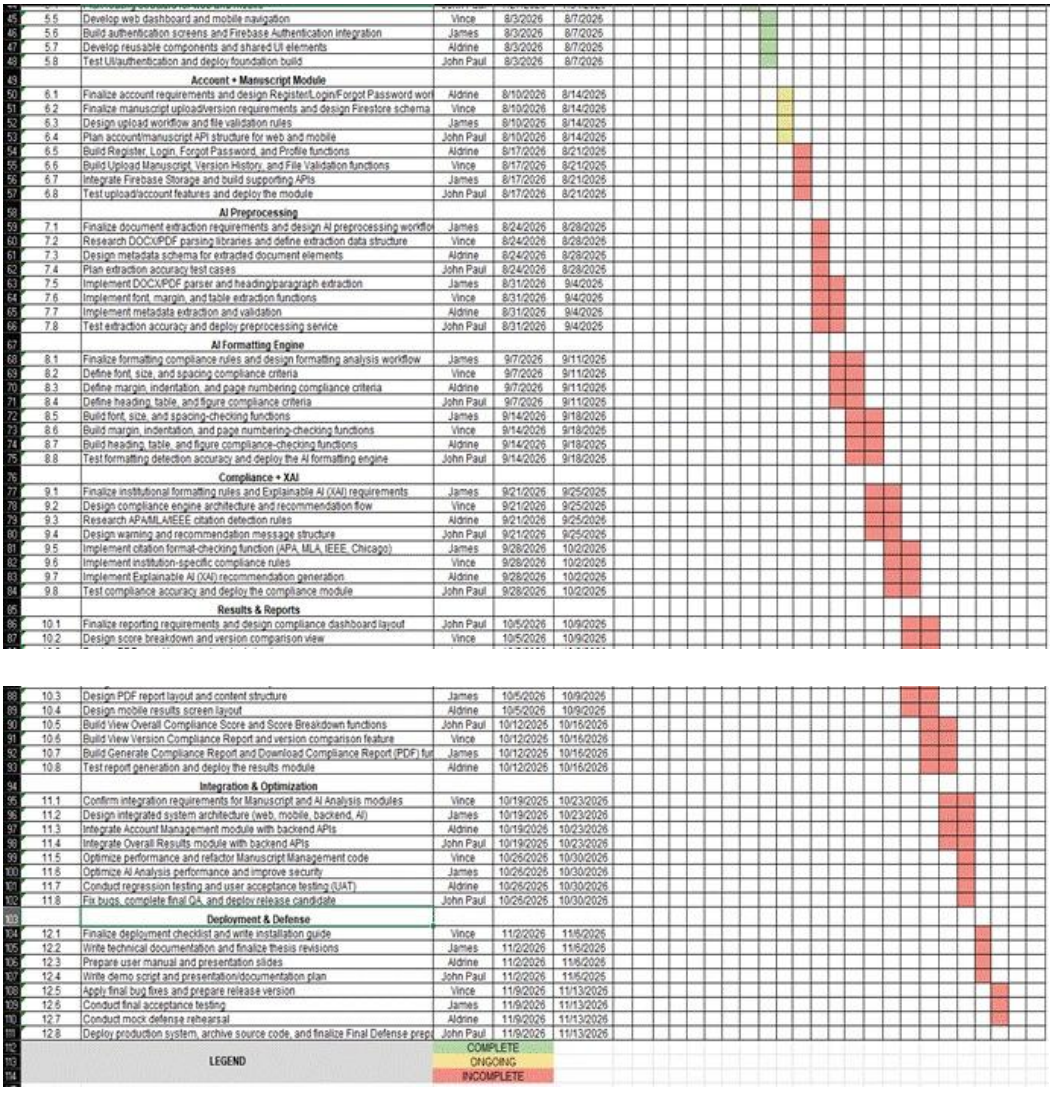


Task ID	Task Name	Task Lead	Start Date	End Date
	Establish Project Foundation and Gather Requirements			
1.1	Define project objectives and rationale for Chapter I	Vince	6/1/2026	6/5/2026
1.2	Formulate scope, limitations, and significance of the study	James	6/1/2026	6/5/2026
1.3	Draft flow of the study and definition of terms	Aldrine	6/1/2026	6/5/2026
1.4	Set up documentation workspace and drafting templates for Chapters I and II	John Paul	6/1/2026	6/5/2026
1.5	Research and compile 10 local related literature/study journals	Vince	6/8/2026	6/12/2026
1.6	Research and compile 10 international related literature/study journals	James	6/8/2026	6/12/2026
1.7	Validate completeness and accuracy of gathered literature sources	Aldrine	6/8/2026	6/12/2026
1.8	Organize research materials and prepare Chapter I draft for adviser review	John Paul	6/8/2026	6/12/2026
	Complete Literature Synthesis and Finalize Related Studies			
2.1	Synthesize the 10 local related literature journals	Vince	6/15/2026	6/19/2026
2.2	Synthesize the 10 international related literature journals	James	6/15/2026	6/19/2026
2.3	Identify supporting theories, concepts, and framework	Aldrine	6/15/2026	6/19/2026
2.4	Research and evaluate 10 existing systems in the market	John Paul	6/15/2026	6/19/2026
2.5	Create Theoretical Background and Review of Related Literature	Vince	6/22/2026	6/26/2026
2.6	Create Review of Related Studies and Comparative Matrix	James	6/22/2026	6/26/2026
2.7	Determine and document the Software Engineering Methodology	Aldrine	6/22/2026	6/26/2026
2.8	Validate Comparative Matrix and prepare Chapter II for adviser review	John Paul	6/22/2026	6/26/2026
	Finalize System Analysis, Initial Design, and Prepare for Oral Defense			
3.1	Create Business Model Canvas and Program Workflow	Vince	6/29/2026	7/3/2026
3.2	Create Validation Board and Gantt Chart	James	6/29/2026	7/3/2026
3.3	Create Functional Decomposition Diagram and Use Case Diagram/DFD	Aldrine	6/29/2026	7/3/2026
3.4	Create User Interface Design and Storyboard	John Paul	6/29/2026	7/3/2026
3.5	Create Entity Relationship Diagram and Data Dictionary	Vince	7/6/2026	7/10/2026
3.6	Create Network Model and Network Topology	James	7/6/2026	7/10/2026
3.7	Prepare design assets and diagram files as development references	Aldrine	7/6/2026	7/10/2026
3.8	Validate diagram consistency and finalize proposal documentation for oral defen	John Paul	7/6/2026	7/10/2026
	Complete Development Environment			
4.1	Finalize technology stack and write Software/Hardware Specification	Vince	7/13/2026	7/17/2026
4.2	Finalize module architecture and API specifications	James	7/13/2026	7/17/2026
4.3	Create List of Modules and finalize backend architecture	Aldrine	7/13/2026	7/17/2026
4.4	Finalize Repository architecture and Firebase design	John Paul	7/13/2026	7/17/2026
4.5	Set up GitHub repository (main, dev, feature branches) and project folder structur	Vince	7/20/2026	7/24/2026
4.6	Install Node.js/Express, FastAPI, and initialize backend server	James	7/20/2026	7/24/2026
4.7	Set up React.js/React Native, Tailwind CSS, and Android Studio/VS Code environ	Aldrine	7/20/2026	7/24/2026
4.8	Configure Firebase project and environment variables; verify builds and Git workf	John Paul	7/20/2026	7/24/2026
	Web & Mobile Foundation			
5.1	Finalize UI requirements and authentication flow specifications	Vince	7/27/2026	7/31/2026
5.2	Design responsive UI layout and navigation structure	James	7/27/2026	7/31/2026
5.3	Design reusable component library	Aldrine	7/27/2026	7/31/2026
5.4	Plan routing structure for web and mobile	John Paul	7/27/2026	7/31/2026

Table 5.1 presents the initial portion of the PaperPilot Gantt Chart, covering the Establish Project Foundation and Gather Requirements (Tasks 1.1 to 1.8), Complete Literature Synthesis and Finalize Related Studies (Tasks 2.1 to 2.8), and Finalize System Analysis and Initial Design (Tasks 3.1 to 3.8) phases, running from June 1 to July 10, 2026, and covering activities from defining project objectives and scope to compiling literature, building the theoretical background, and producing diagrams, storyboards, and other design outputs, all marked complete. The table also includes the Complete Development Environment and Begin Initial Coding phase (Tasks 4.1 to 4.8), scheduled July 13 to July 24, 2026, covering the preparation of dev tools, design of the repository and app structure, and the setup of GitHub, Firebase, and the development environment, with Tasks 4.1 to 4.4 marked complete and Tasks 4.5 to 4.8 marked incomplete, and concludes with the Development Foundation Finalization phase (Tasks 5.1 to 5.8), scheduled July 27 to August 7, 2026, covering the finalization of the frontend structure, navigation, and authentication framework, and the development of the web dashboard, mobile navigation, and reusable UI components, with Tasks 5.1 to 5.4 marked incomplete and Tasks 5.5 to 5.8 marked complete.

Table 5.2

GANTT CHART CONT'D



45	5.5	Develop web dashboard and mobile navigation	Vince	8/3/2026	8/7/2026
46	5.6	Build authentication screens and Firebase Authentication integration	James	8/3/2026	8/7/2026
47	5.7	Develop reusable components and shared UI elements	Aldrine	8/3/2026	8/7/2026
48	5.8	Test UI/authentication and deploy foundation build	John Paul	8/3/2026	8/7/2026
49		**Account + Manuscript Module**			
50	6.1	Finalize account requirements and design Register/Login/Forgot Password worl	Aldrine	8/10/2026	8/14/2026
51	6.2	Finalize manuscript upload/version requirements and design Firestore schema	Vince	8/10/2026	8/14/2026
52	6.3	Design upload workflow and file validation rules	James	8/10/2026	8/14/2026
53	6.4	Plan account/manuscript API structure for web and mobile	John Paul	8/10/2026	8/14/2026
54	6.5	Build Register, Login, Forgot Password, and Profile functions	Aldrine	8/17/2026	8/21/2026
55	6.6	Build Upload Manuscript, Version History, and File Validation functions	Vince	8/17/2026	8/21/2026
56	6.7	Integrate Firebase Storage and build supporting APIs	James	8/17/2026	8/21/2026
57	6.8	Test upload/account features and deploy the module	John Paul	8/17/2026	8/21/2026
58		**AI Preprocessing**			
59	7.1	Finalize document extraction requirements and design AI preprocessing workflo	James	8/24/2026	8/28/2026
60	7.2	Research DOCX/PDF parsing libraries and define extraction data structure	Vince	8/24/2026	8/28/2026
61	7.3	Design metadata schema for extracted document elements	Aldrine	8/24/2026	8/28/2026
62	7.4	Plan extraction accuracy test cases	John Paul	8/24/2026	8/28/2026
63	7.5	Implement DOCX/PDF parser and heading/paragraph extraction	James	8/31/2026	9/4/2026
64	7.6	Implement font, margin, and table extraction functions	Vince	8/31/2026	9/4/2026
65	7.7	Implement metadata extraction and validation	Aldrine	8/31/2026	9/4/2026
66	7.8	Test extraction accuracy and deploy preprocessing service	John Paul	8/31/2026	9/4/2026
67		**AI Formatting Engine**			
68	8.1	Finalize formatting compliance rules and design formatting analysis workflow	James	9/7/2026	9/11/2026
69	8.2	Define font, size, and spacing compliance criteria	Vince	9/7/2026	9/11/2026
70	8.3	Define margin, indentation, and page numbering compliance criteria	Aldrine	9/7/2026	9/11/2026
71	8.4	Define heading, table, and figure compliance criteria	John Paul	9/7/2026	9/11/2026
72	8.5	Build font, size, and spacing-checking functions	James	9/14/2026	9/18/2026
73	8.6	Build margin, indentation, and page numbering-checking functions	Vince	9/14/2026	9/18/2026
74	8.7	Build heading, table, and figure compliance-checking functions	Aldrine	9/14/2026	9/18/2026
75	8.8	Test formatting detection accuracy and deploy the AI formatting engine	John Paul	9/14/2026	9/18/2026
76		**Compliance + XAI**			
77	9.1	Finalize institutional formatting rules and Explainable AI (XAI) requirements	James	9/21/2026	9/25/2026
78	9.2	Design compliance engine architecture and recommendation flow	Vince	9/21/2026	9/25/2026
79	9.3	Research APA/MLA/IEEE citation detection rules	Aldrine	9/21/2026	9/25/2026
80	9.4	Design warning and recommendation message structure	John Paul	9/21/2026	9/25/2026
81	9.5	Implement citation format-checking function (APA, MLA, IEEE, Chicago)	James	9/28/2026	10/2/2026
82	9.6	Implement institution-specific compliance rules	Vince	9/28/2026	10/2/2026
83	9.7	Implement Explainable AI (XAI) recommendation generation	Aldrine	9/28/2026	10/2/2026
84	9.8	Test compliance accuracy and deploy the compliance module	John Paul	9/28/2026	10/2/2026
85		**Results & Reports**			
86	10.1	Finalize reporting requirements and design compliance dashboard layout	John Paul	10/5/2026	10/9/2026
87	10.2	Design score breakdown and version comparison view	Vince	10/5/2026	10/9/2026
88	10.3	Design PDF report layout and content structure	James	10/5/2026	10/9/2026
89	10.4	Design mobile results screen layout	Aldrine	10/5/2026	10/9/2026
90	10.5	Build View Overall Compliance Score and Score Breakdown functions	John Paul	10/12/2026	10/16/2026
91	10.6	Build View Version Compliance Report and version comparison feature	Vince	10/12/2026	10/16/2026
92	10.7	Build Generate Compliance Report and Download Compliance Report (PDF) fur	James	10/12/2026	10/16/2026
93	10.8	Test report generation and deploy the results module	Aldrine	10/12/2026	10/16/2026
94		**Integration & Optimization**			
95	11.1	Confirm integration requirements for Manuscript and AI Analysis modules	Vince	10/19/2026	10/23/2026
96	11.2	Design integrated system architecture (web, mobile, backend, AI)	James	10/19/2026	10/23/2026
97	11.3	Integrate Account Management module with backend APIs	Aldrine	10/19/2026	10/23/2026
98	11.4	Integrate Overall Results module with backend APIs	John Paul	10/19/2026	10/23/2026
99	11.5	Optimize performance and refactor Manuscript Management code	Vince	10/26/2026	10/30/2026
100	11.6	Optimize AI Analysis performance and improve security	James	10/26/2026	10/30/2026
101	11.7	Conduct regression testing and user acceptance testing (UAT)	Aldrine	10/26/2026	10/30/2026
102	11.8	Fix bugs, complete final QA, and deploy release candidate	John Paul	10/26/2026	10/30/2026
103		**Deployment & Defense**			
104	12.1	Finalize deployment checklist and write installation guide	Vince	11/2/2026	11/6/2026
105	12.2	Write technical documentation and finalize thesis revisions	James	11/2/2026	11/6/2026
106	12.3	Prepare user manual and presentation slides	Aldrine	11/2/2026	11/6/2026
107	12.4	Write demo script and presentation/documentation plan	John Paul	11/2/2026	11/6/2026
108	12.5	Apply final bug fixes and prepare release version	Vince	11/9/2026	11/13/2026
109	12.6	Conduct final acceptance testing	James	11/9/2026	11/13/2026
110	12.7	Conduct mock defense rehearsal	Aldrine	11/9/2026	11/13/2026
111	12.8	Deploy production system, archive source code, and finalize Final Defense prep	John Paul	11/9/2026	11/13/2026
112			COMPLETE		
113		LEGEND	ONGOING		
114			INCOMPLETE		

Table 5.2 presents the continuation of the PaperPilot Gantt Chart, beginning with the Complete Backend Services Setup phase (Tasks 6.1 to 6.8), scheduled August 10 to 21, 2026, covering the design and development of account authentication (Register, Login, Forgot Password) and the Users data layer, with Tasks 6.1 to 6.4 marked ongoing and Tasks 6.5 to 6.8 marked incomplete. This is followed by the Build Core Account Management Features (Tasks 7.1 to 7.8, August 24 to September 4) and Begin Manuscript Management Module (Tasks 8.1 to 8.8, September 7 to 18) phases, covering profile and account functions, and manuscript upload and file validation, respectively. The Complete Manuscript Management, Begin AI Analysis phase (Tasks 9.1 to 9.8, September 21 to October 2) covers manuscript versioning and initial formatting validation, while the Complete AI Analysis Structure Module phase (Tasks 10.1 to 10.8, October 5 to 16) covers margin, page break, table, and citation format checking. The table concludes with the Build Overall Results Module phase (Tasks 11.1 to 11.8, October 19 to 30),

covering compliance scoring and report generation, and the Final Validation, Documentation and Defense Prep phase (Tasks 12.1 to 12.8, November 2 to 9), covering final documentation, testing, and defense preparation. All tasks from 6.5 onward are marked incomplete under the color-coded legend.

Functional Decomposition Diagram

This section shows the functional decomposition diagram, a method for decomposing a complicated system or process into smaller, easier-to-manage components or functions. This diagram helps to show the primary functions and their sub-functions for PaperPilot, giving users a clear understanding of how each component works together to support the program's overall functionality.

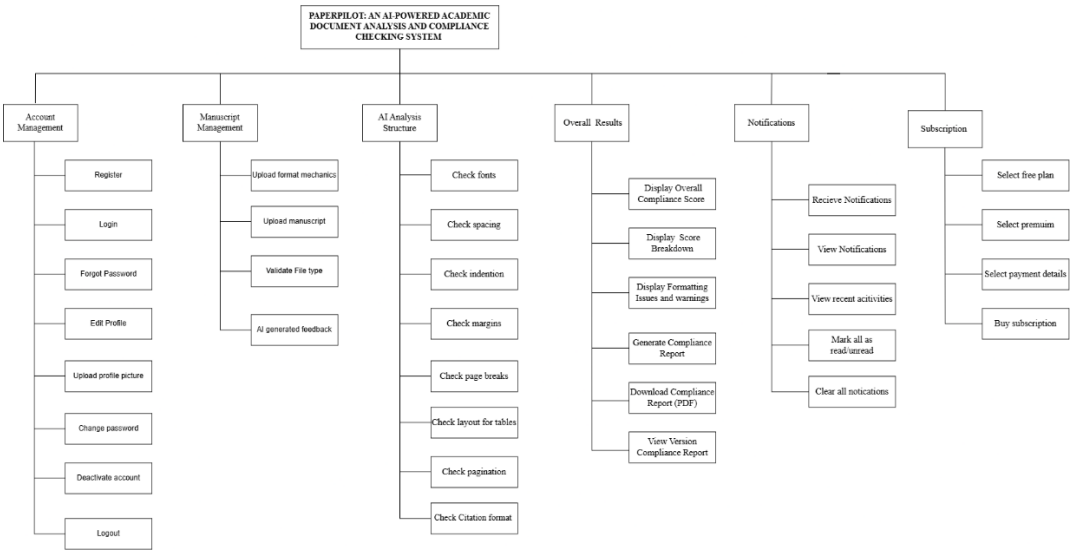


Figure 4:**Functional Decomposition Diagram**

Figure 4 presents the major modules and functions of PaperPilot as illustrated in the Functional Decomposition Diagram (FDD), namely Account Management, Manuscript Management, AI Analysis Structure, Overall Results, Notification, and Subscription. The Account Management module enables researchers to register, log in, recover forgotten passwords, edit their profiles, upload profile pictures, change passwords, deactivate their accounts, and log out, providing secure access and effective user account management. The Manuscript Management module allows researchers to upload formatting templates, upload manuscripts, validate file types, re-upload formatting templates, re-upload revised manuscripts, and receive AI-generated feedback, supporting an organized workflow for document submission, revision, and analysis. The AI Analysis Structure module automatically evaluates uploaded manuscripts by checking fonts, spacing, indentation, margins, page breaks, table layouts, pagination,

and citation format to ensure compliance with the required academic formatting standards. The Overall Results module presents the outcomes of the analysis by displaying the overall compliance score, providing a detailed score breakdown, identifying formatting issues and warnings, generating compliance reports, allowing researchers to download compliance reports in PDF format, and viewing version compliance reports to monitor improvements across manuscript revisions. The Notification module enables researchers to receive notifications, view notifications, review recent activities, mark notifications as read or unread, and clear all notifications, ensuring they remain informed about system updates and manuscript analysis progress. Finally, the Subscription module allows researchers to select either the free or premium plan, provide payment details, and purchase a subscription, enabling access to system features according to their chosen subscription plan.

Analysis Design Phase

During the Analysis-Design phase of PaperPilot, the focus is on understanding user requirements and designing the system's functional structure. This phase involves identifying project goals, analyzing user needs, and outlining technical requirements to define the system's essential features and expected performance. Key functions including AI-powered research analysis, document upload and management, automated compliance checking against APA, MLA, and IEEE formatting standards, account management, and dashboard notifications are addressed in detail. Key outputs in this phase include the use case diagram, user interface, storyboards, database design, entity-relationship diagram, and data dictionary. These elements provide a clear foundation for the system's development, ensuring that PaperPilot effectively supports the needs of students, researchers, and academic institutions involved in the preparation and submission of research, thesis, and capstone manuscripts.

Use Case Diagram

This section presents the use case diagram, which provides a visual representation of the functional requirements and interactions of a system from the perspective of its users. For PaperPilot, the use case diagram outlines various scenarios or "use cases" where the Researcher interacts with the application to achieve specific goals related to academic document uploading, formatting compliance checking, and compliance report generation.

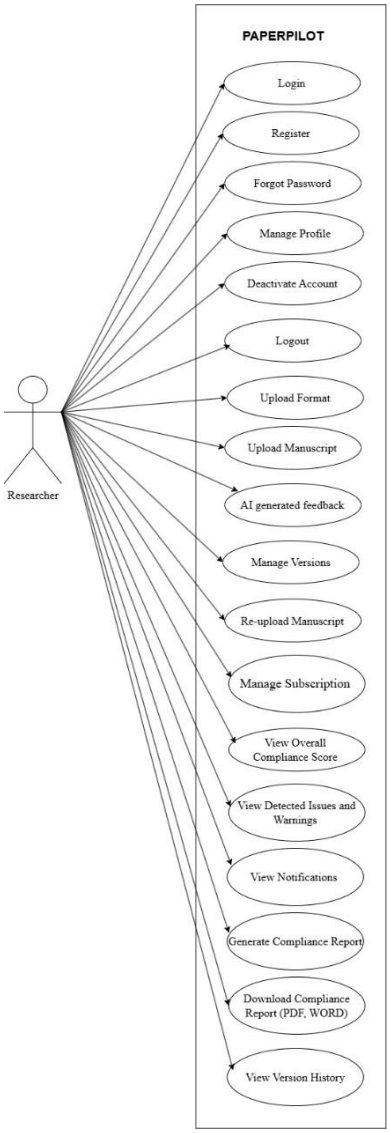


Figure 5: **Use Case Diagram**

Figure 5 presents the Use Case Diagram of PaperPilot, illustrating the interaction between the system and its sole actor, the Researcher. The diagram identifies the primary functions available to the researcher, including account management, manuscript management, AI-assisted document analysis, compliance result viewing, notification access, subscription management, and report generation. The researcher can perform account-related functions such as Login, Register, Forgot Password, Manage Profile, Deactivate Account, and Logout to securely access and manage the system. For manuscript management, the researcher can Upload Format, Upload Manuscript, Manage Versions, and Re-upload Manuscript, enabling the submission of academic documents and the management of revised versions throughout the research process. After the manuscript is analyzed, the system provides AI-generated Feedback,

allowing the researcher to understand detected formatting and compliance issues and receive recommendations for improvement. The researcher can also View Overall Compliance Score, View Detected Issues and Warnings, View Notifications, Generate Compliance Report, Download Compliance Report (PDF or Word), View Version History, and Manage Subscription, enabling them to monitor document compliance, review analysis results, track manuscript revisions, manage subscription plans, and generate downloadable reports for documentation and further revision.

User Interface Design

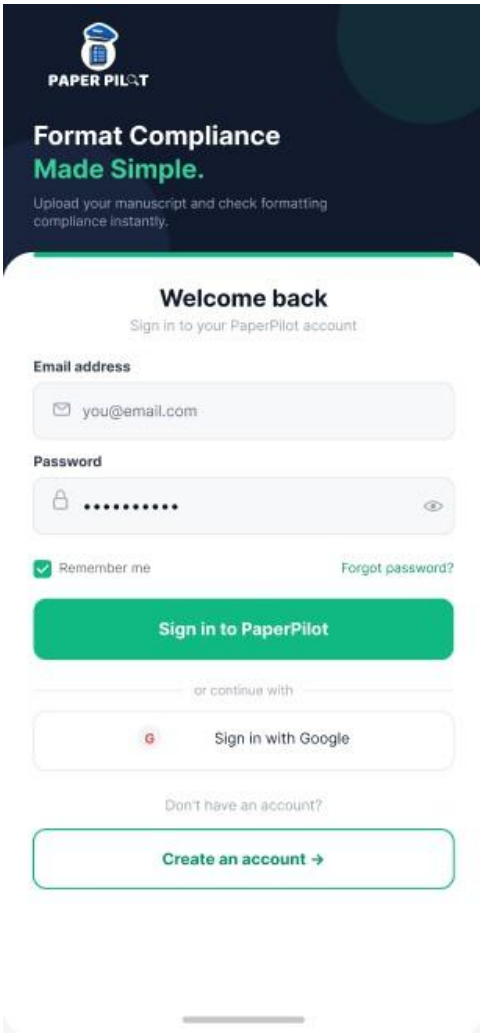


Figure 6: **Mobile Login Page**

Figure 6 shows the Login Page of the PaperPilot system. This page allows users to securely access their accounts by entering their registered email address and password. Once the user clicks the "Sign in to PaperPilot" button, the system validates the entered credentials and, if authenticated successfully, grants access to the user's dashboard. If the credentials are incorrect, the system displays an error message and prompts the user to enter the correct login information. Users who do not yet have an account may register by selecting the "Create an account" option, while those who have forgotten their password can recover their account through the "Forgot password?" feature. The system also provides the option to sign in using a Google account for a more convenient authentication process.

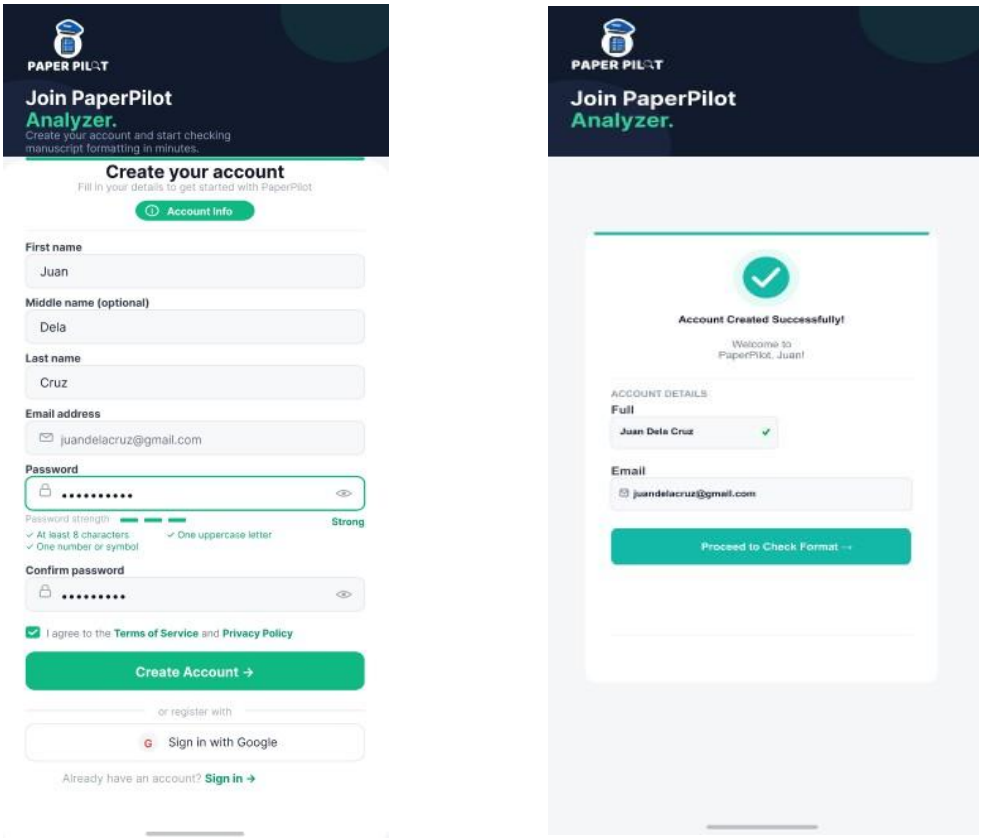


Figure 6.1: **Mobile Register Account Page**

Figure 6.1 shows the Register Account Page of the PaperPilot system. This page allows new users to create an account by providing the required registration information, such as their full name, email address, and password. The system validates the entered information to ensure that all required fields are completed correctly and that the email address has not already been registered. After the user successfully creates an account by clicking the "Create an account" button, the system authenticates the newly

registered account and redirects the user to the Upload Manuscripts page, where they can begin uploading their research documents for analysis. If any required information is incomplete or invalid, the system displays appropriate error messages and prompts the user to correct the input before proceeding.

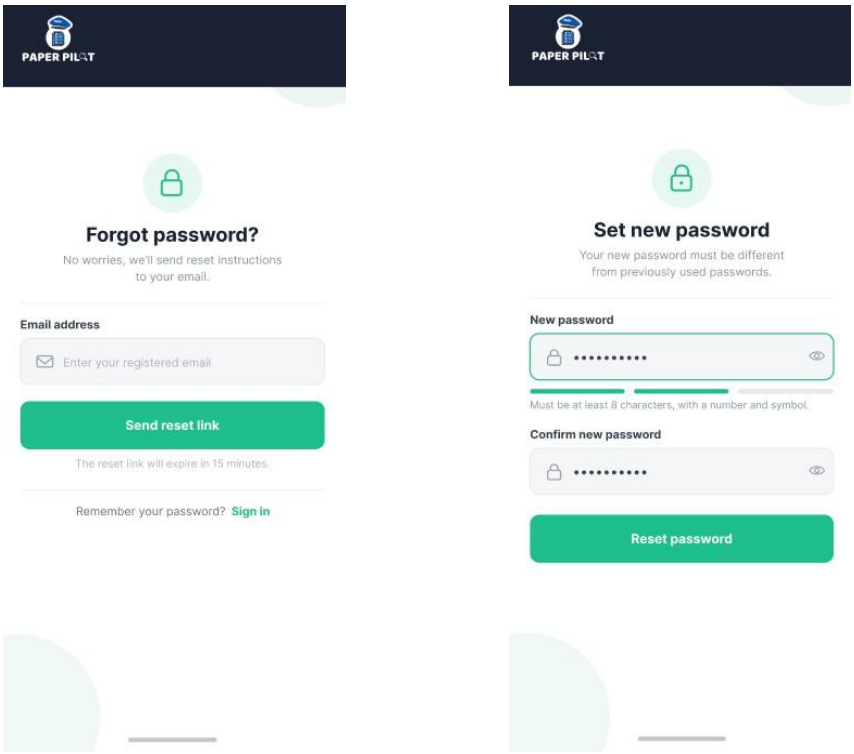


Figure 6.2: **Mobile Forgot Password Page**

Figure 6.2 shows the Forgot Password Page of the PaperPilot system. This page enables users to recover access to their accounts by entering their registered email address. After submitting a valid email address, the system verifies the account and sends a password reset link or verification instructions to the user's email. Users can then follow the provided instructions to create a new password and regain access to their account. If the entered email address is not associated with any existing account, the system displays an appropriate error message and prompts the user to enter a valid registered email address.

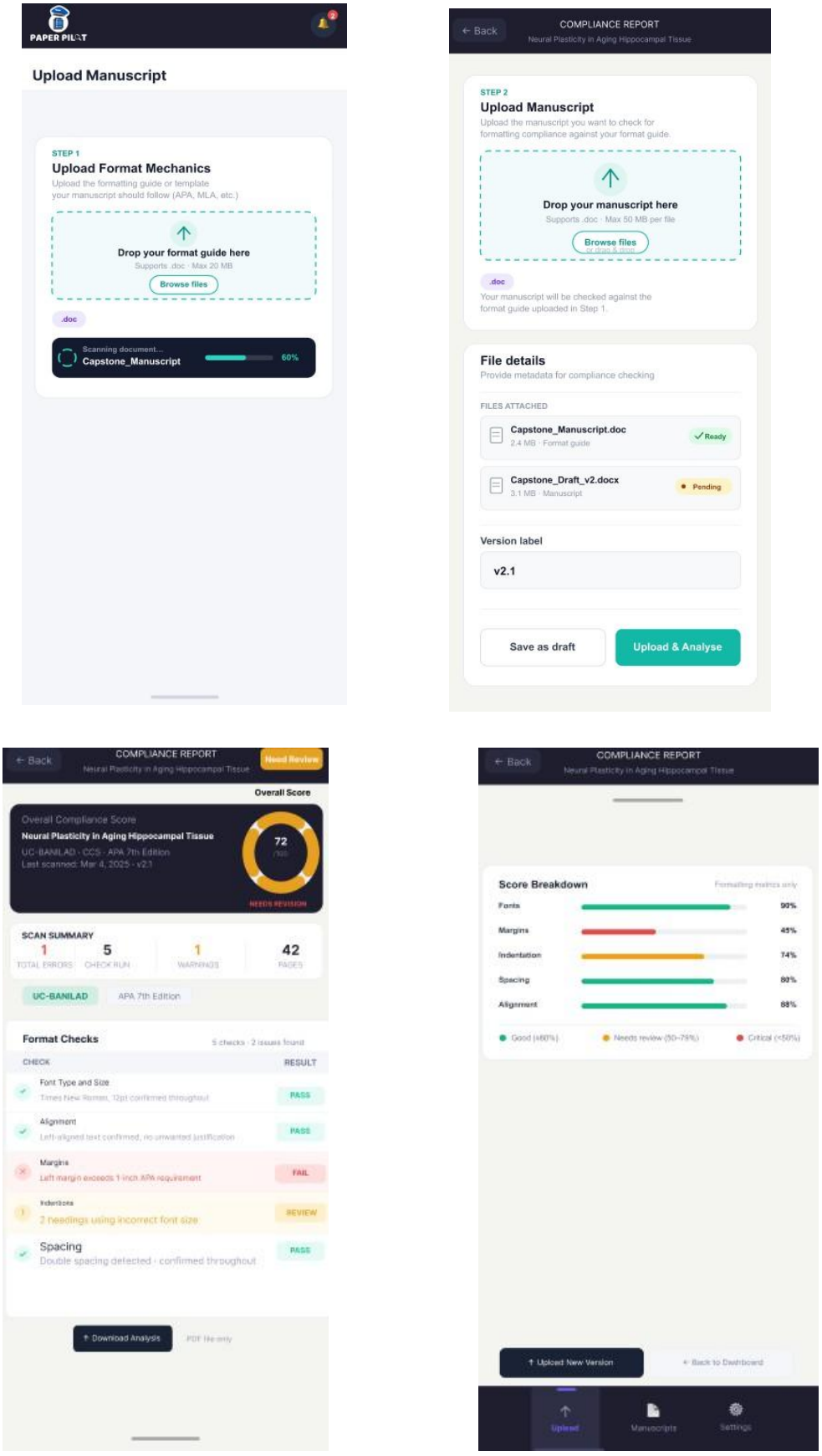
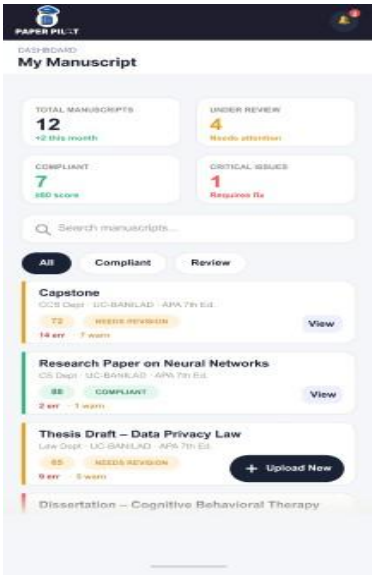
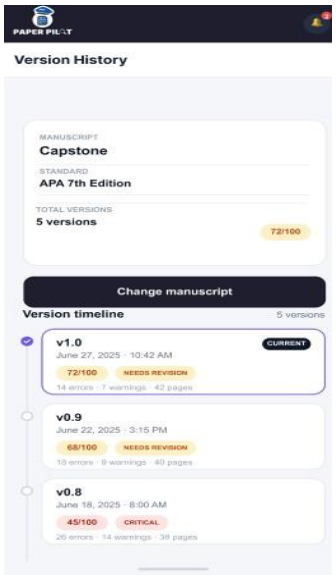


Figure 6.3: **Mobile Upload Manuscript Page**

Figure 6.3 shows the Upload Manuscript Page of the PaperPilot mobile application. This page enables users to upload both a formatting guide and a research manuscript for automated compliance analysis using their mobile devices. Users first upload the required formatting guide or institutional template, followed by the research manuscript to be evaluated. The page also displays the uploaded files, their upload status, and allows users to enter a version label before proceeding with the analysis. Once all required files are ready, users can tap the Upload & Analyse button to begin the document analysis. The system then evaluates the manuscript based on the uploaded formatting guide and generates detailed analysis results, helping users identify formatting issues and improve their academic documents before uploading revised



versions when necessary.



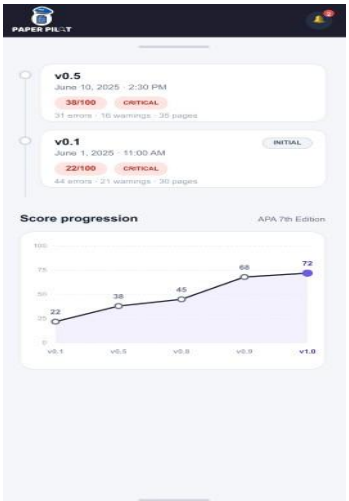




Figure 6.4: **Mobile My Manuscript and Version History Page**

Figure 6.4 shows the My Manuscripts Page and Version History of the PaperPilot mobile application. This page displays all manuscripts previously uploaded by the user, along with their corresponding compliance scores and review statuses. Users can search and filter their manuscripts and select the "View" button to access the manuscript's version history. The version history presents all uploaded versions, allowing users to monitor their compliance progress over time through score trends and detailed analysis results for each version, including the overall compliance score, formatting issues, and score breakdown. Users may also upload a new version of the manuscript after making the necessary revisions.

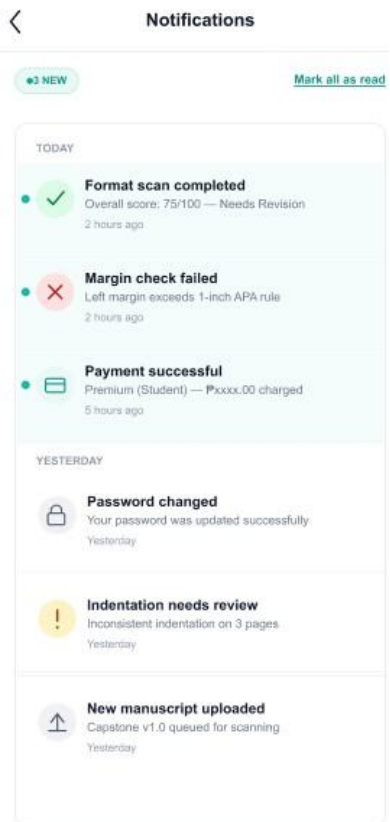


Figure 6.5: **Mobile Notification Page**

Figure 6.5 shows the Notifications Page of the PaperPilot mobile application. This page displays the user's recent activities and system-generated notifications, such as manuscript upload confirmations, analysis completion, compliance results, and status updates. Notifications are presented in chronological order, allowing users to quickly monitor the progress of their manuscripts and stay informed of important system events. Users can tap a notification to view additional details related to the activity.

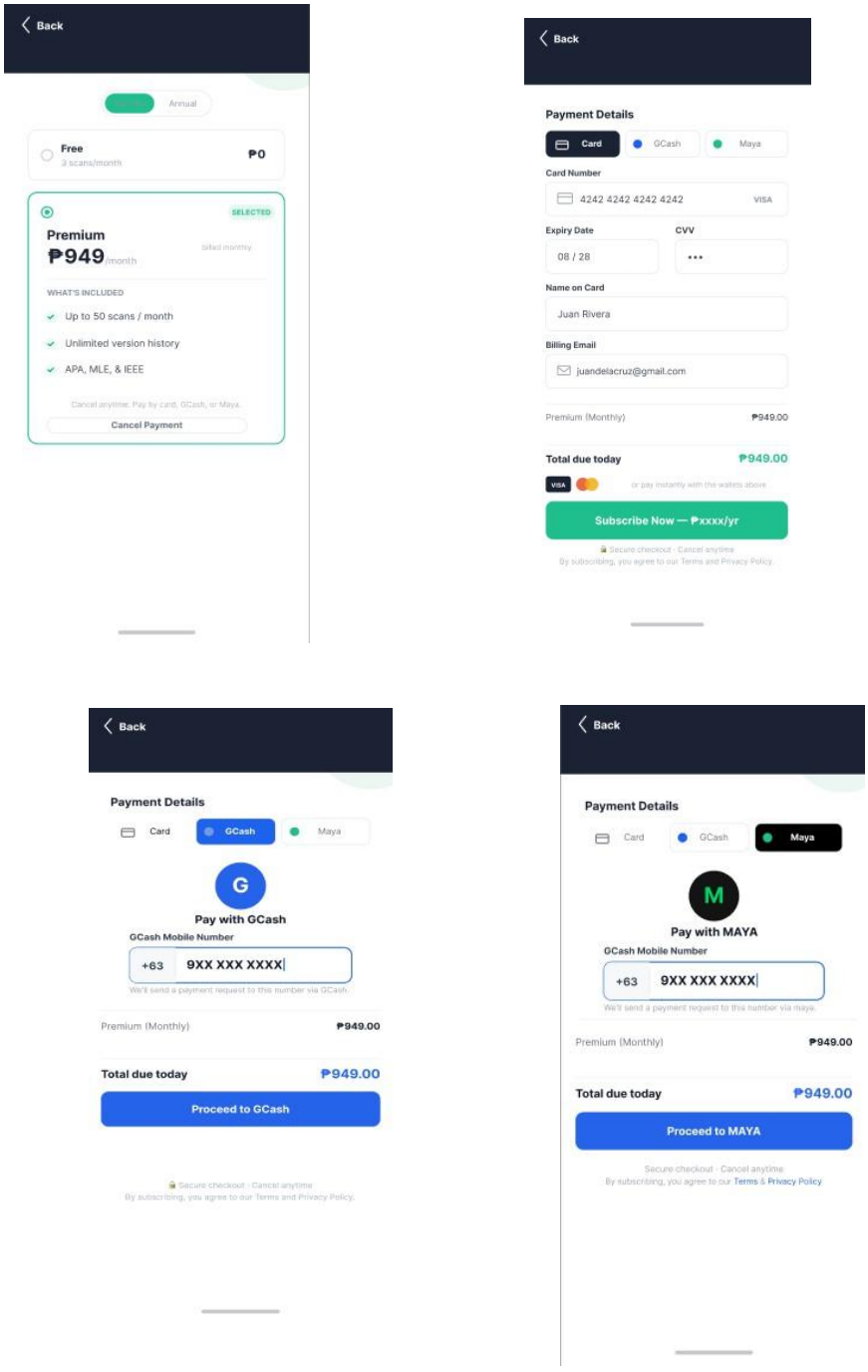


Figure 6.6: **Mobile Subscription Plan Page**

Figure 6.6 shows the Subscription Plan Page of the PaperPilot mobile application. This page allows users to view and manage their subscription plans by choosing between the Free and Premium options. The Premium plan displays the available features, including unlimited manuscript scans, unlimited version history, advanced AI formatting feedback, and priority support. Users may subscribe by selecting their preferred payment method, such as Card, GCash, or Maya, entering the required payment details, and confirming the subscription. After successful payment, the system activates the selected plan and grants access to the corresponding premium features.

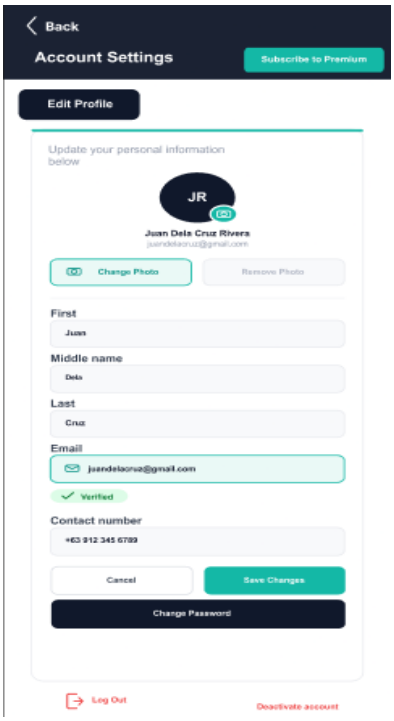


Figure 6.7: **Mobile Account Settings Page**

Figure 6.7 shows the Account Settings Page of the PaperPilot mobile application. This page allows users to manage their account information by updating their profile details, including their profile picture, name, email address, and contact number. Users can save any changes made to their personal information or change their account password through the "Change Password" option. The page also provides options to log out, deactivate the account, and subscribe to the Premium plan to access additional system features.

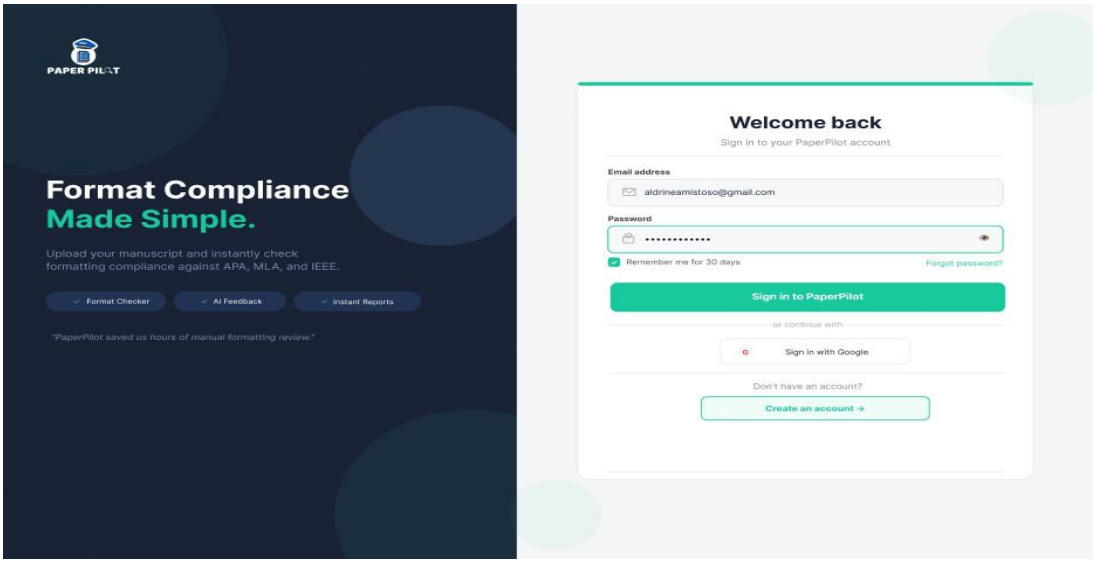
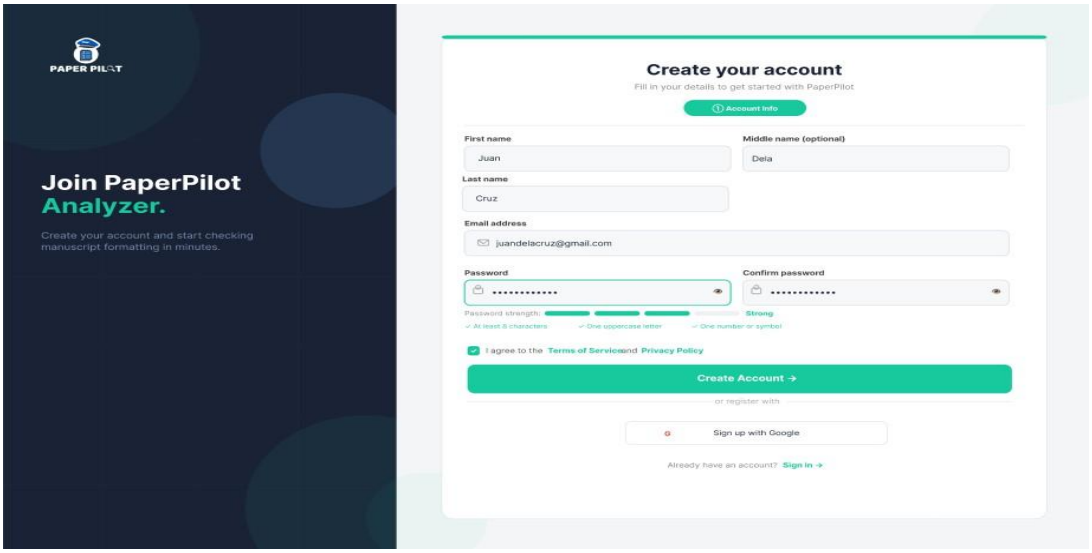


Figure 6.8: **Web Login Page**

Figure 6.8 shows the Web Login Page of the PaperPilot system. This page allows users to securely access their accounts by entering their registered email address and password. Upon clicking the "Sign In" button, the system validates the entered credentials and grants access to the user's account if authentication is successful. If the credentials are invalid, the system displays an error message and prompts the user to try again. Users who do not have an account can select the "Create an Account" option to register, while those who have forgotten their password can use the "Forgot Password?" feature to recover their account. The page also provides a "Sign in with Google" option, allowing users to authenticate using their Google account for a faster and more convenient login experience.



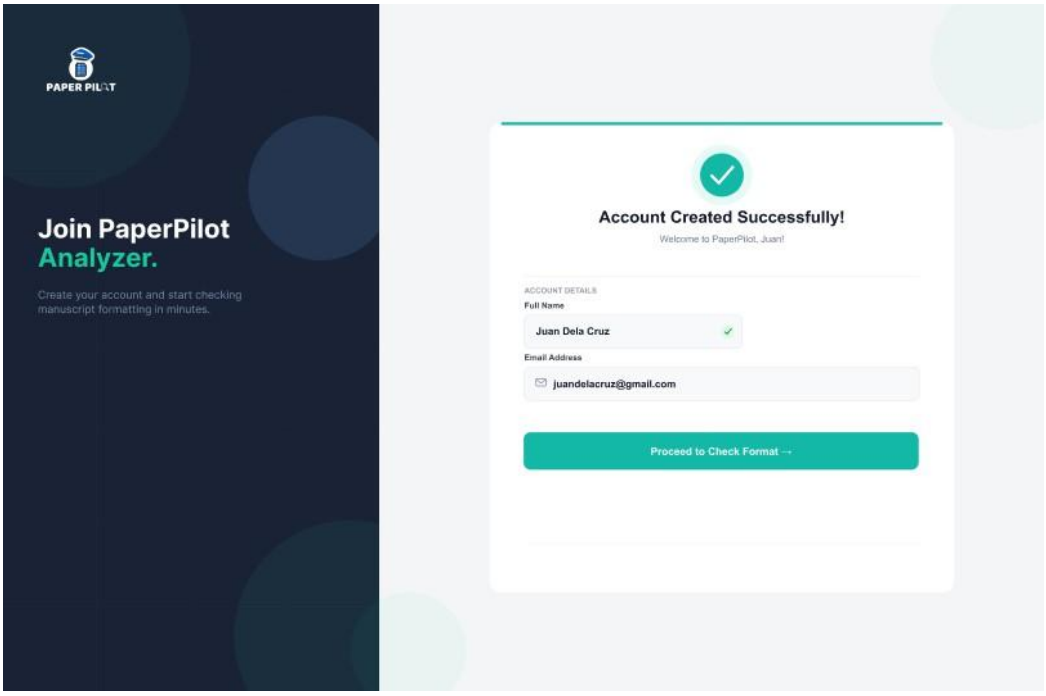


Figure 6.9: **Web Register Page**

Figure 6.9 shows the Web Register Page of the PaperPilot system. This page enables new users to create an account by providing the required registration information, including their full name, email address, and password. The system validates the entered information to ensure that all required fields are completed correctly and that the email address has not already been registered. Upon successful account creation, the system displays a confirmation message and redirects the user to the Upload Manuscript page, where they can begin uploading research documents for compliance analysis. Users may also register using the "Sign up with Google" option for a faster and more convenient registration process. If any required information is incomplete or invalid, the system displays appropriate error messages and prompts the user to correct the input before proceeding.

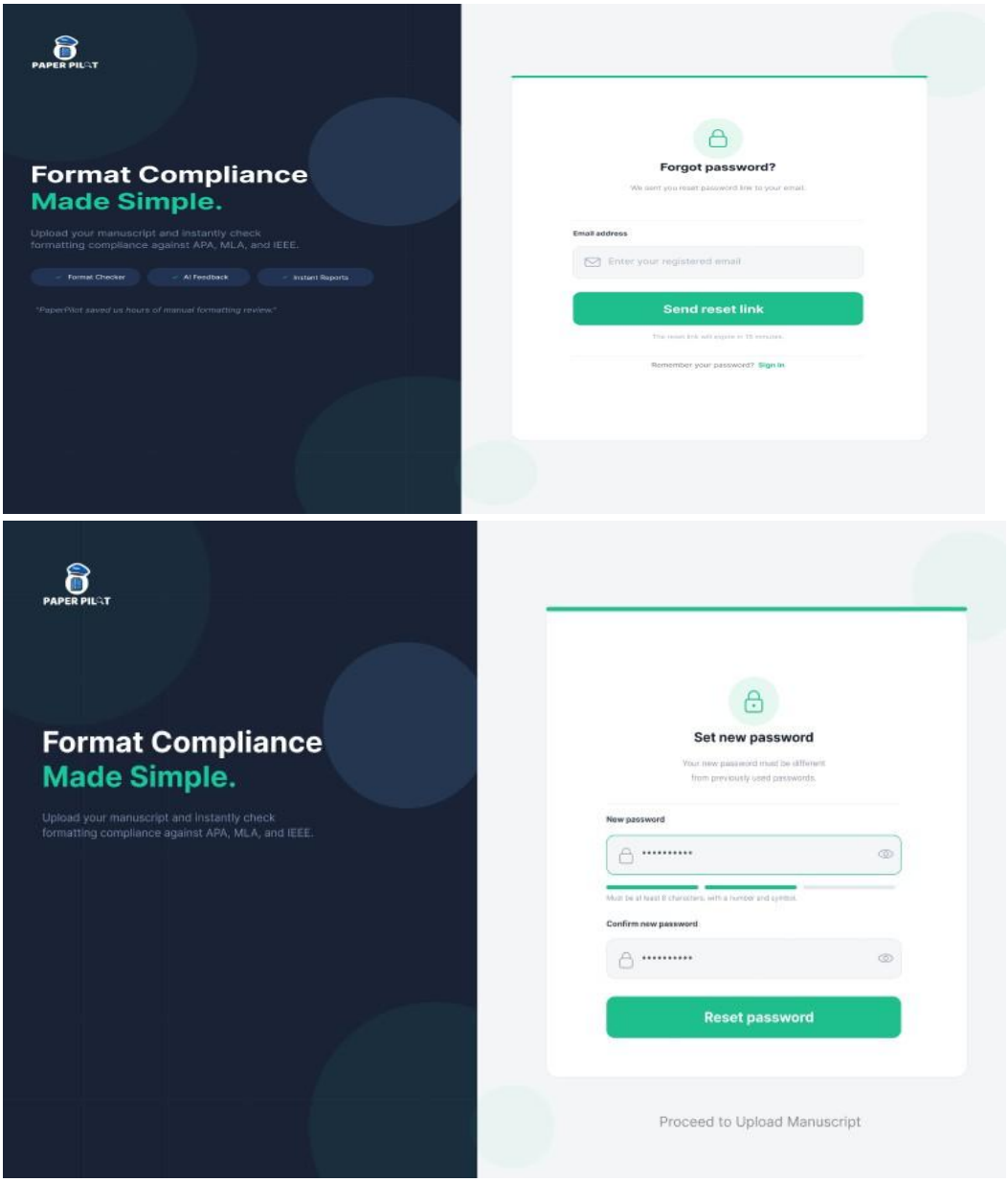
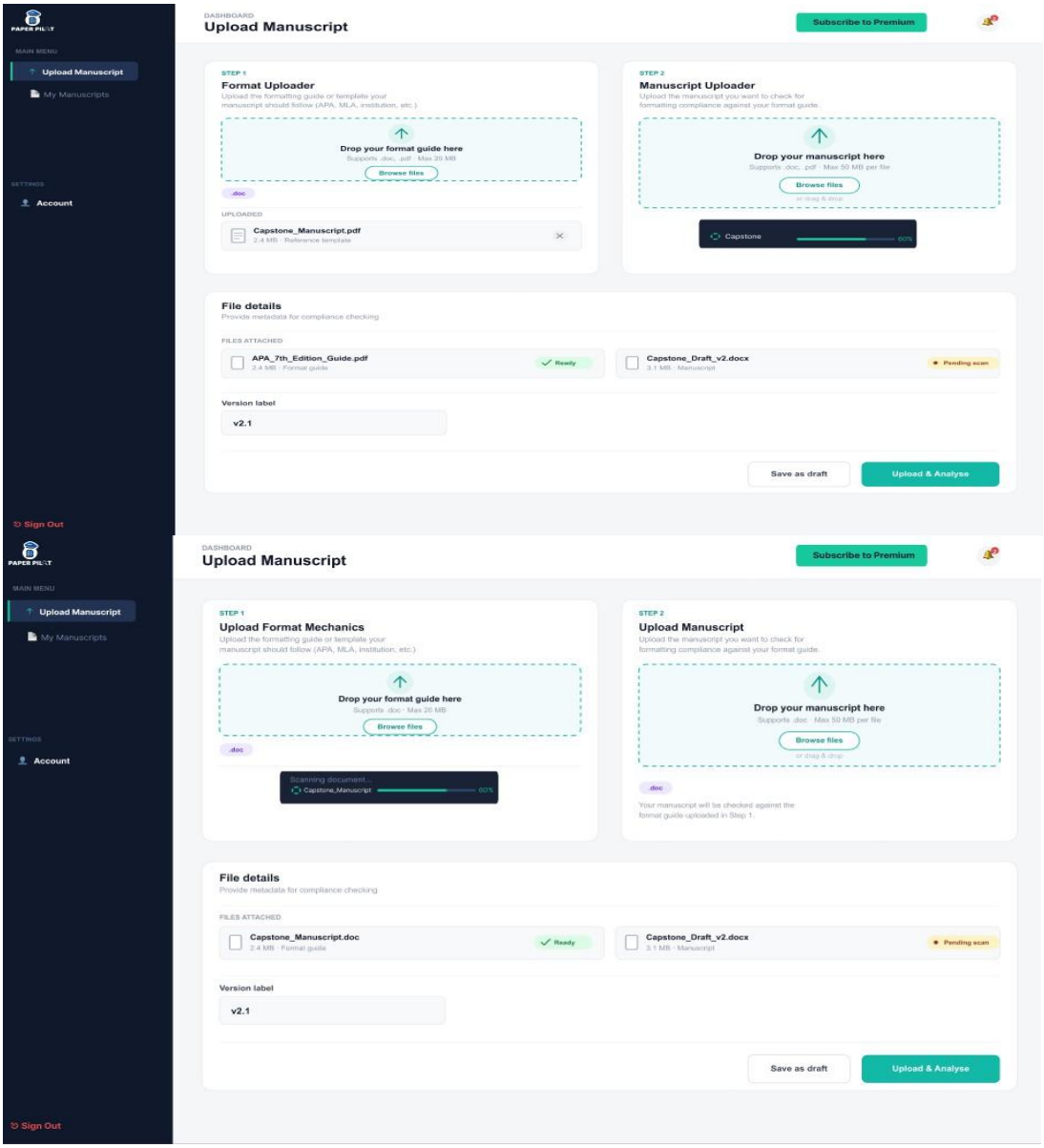


Figure 6.10: **Web Forgot Password Page**

Figure 6.10 shows the Web Forgot Password Page of the PaperPilot system. This page allows users to recover access to their accounts by entering their registered email address. After submitting a valid email address, the system sends a password reset link to the user's email. By accessing the link, users are redirected to the Set New Password page, where they can enter and confirm a new password. Upon successfully resetting the password, the system displays a confirmation message and redirects the user to the Login Page, allowing them to sign in using their newly created password. If the entered email address is not associated with an existing account, the system displays an appropriate error message and prompts the user to enter a valid registered email address.


PAPER PILOT
MAIN MENU
Upload Manuscript
My Manuscripts
SETTINGS
Account
Sign Out
DASHBOARD
Upload Manuscript
Subscribe to Premium
STEP 1
Format Uploader
Upload the formatting guide or template your manuscript should follow (APA, MLA, institution, etc.)
Drop your format guide here
Supports .doc, .pdf · Max 20 MB
Browse files
.doc
UPLOADED
Capstone_Manuscript.pdf
2.4 MB · Reference template
STEP 2
Manuscript Uploader
Upload the manuscript you want to check for formatting compliance against your format guide.
Drop your manuscript here
Supports .doc, .pdf · Max 50 MB per file
Browse files
or drag & drop
Capstone
60%
File details
Provide metadata for compliance checking.
FILES ATTACHED
APA_7th_Edition_Guide.pdf
2.4 MB · Format guide
Ready
Capstone_Draft_v2.docx
3.1 MB · Manuscript
Pending scan
Version label
v2.1
Save as draft
Upload & Analyse
PAPER PILOT
MAIN MENU
Upload Manuscript
My Manuscripts
SETTINGS
Account
Sign Out
DASHBOARD
Upload Manuscript
Subscribe to Premium
STEP 1
Upload Format Mechanics
Upload the formatting guide or template your manuscript should follow (APA, MLA, institution, etc.)
Drop your format guide here
Supports .doc · Max 20 MB
Browse files
.doc
Scanning document...
Capstone_Manuscript
60%
STEP 2
Upload Manuscript
Upload the manuscript you want to check for formatting compliance against your format guide.
Drop your manuscript here
Supports .doc · Max 50 MB per file
Browse files
or drag & drop
.doc
Your manuscript will be checked against the format guide uploaded in Step 1.
File details
Provide metadata for compliance checking.
FILES ATTACHED
Capstone_Manuscript.doc
2.4 MB · Format guide
Ready
Capstone_Draft_v2.docx
3.1 MB · Manuscript
Pending scan
Version label
v2.1
Save as draft
Upload & Analyse

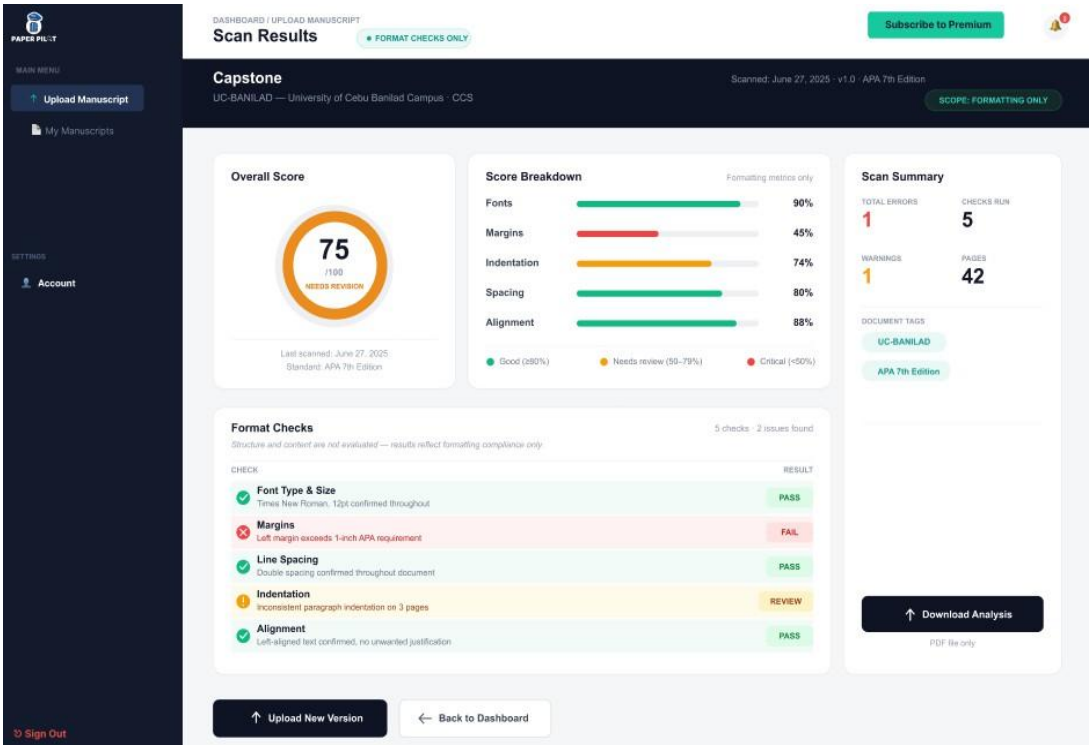


Figure 6.11: **Web Upload Manuscript Page**

Figure 6.11 shows the Upload Manuscript Page of the PaperPilot web application. This page enables users to upload both a formatting guide and a research manuscript for automated compliance analysis. In the first step, users upload the required formatting guide or institutional template that serves as the reference for evaluating the manuscript. In the second step, users upload the research manuscript to be analyzed against the selected formatting guide. The page also displays the uploaded files, their upload status, and allows users to specify a version label before proceeding with the analysis. Once all required files are ready, users can click the Upload & Analyse button to initiate the document analysis. The system then evaluates the manuscript based on the uploaded formatting guide and generates detailed analysis results, enabling users to identify formatting issues and improve their academic documents before submitting revised versions if necessary.

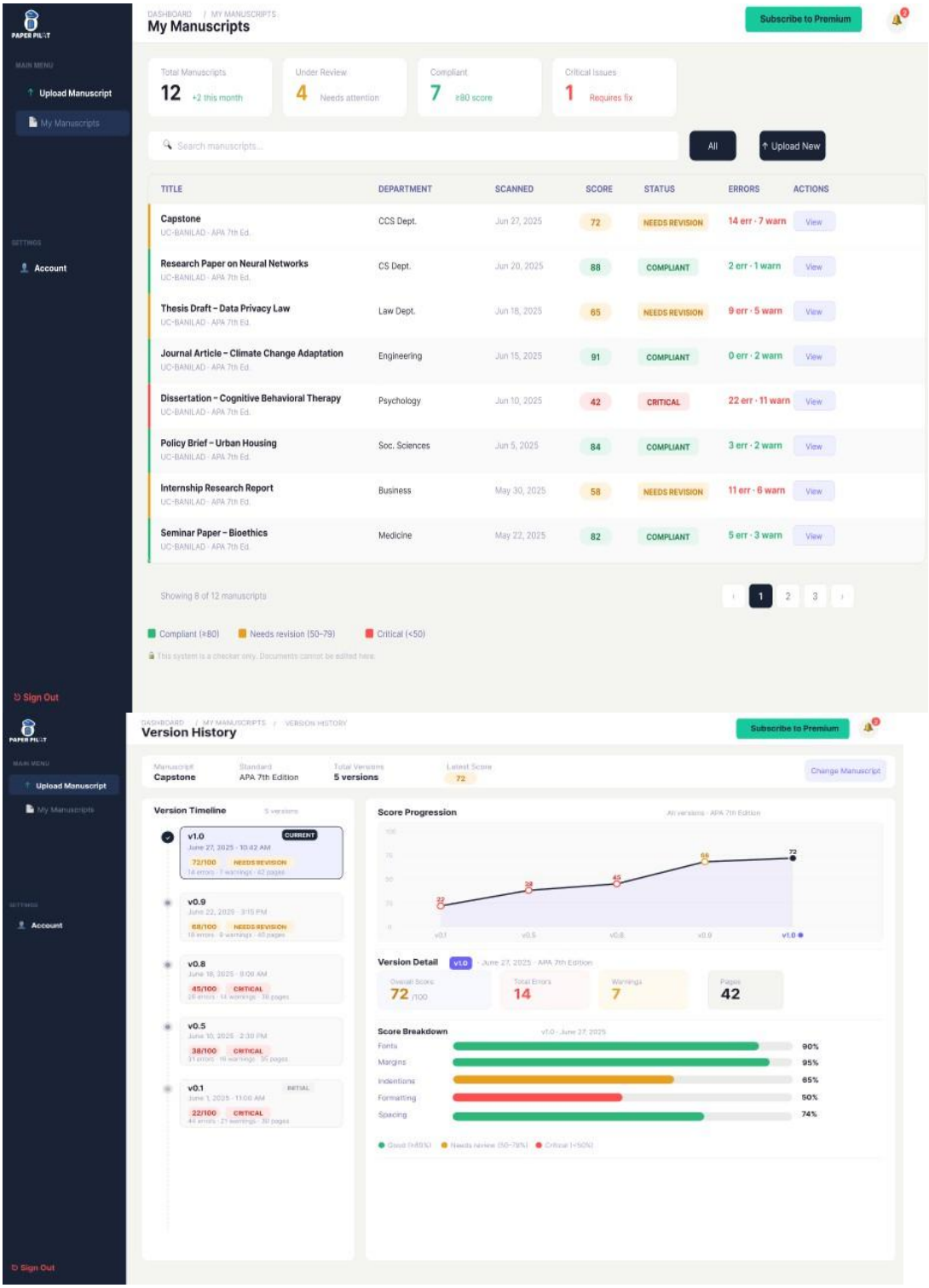
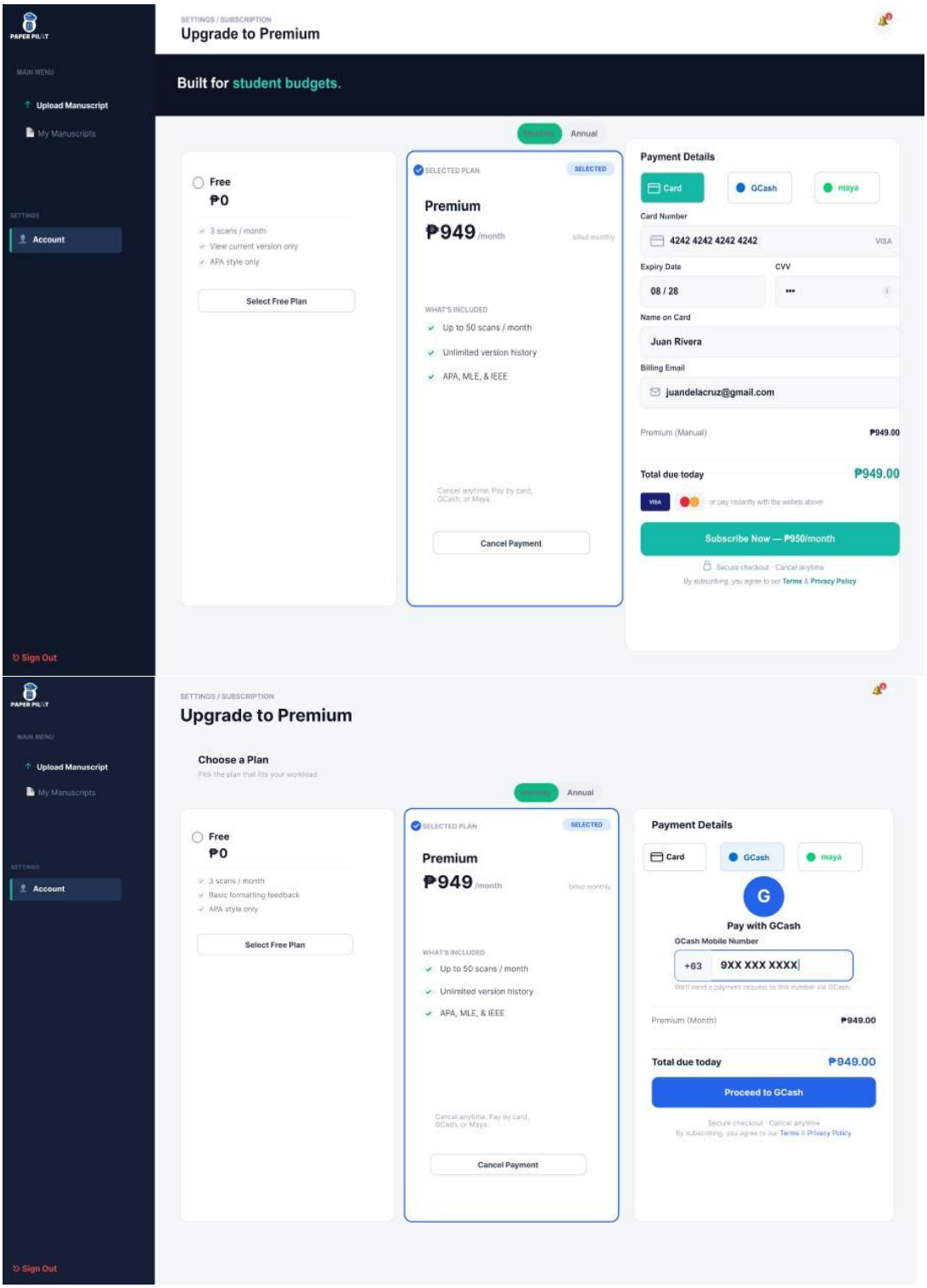


Figure 6.12: **Web My Manuscript and Version History Page**

Figure 6.12 shows the My Manuscripts and Version History Page of the PaperPilot web application. This page displays all manuscripts previously uploaded by the user, together with their corresponding compliance scores and review statuses. Users can search and manage their manuscripts and select a specific manuscript to view its version history. The version history presents all uploaded versions, allowing users to

track compliance progress over time through detailed analysis results, including the overall compliance score, formatting issues, and score breakdown. Users may also upload a new version of the manuscript after making the necessary revisions to improve compliance.



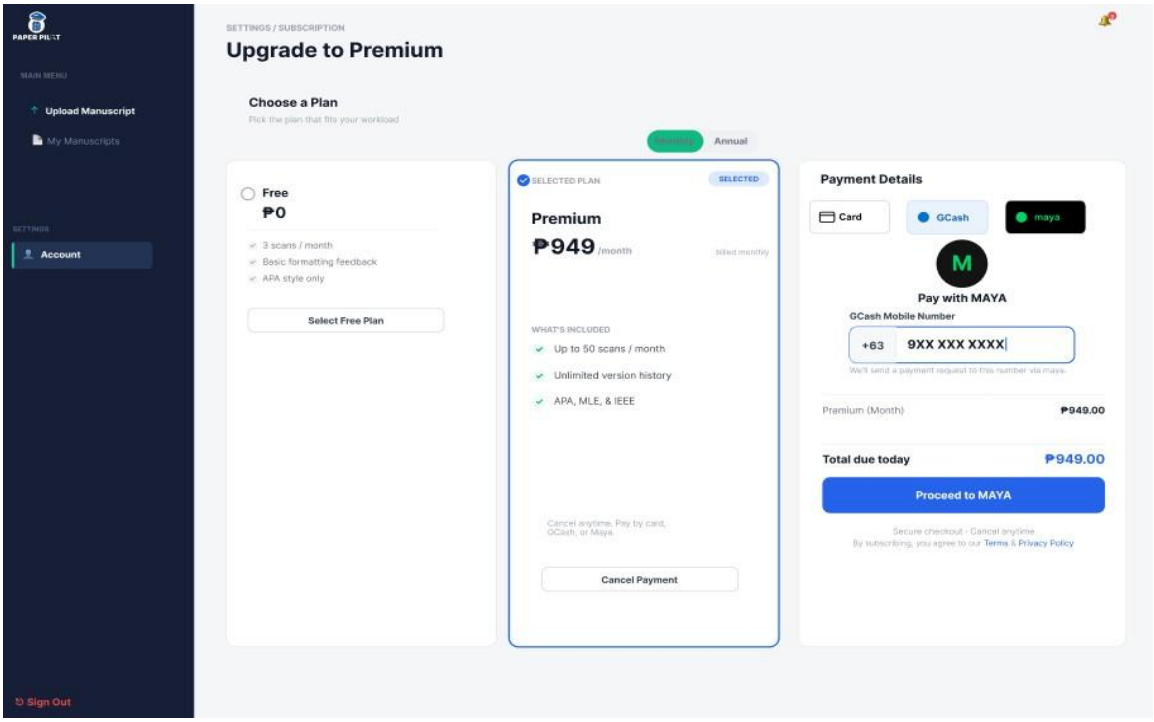


Figure 6.13: **Web Subscription Plan Page**

Figure 6.13 shows the Subscription Plan Page of the PaperPilot web application. This page allows users to view and manage their subscription plans by choosing between the Free and Premium options. The Premium plan provides additional features, including unlimited manuscript

scans, unlimited version history, advanced AI formatting feedback, and priority support. Users can subscribe by selecting their preferred payment method, such as Card, GCash, or Maya, entering the required payment details, and confirming the payment. Upon successful payment, the system activates the selected subscription plan and grants access to the corresponding premium features.

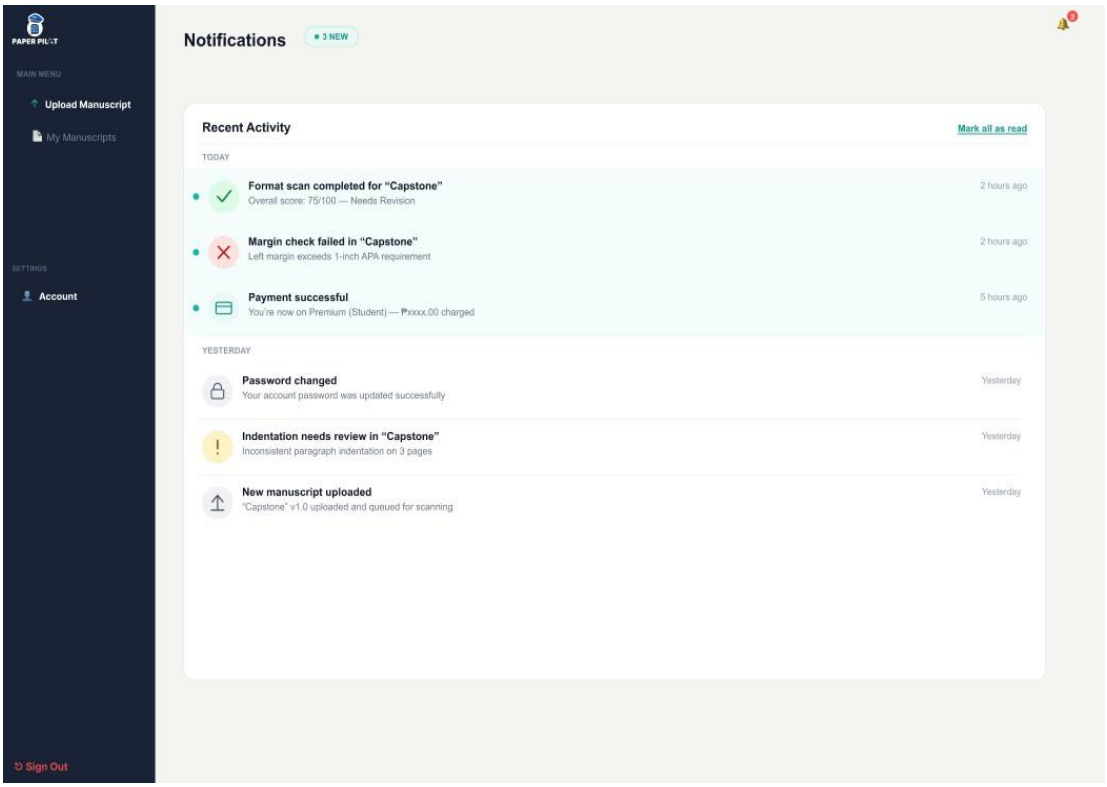


Figure 6.14: **Web NotificationPage**

Figure 6.14 shows the Notifications Page of the PaperPilot web application. This page displays the user's recent activities and system-generated notifications, including manuscript upload confirmations, analysis completion, compliance results, subscription updates, and other important account-related events. Notifications are arranged in chronological order, allowing users to easily monitor the status of their manuscripts and remain informed of significant system activities. Users can select a notification to view additional details related to the corresponding event.

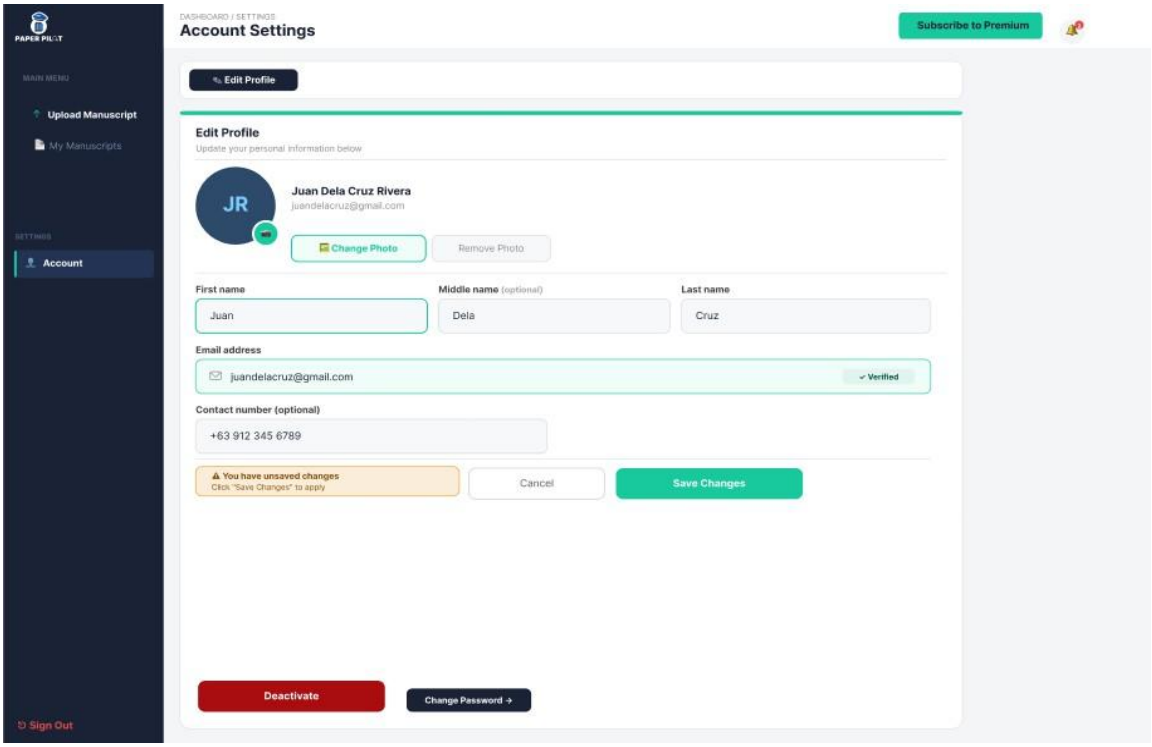


Figure 6.15: **Web Account Settings Page**

Figure 6.15 shows the Account Settings Page of the PaperPilot web application. This page allows users to manage their account information by updating their profile details, including their profile picture, name, email address, and contact number. Users can save changes to their personal information, change their account password, or deactivate their account when necessary. The page also provides options to sign out of the system and subscribe to the Premium plan to access additional features and services.

Storyboard

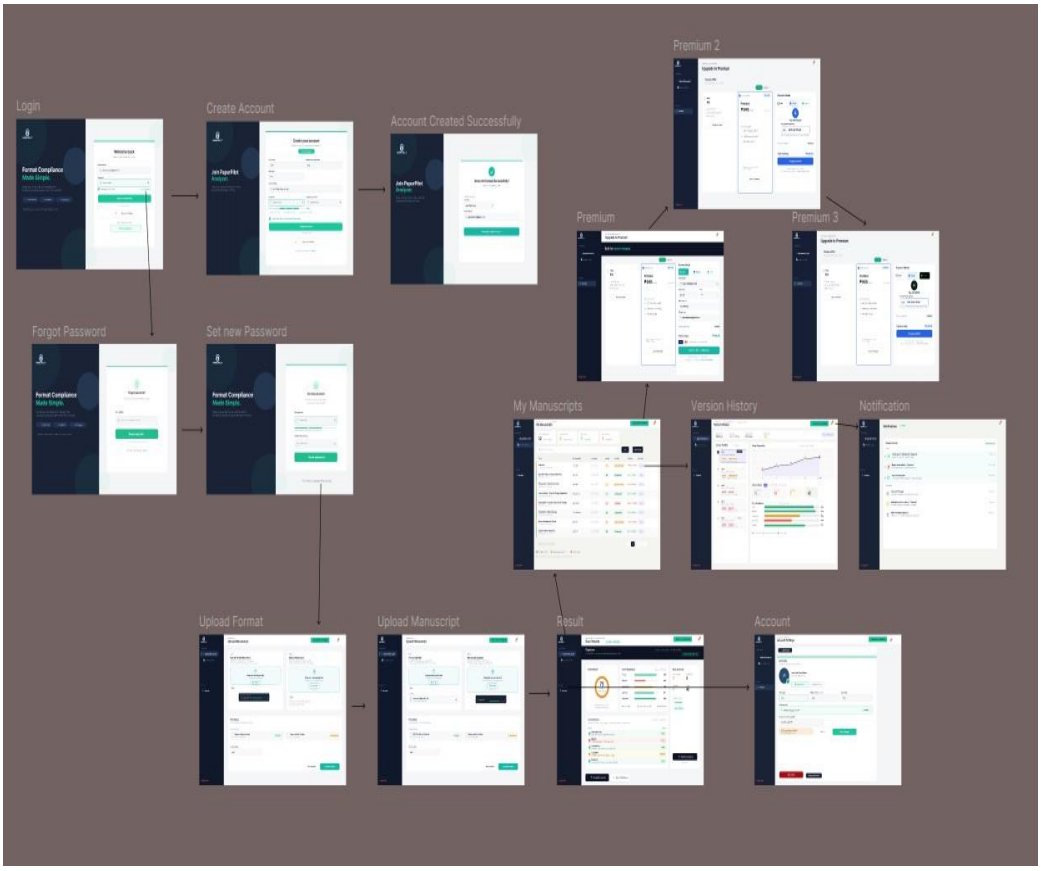


Figure 7: **Storyboard of the PaperPilot Mobile Application**

Figure 7.1 illustrates the storyboard of the PaperPilot web application, an AI-powered research manuscript compliance analyzer developed to assist students, researchers, and academic institutions in evaluating research manuscripts based on academic formatting standards. The application provides a structured workflow that enables users to manage their accounts, upload formatting guides and manuscripts, perform automated compliance analysis, and monitor manuscript versions through a web-based platform.

The storyboard begins with the account registration process, where new users can create an account by providing the required information or by registering through their Google account. Existing users can log in using their registered email address and password or the Sign in with Google option. If users forget their password, they can recover their account through the Forgot Password feature by resetting their password via their registered email.

After logging in, users are directed to the Upload Manuscript page, where they

first upload a formatting guide or institutional template that serves as the reference for compliance checking.

They then upload their research manuscript in a supported file format and specify a version label before initiating the analysis. The system validates the uploaded files and performs an AI-powered compliance analysis by comparing the manuscript against the uploaded formatting guide. Once the analysis is complete, the system generates detailed analysis results, including the overall compliance score, formatting evaluation, citation checking results, and AI-generated feedback with explanations of detected issues. Users can review the analysis results and upload revised versions of their manuscripts to monitor improvements over time.

The application also includes a My Manuscripts module, where users can view all previously uploaded manuscripts together with their compliance scores and review statuses. Selecting a manuscript opens its Version History, allowing users to monitor compliance progress across multiple uploaded versions and compare analysis results over time.

To keep users informed, the Notifications page displays recent activities and system-generated updates, including manuscript uploads, completed analyses, compliance results, and other important account-related events. The Account Settings page enables users to update their personal information, change their password, manage their profile, deactivate their account, or subscribe to the Premium plan.

For users who require additional features, the application provides a Subscription Plan page where they can choose between the Free and Premium plans. Premium subscriptions unlock enhanced capabilities such as unlimited manuscript scans, unlimited manuscript versions, advanced AI formatting feedback, and priority support. Users can complete their subscription securely using Card, GCash, or Maya payment methods.

Overall, the storyboard demonstrates the complete workflow of the PaperPilot mobile application, highlighting its user-friendly interface and AI-assisted features for manuscript upload, compliance analysis, version management, account management, notifications, and subscription management. By automating document analysis and providing detailed compliance feedback, the system assists users in preparing research manuscripts that adhere to institutional and academic formatting requirements.

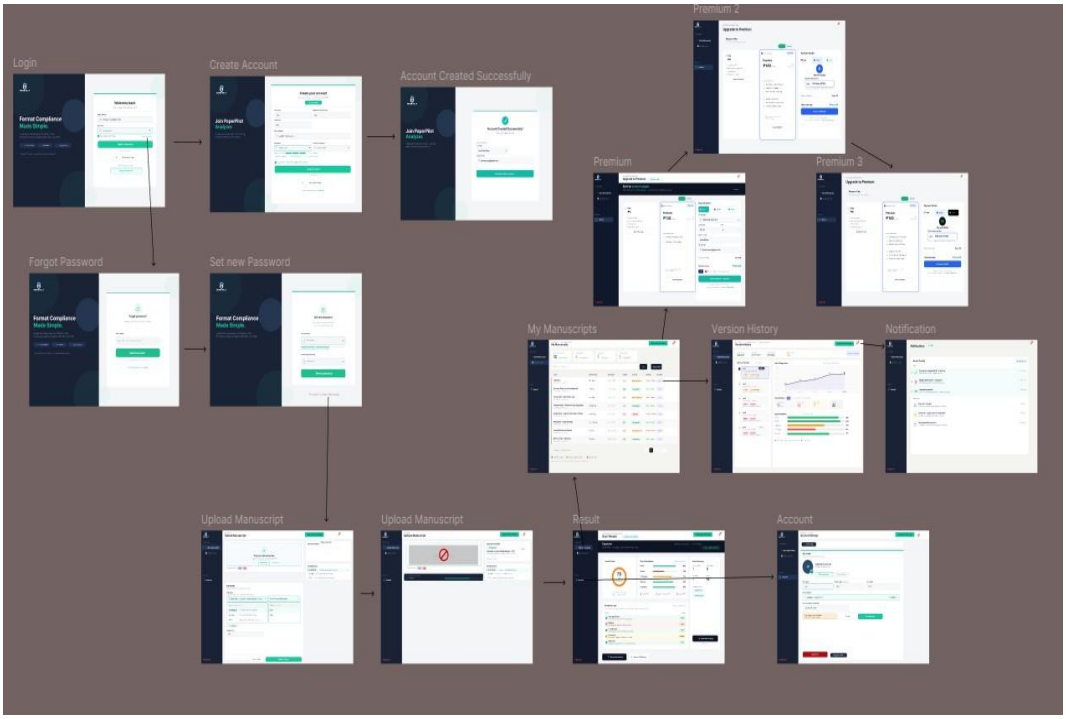


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The storyboard begins with the account registration process, where new users can create an account by providing the required information or by registering through their Google account. Existing users can log in using their registered email address and password or the Sign in with Google option. If users forget their password, they can recover their account through the Forgot Password feature by resetting their password via their registered email.

After logging in, users are directed to the Upload Manuscript page, where they can upload a supported research document, select the appropriate institution and course or department, and initiate the compliance analysis. The system then scans the manuscript and generates a Compliance Report that displays the overall compliance score, scan summary, formatting results, and a detailed score breakdown. Based on the analysis, users can review the identified formatting issues, download the report, and

upload a new version of the manuscript to track improvements.

The application also includes the My Manuscripts page, where users can access all previously uploaded manuscripts and view their Version History to compare compliance results across different uploaded versions. Additional features include the Notifications page for viewing recent system activities and analysis updates, the Account Settings page for managing profile information and account security, and the Subscription Plan page, where users can upgrade to the Premium plan using Card, GCash, or Maya payment methods.

Overall, the storyboard presents the complete workflow of the PaperPilot web application, demonstrating its capabilities in manuscript management, automated compliance analysis, version tracking, subscription management, and account administration. The system provides users with an efficient and accessible platform for evaluating research manuscripts and ensuring compliance with institutional and academic formatting requirements.

Database Design

This section presents the database design used to store, manage, and organize the data required by the system. It defines the data structure, relationships, and rules that govern how information is stored and accessed within the database management system. The database supports the management of user accounts, uploaded academic documents, document analysis results, compliance reports, supported academic writing styles (APA, MLA, and IEEE), Explainable Artificial Intelligence (XAI) explanations, and system notifications. This design ensures that all data is well organized, securely managed, and efficiently accessible across the system's modules.

Entity Relationship Diagram

The entity relationship diagram visually shows how entities particularly people, objects, or concepts are related in a database. It defines each entity, its attributes, and the connections between them, helping to explain the logical structure of the database. This makes it easier to understand and trace how data is organized and linked across the system.

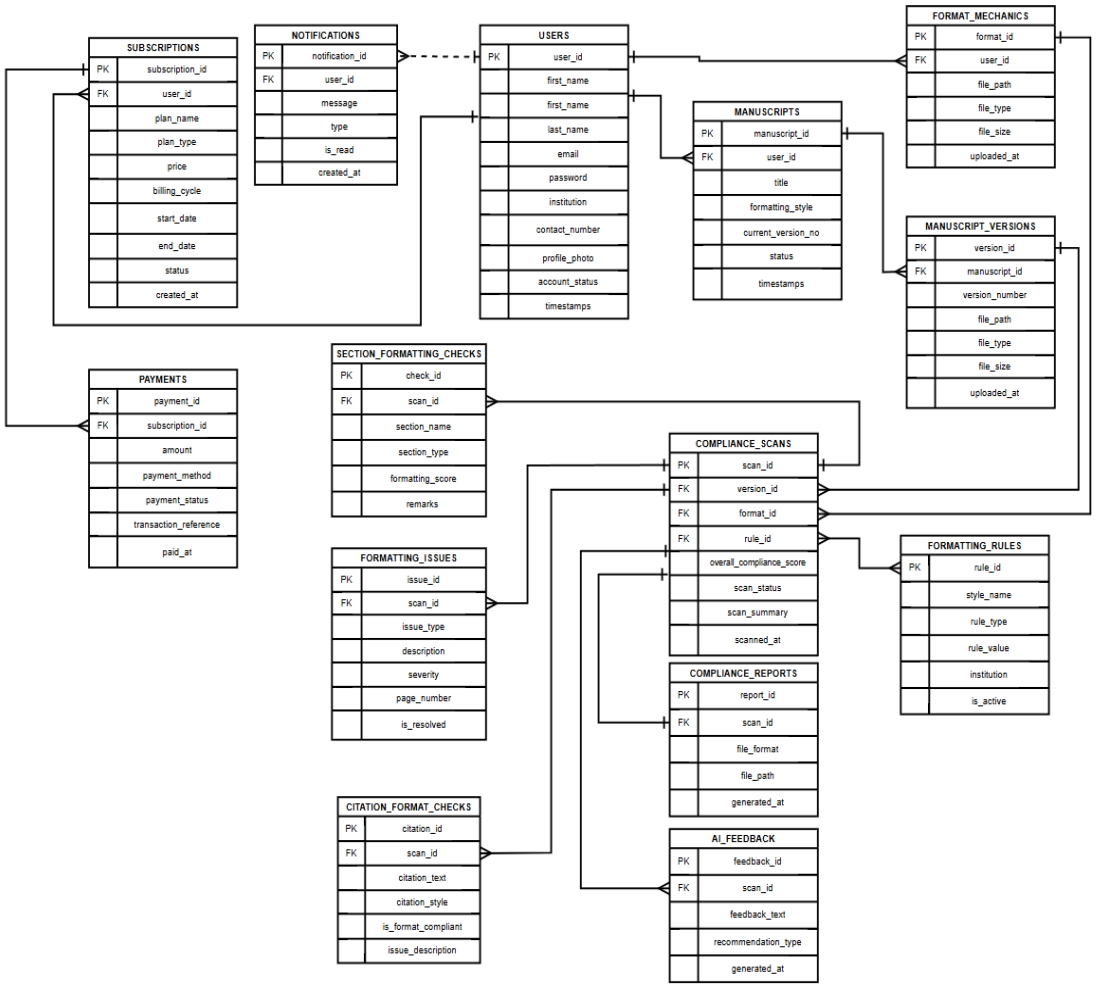


Figure 8 : **Entity Relationship Diagram**

Figure 8 illustrates the complete database architecture designed to support the core modules and services of the PaperPilot platform. The Entity-Relationship Diagram (ERD) models the logical data structure, showcasing how different components of the system interact across manuscript submission, format mechanics management, formatting compliance checking, and feedback delivery. At the core of the ERD is the Users entity, which encapsulates all platform participants under a single Researcher role. This table captures essential identity and access attributes like user_id, email, password, and account_status.

Connected to the user structure is the Manuscripts entity, which stores each submitted work's title, formatting_style, current_version_no, and status, allowing every manuscript to be traced back to its owning researcher. Version handling is managed through the Manuscript_Versions entity, which records file_path, file_type, file_size,

and uploaded_at for every re-upload, preserving a complete history of manuscript revisions over time. Also connected directly to the Users entity is the Format_Mechanics entity, which stores the institutional formatting guide or template a researcher uploads, along with its file_path, file_type, file_size, and uploaded_at. Because this entity is linked to Users rather than to a single manuscript, a researcher can upload a format mechanics file once and reuse it across any number of manuscripts or academic documents without having to re-upload it for each submission.

Formatting compliance checking is centered on the Compliance_Scans entity, which links a specific manuscript version, its corresponding format mechanics file, and the applicable formatting rule, and stores the overall_compliance_score, scan_status, and scan_summary generated for each scan. This entity branches out into several supporting tables that record the specifics of each check. The Section_Formatting_Checks entity captures section-level formatting_score and remarks, the Formatting_Issues entity logs individual violations such as issue_type, severity, and page_number, and the Citation_Format_Checks entity evaluates citation_text against the expected citation_style to determine is_format_compliant. These checks are validated against the Formatting_Rules entity, which defines the style_name, rule_type, rule_value, and institution-specific settings applied whenever a scan is performed.

Once a scan is completed, its results can be compiled into the Compliance_Reports entity, which produces downloadable outputs through file_format and file_path, while the AI_Feedback entity stores the feedback_text and recommendation_type generated to guide the researcher's revisions. The Notifications entity ensures researchers receive timely alerts regarding scan completion, detected issues, and report generation. To support the platform's subscription-based access, the Subscriptions entity records each researcher's plan_name, plan_type, billing_cycle, start_date, and status, while the Payments entity logs the corresponding transaction details such as amount, payment_method, payment_status, and paid_at for every payment made under a subscription.

Overall, the PaperPilot ERD outlines a modular architecture that governs the flow of manuscript submission, reusable format mechanics management, version tracking, formatting compliance analysis, AI-assisted feedback, subscription management, and reporting. By establishing clear relationships between logically grouped entities, the system ensures data integrity, traceability, and accuracy throughout its core processes.

Data Dictionary

This section presents the data dictionary for the "Paperpilot" application that would document all the data elements used within the system, providing detailed descriptions of each element, its attributes, and other relevant information.

Table 6

USER

Column Name	Data Type	Field Size	Null	Description
user_id	INT	11	NO	Primary Key - Unique identifier for each user
first_name	VARCHAR	100	NO	User first name
last_name	VARCHAR	100	NO	User last name
email	VARCHAR	255	NO	User email address
password	VARCHAR	255	NO	Encrypted user password

Table 6.1

USERS CONT'D CONT'D

Institution	VARCHAR	255	YES	User institution or university
contact_number	VARCHAR	20	YES	User contact number
profile_photo	VARCHAR	255	YES	Profile photo path
account_status	VARCHAR	50	NO	Current account status
Timestamps	VARCHAR	-	NO	Record creation/update timestamp

Table 6 manages researcher account information used to access the PaperPilot system. It stores personal details, login credentials, institution information, account status, and profile data to support secure authentication and user management.

Table 7

SUBSCRIPTIONS

Column Name	Data Type	Field Size	Null	Description
subscription_id	INT	11	NO	Primary Key
user_id	INT	11	NO	ForeignKey referencing USERS
plan_name	VARCHAR	100	NO	Subscription plan name
plan_type	VARCHAR	50	NO	Subscription type
price	DECIMAL	10,2	NO	Subscription price
billing_cycle	VARCHAR	50	NO	Monthly or yearly billing
start_date	DATE	-	NO	Subscription start date
end_date	DATE	-	YES	Subscription end date
status	VARCHAR	50	NO	Subscription status
created_at	DATETIME	-	NO	Record creation timestamp

Table 7 manages subscription records associated with user accounts. It stores plan details, billing information, subscription period, and status to support subscription-based access and service management.

Table 8

PAYMENTS

Column Name	Data Type	Field Size	Null	Description
payment_id	INT	11	NO	Primary Key
subscription_id	INT	11	NO	Foreign Key referencing SUBSCRIPTIONS
amount	DECIMAL	10,2	NO	Payment amount
payment_method	VARCHAR	50	NO	Payment method used
payment_status	VARCHAR	50	NO	Payment status

transaction_reference	VARCHAR	255	YES	Transaction reference number
paid_at	DATETIME	-	YES	Payment date and time

Table 8 stores payment transactions related to subscriptions. It records payment amounts, methods, transaction references, payment status, and payment dates for billing and financial tracking.

Table 9

NOTIFICATIO

NS

Column Name	Data Type	Field Size	Null	Description
notification_id	INT	11	NO	Primary Key
user_id	INT	11	NO	Foreign Key referencing USERS
message	TEXT	-	NO	Notification message
type	VARCHAR	50	NO	Notification type
is_read	BOOLEAN	-	NO	Read or unread status
created_at	DATETIME	-	NO	Notification creation timestamp

Table 9 manages system notifications sent to users. It stores notification messages, notification types, read status, and timestamps to support user alerts and updates.

Table 10

MANUSCRIPTS

Column Name	Data Type	Field Size	Null	Description
manuscript_id	INT	11	NO	Primary Key
user_id	INT	11	NO	Foreign Key referencing USERS
title	VARCHAR	255	NO	Manuscript title

formatting_style	VARCHAR	100	YES	Selected formatting style
current_version_no	INT	11	NO	Current manuscript version

status	VARCHAR	50	NO	Current manuscript status
timestamps	DATETIME	-	NO	Record creation/update timestamp

Table 10 stores academic documents uploaded by researchers. It records ownership, title, formatting style, current version, and processing status for document management.

Table 11

MANUSCRIPTS VERSIONS

Column Name	Data Type	Field Size	Null	Description
version_id	INT	11	NO	Primary Key
manuscript_id	INT	11	NO	Foreign Key referencing MANUSCRIPTS
version_number	INT	11	NO	Version number
file_path	VARCHAR	255	NO	File storage path
file_type	VARCHAR	20	NO	File type
file_size	BIGINT	-	YES	File size in bytes
uploaded_at	DATETIME	-	NO	Upload timestamp

Table 11 stores version history for uploaded academic documents. It tracks file versions, file names, file types, file sizes, and upload dates to support version management.

Table 12

FORMATTING RULES

Column Name	Data Type	Field Size	Null	Description
rule_id	INT	11	NO	Primary Key
style_name	VARCHAR	100	NO	Formatting style name
rule_type	VARCHAR	100	NO	Rule category
rule_value	VARCHAR	255	NO	Required formatting value

institution	VARCHAR	255	YES	Institution associated with the rule

is_active	BOOLEAN	-	NO	Indicates whether the rule is active

Table 12 stores institution-specific formatting and compliance rules. It contains rule names, rule types, required values, institution references, and activation status used during compliance evaluation.

Table 13

COMPLIANCE SCAN

Column Name	Data Type	Field Size	Null	Description
scan_id	INT	11	NO	Primary Key
version_id	INT	11	NO	Foreign Key referencing MANUSCRIPT_VERSIONS
rule_id	INT	11	NO	Foreign Key referencing FORMATTING_RULES
overall_compliance_score	DECIMAL	5,2	NO	Overall compliance percentage
scan_status	VARCHAR	50	NO	Scan status
scan_summary	TEXT	-	YES	Summary of scan results
scanned_at	DATETIME	-	NO	Scan timestamp

Table 13 stores the overall results of manuscript compliance evaluations. It records scanned manuscript versions, applied rules, compliance scores, scan status, summaries, and scan dates.

Table 14

SECTION_FORMATTING_CHEC

KS

Column Name	Data Type	Field Size	Null	Description
check_id	INT	11	NO	Primary Key
scan_id	INT	11	NO	Foreign Key referencing COMPLIANCE_SCANS

Table 14.1

SECTION_FORMATTING_CHECKS CONT'D

section_name	VARCHAR	255	NO	Document section name
section_type	VARCHAR	100	NO	Section category
formatting_score	DECIMAL	5,2	NO	Section-level formatting compliance score
remarks	TEXT	-	YES	Additional remarks

Table 14 stores section-level formatting evaluation results for scanned academic documents. It records section names, formatting scores, and remarks related to per-section formatting verification.

Table 15

FORMATTIN

G ISSUES

Column Name	Data Type	Field Size	Null	Description
issue_id	INT	11	NO	Primary Key
scan_id	INT	11	NO	Foreign Key referencing COMPLIANCE_SCANS
issue_type	VARCHAR	100	NO	Type of formatting issue
description	TEXT	-	NO	Issue description
severity	VARCHAR	20	NO	Issue severity
page_number	INT	11	YES	Page where issue was found
is_resolved	BOOLEAN	-	NO	Indicates whether the issue is resolved

Table 15 stores detailed formatting violations detected during compliance scanning. It records issue types, descriptions, severity levels, page numbers, and resolution status.

Table 16

CITATION FORMATS CHECK

Column Name	Data Type	Field Size	Null	Description
citation_id	INT	11	NO	Primary Key

Table 16.1

CITATION FORMAT CHECKS CONT'D

scan_id	INT	11	NO	Foreign Key referencing COMPLIANCE_SCANS
citation_text	TEXT	-	NO	Citation content
citation_style	VARCHAR	100	NO	Citation style
is_format_compliant	BOOLEAN	-	NO	Indicates whether citation format is compliant
issue_description	TEXT	-	YES	Citation formatting issue description

Table 16 stores citation formatting validation results for scanned manuscripts. It records citation text, citation style, compliance status, and issue descriptions related to citation formatting.

Table 17

COMPLIANCE REPORT

Column Name	Data Type	Field Size	Null	Description
report_id	INT	11	NO	Primary Key
scan_id	INT	11	NO	Foreign Key referencing COMPLIANCE_SCANS
file_format	VARCHAR	20	NO	Report file format
file_path	VARCHAR	255	NO	Report storage path
generated_at	DATETIME	-	NO	Report generation timestamp

Table 17 stores generated compliance reports for scanned manuscripts. It records report files, storage paths, and generation timestamps for report management and retrieval.

Table 18

AI FEEDBACK

Column Name	Data Type	Field Size	Null	Description
feedback_id	INT	11	NO	Primary Key
scan_id	INT	11	NO	Foreign Key referencing COMPLIANCE_SCANS
feedback_text	TEXT	-	NO	Generated feedback
recommendation_type	VARCHAR	100	NO	Type of recommendation
generated_at	DATETIME	-	NO	Feedback generation timestamp

Table 18 stores automated feedback and recommendations generated after compliance scanning. It records feedback text, recommendation types, and generates timestamps to assist researchers in improving manuscript compliance.

Network Design

Network design is the planning and deployment of the computer network infrastructure of a system. This process involves recognizing the network requirements, determining the needed components and connections, designing the network model, and establishing the network topology. This section introduces the network design of the PaperPilot system, including the network model and topology. The system connects users to the internet via personal computers, laptops and mobile devices. The application can be accessed by students, moderators and administrators at any time and from any place with an internet connection. As a centralised hub, the network connects client devices to the web server, application server, AI processing services, and cloud database, allowing for efficient communication, real-time data processing, secure authentication, document management, and retrieval of academic information within the PaperPilot platform.

Network Model

The network model diagram shows how data is transferred from one function to another within the PaperPilot system. The network model aims to optimize data transfer to satisfy user requests efficiently. Entities in the system are represented graphically, with some components accessible through different paths, enhancing the flexibility and overall performance of the application.

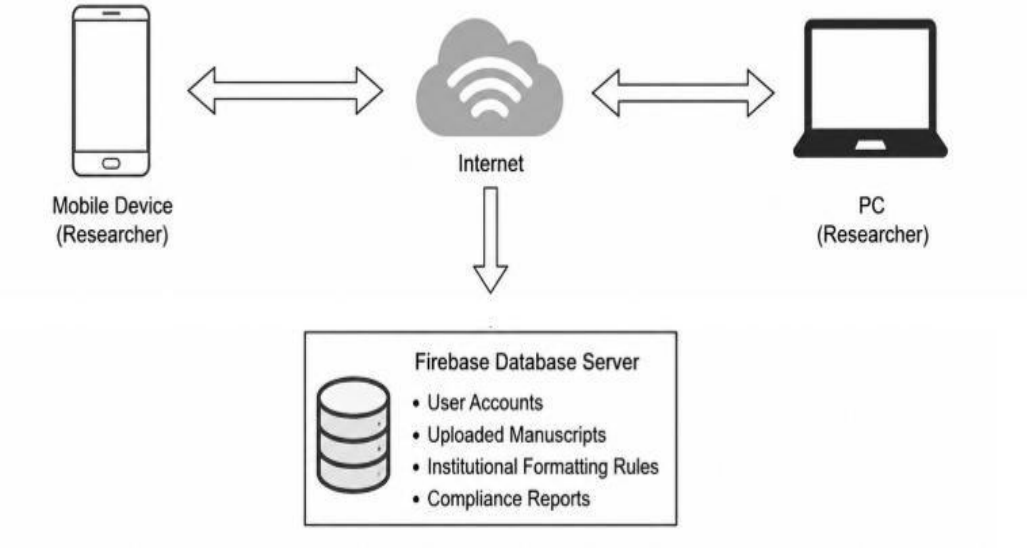


Figure 9: Network Model

Figure 9 shows how devices connect and communicate through the Internet. PaperPilot data is stored in a centralized database server where user documents and information are securely maintained. When students access the system using mobile devices or personal computers, their data is transmitted through the internet for processing and storage. To upload, retrieve, or modify documents, the device must be connected to the internet. Thus, the system allows users to efficiently store and access their academic documents anytime.

Network Topology

This section presents the network architecture of PaperPilot, illustrating how the system components interact to support institutional manuscript compliance checking. The architecture integrates a cloud server, database, web application, and mobile application to facilitate secure communication and centralized data management.

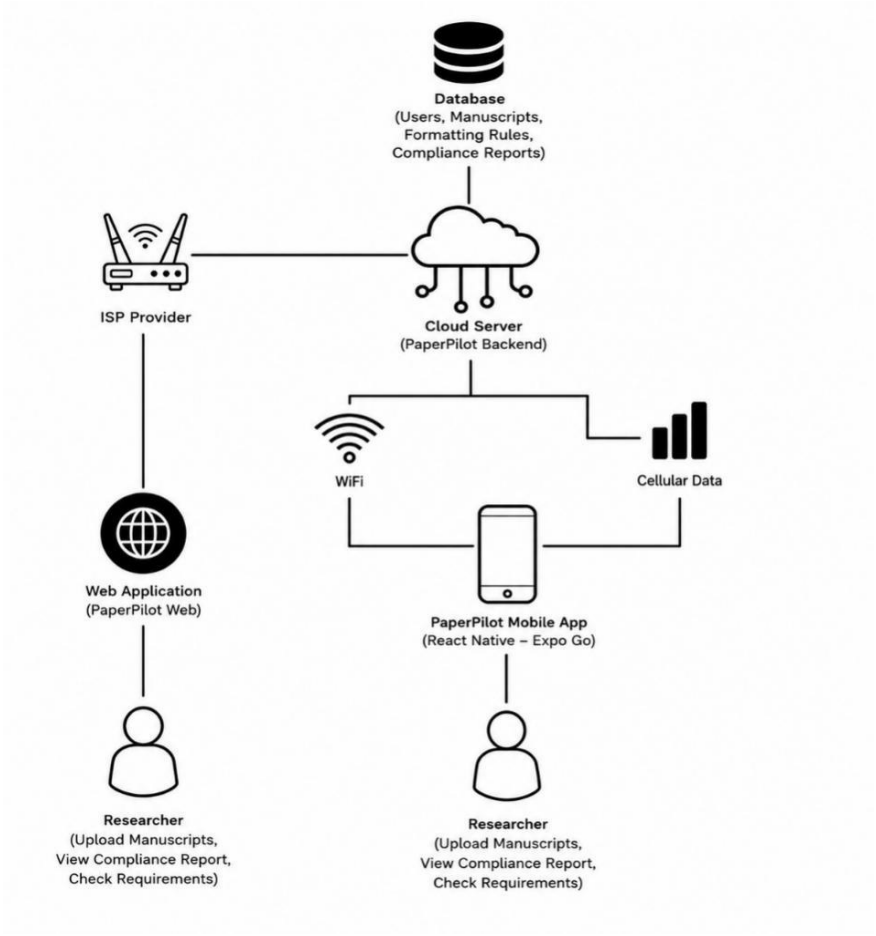


Figure 9.1: Network Topology

Figure 9.1 illustrates the network topology of the PaperPilot system, which follows a client-server architecture to support automated manuscript compliance checking based on institutional formatting standards. The topology consists of six main components: the database, cloud server, internet service provider (ISP), web application, mobile application, and researcher users. The cloud server functions as the central processing unit, handling requests from both the web and mobile applications while communicating with the database. The database stores user accounts, uploaded academic documents, institutional formatting rules, and generated compliance reports. Researchers can access PaperPilot through either the web application or the mobile application using an internet connection via an ISP, Wi-Fi, or cellular data. Through these platforms, users can upload academic documents, verify compliance with institutional formatting requirements, and view generated compliance reports. This client-server architecture provides secure, centralized, and efficient communication between the client applications, cloud server, and database, ensuring reliable document processing and compliance verification across both web and mobile platforms.

Deployment/Build Phase

This section is where the proposed PaperPilot system is developed and implemented based on the approved design. It serves as a guide for the development process and includes the technologies used in building the system, as presented in the Technology Stack Diagram, along with their corresponding functions. The hardware and software requirements necessary for the system are also specified to ensure proper development and operation. The outputs of this phase include the fully developed system features, which are prepared for testing and further evaluation.

Technology Stack Diagram

The technology stack diagram outlines the essential tools and frameworks needed for building the system. The technologies are categorized according to whether they are for mobile development, web development, backend services, or enhancements in the future. It helps to visualize how different components work together to support the whole system architecture.

Table 19

PAPERPILOT TECHNOLOGY STACK

Mobile Development	**Web Development**	**Future Development**
React Native Expo Go	React.js Node.js Express.js FastApi Tailwind CSS	IOS Platform

Machine Learning/AI	Server
Python Rule based Engine Gemini (API)	Google Firebase OnRender

Table 19 presents that PaperPilot is an ai powered academic document compliance-checking system designed to help students and academic institutions ensure that academic documents comply with institutional formatting standards. The system is developed using a modern and scalable technology stack that supports web-based document processing, automated compliance verification, secure cloud services, and future system expansion.

For web development, PaperPilot is built using React to create a responsive and interactive user interface, while Tailwind CSS is used to develop a clean and user-friendly design. Node.js and Express.js serve as the backend technologies, handling server-side processing, API communication, user authentication, document management, and overall system functionality. FastAPI and Python are utilized to implement the compliance-checking engine, which analyzes uploaded academic documents and verifies formatting elements such as font style and size, margins, spacing, headings, and page numbering based on institution-specific standards. Google Firebase provides cloud-based database management, authentication, file storage, and secure data synchronization across the system.

For mobile development, React Native with Expo Go is utilized to develop a mobile application that allows students to upload academic documents, view compliance reports, and monitor formatting corrections using Android devices through a single codebase

For future development, the system is designed to support iOS platform compatibility to increase accessibility across Apple devices. TensorFlow is also planned for future integration to enable machine learning capabilities, such as recognizing varying institutional formatting patterns and improving the Accuracy and adaptability of the compliance-checking engine while maintaining the system's focus on formatting compliance rather than research content.

Software Specification

This section outlines the required software for developing and deploying the PaperPilot system, including platforms, programming languages, tools, and environments needed to build and maintain both web and mobile access.

Table 20

SOFTWARE SPECIFICATION

Component	**Details**
Operating System	Compatible with Windows, Android
Text Editor/IDE	Visual Studio Code, Android Studio

Table 20.1

SOFTWARE SPECIFICATION CONT'D

Programming Language	JavaScript, TypeScript, Kotlin
Web Browser	Google Chrome
Mobile Platform	Android
Server	Google Firebase
Framework	Node.js with Express.js, React.js, ReactNative

Table 20 outlines the software specifications used in the development and deployment of the PaperPilot system. The platform is designed to support multiple operating systems including Windows, macOS, Android, and iOS to ensure accessibility across desktop and mobile environments. Development is primarily conducted using Visual Studio Code and Android Studio, which provide the necessary tools, debugging support, and development environment for both web and mobile application creation. JavaScript and TypeScript are utilized as the primary programming languages for frontend and backend development, while Kotlin supports Android-related mobile functionalities and compatibility. The web platform is developed using React.js to provide an interactive and dynamic user interface, while Node.js with Express.js manages backend services, API requests, authentication, and server-side processing. These technologies enable efficient communication between the client-side interface,

backend services, and cloud-based resources. Google Chrome and Microsoft Edge are utilized as the primary supported web browsers to ensure compatibility, responsive performance, and accessibility for users accessing the PaperPilot web application. For mobile accessibility, the system supports Android devices and iOS devices through mobile browser access, allowing students and users to utilize the platform conveniently on smartphones and tablets. Google Firebase serves as the cloud-based backend server and database provider, supporting real-time data storage, authentication, cloud synchronization, and secure management of user information and academic documents. The integration of these technologies enables the system to support academic document formatting support, document management, compliance checking, and AI-powered analysis features efficiently. Planned future developments for the PaperPilot system include expanded iOS support, enhanced AI-powered academic assistance capabilities, and additional intelligent processing features to further improve system scalability, accessibility, and overall performance for students, moderators, and administrators involved in academic research writing and compliance monitoring.

Hardware Specification

This section outlines the hardware requirements essential for the development and deployment of the PaperPilot system. It includes computing devices, memory, storage, and mobile hardware needed for smooth system performance during development and usage.

Table 21

HARDWARE SPECIFICATION

Pc	
Component	**Details**
CPU	A multi-core processor, such as Intel Core i5 or AMD Ryzen 5 and above
Memory	Minimum 8GB RAM (16GB recommended)
Storage	At least 256GB SSD for faster performance and data handling
MOBILE	
CPU	64-bit processor (e.g., Snapdragon or equivalent)

Memory	Minimum 4GB RAM
Storage	At least 64GB internal storage

Table 21 presents the hardware specifications required for developing and operating the PaperPilot system on both PCs and mobile devices. For PC use, the system requires a 64-bit or 32-bit processor with a clock speed of at least 2.5 GHz, such as an Intel Core i5 (10th Generation or newer) or AMD Ryzen 5 (3000 Series or higher), to efficiently support web application development, backend processing, document compliance verification, and cloud-based services. The computer should have a minimum of 8 GB RAM to ensure smooth multitasking and stable system performance during development and operation. In addition, the system requires either integrated or dedicated graphics, such as Intel UHD Graphics, AMD Graphics, or NVIDIA Graphics, to support the graphical user interface and overall application performance.

For mobile use, the device should be equipped with an octa-core processor running at 2.0 GHz or faster to provide responsive performance when accessing the PaperPilot application. A minimum of 4 GB RAM is required to ensure smooth navigation, document uploads, and viewing of compliance reports. The device should also have at least 64 GB of available storage to accommodate the application, cached data, and downloaded compliance reports. These hardware specifications ensure that PaperPilot operates efficiently across both desktop and mobile platforms while providing reliable academic document formatting verification and compliance checking based on institutional standards.

Program Specification

This section defines the intended functionalities and expected outputs of the PaperPilot system. It outlines the key features of the application and how each function supports researchers in verifying the formatting and citation format compliance of their academic documents. The system is divided into different modules, each responsible for specific tasks such as document management, compliance checking, and analysis reporting.

List of Modules

The modules and features required to run the PaperPilot application efficiently are outlined in this section. Different programmers are given the modules to build the

different features.

Table 22

LIST OF MODULES

Programmer	Modules	Researchers
		Account Management
	Register	*
	Login	*
Amistoso	Forgot Password	*
	Edit Profile	*
	Upload Profile Picture	*
	Change Password	*
	Deactivate Account	*
	Logout	*
Number of point (1 point per module per user)		**1**
Opo	**Manuscript Management**	
	Upload Format Mechanics	*
	Upload Manuscript	*
	Validate File Type	*
	Manage Versions	*
	Upload and Analyze	*

Table 22.1

LIST OF MODULES CONT'D

Number of point (1 point per module per user)		**1**
	AI Analysis Structure	
	Check fonts	*
	Check spacing	*
Navarro	Check indention	*
	Check margins	*
	Check page breaks	*
	Check layout for tables	*
	Check pagination	*
	Check Citation format	*
Number of point (1 point per module per user)		**1**
Pelayo	**Overall Results**	
	View Overall Compliance Score	*
	View Score Breakdown	*
	View Detected Issues and Warnings	*
	Generate Compliance Report	*
	Download Compliance Report (PDF)	*
	View Version Compliance Report	*

Number of point (1 point per module per user)	**1**
Number of Modules per User (equals to total no. of points per user)	4
Total Number of Modules	4

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APPENDICES

APPENDIX A

Transmittal Letter - University of the Philippines

TRANSMITTAL LETTER

June 15, 2026

Dr. Daisy A. Romeo
University of the Philippines
Gorordo Ave, Lahug, Cebu City, 6000,
Philippines

Dear Dr. Romeo:

We are fourth-year Bachelor of Science in Information Technology students from the College of Computer Studies, currently working on our Capstone Project 1 entitled **"PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM."** The project aims to develop a platform that helps students with research writing, formatting, and citation management for their thesis and capstone.

As part of our Capstone Project 1 requirements, we are conducting interviews and surveys to gather information and validate the need for our proposed system. We would like to request the participation of your office in an interview and/or survey to gain insights on the common challenges students face in writing their thesis or capstone, as well as the support services available to them.

We are also currently interns assigned to the UP Information Technology Center. As part of both our internship and capstone requirements, we would appreciate the opportunity to interview your office for professional insights that will help in developing and evaluating our proposed system. The information gathered will be used only for analyzing, designing, and evaluating our system, to ensure it effectively supports students throughout their thesis and capstone writing process. All information will be kept strictly confidential and used only for academic and research purposes. We hope for your favorable consideration and participation in this study.

Respectfully yours

Vince Bello B. Opo
Project Manager

Approved by:

Mr. Joemarie Amparo
Adviser

Mr. Eric P. Ortega
Chairperson, College of Computer Studies
University of Cebu - Banilad Campus

APPENDIX B

Transmittal Letter - CITU Research Office

TRANSMITTAL LETTER

June 15, 2026

Research Office Personnel of Cebu
Institute of Technology
Natalio B. Bacalso Avenue, Cebu City,
Philippines, 6000, Philippines

Dear Research Office Personnel,

We, the fourth-year Bachelor of Science in Information Technology students of the College of Computer Studies, are currently undertaking our Capstone Project 1 entitled **"PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM."** This study aims to develop a platform designed to assist students in prodocing quality thesis and capstone papers by supporting them throughout the research writing process.

We are gathering data for the study and validation of our proposed system through surveys and interviews as part of the requirements for finishing our Capstone Project 1. We kindly ask that your office participate in this. We want to learn more about the typical difficulties students have when writing their capstone projects or theses, as well as the resources offered to student researchers during this process. The information gathered will greatly contribute to the design, development, and evaluation of PAPERPILOT. Specifically, it will help ensure that the system effectively supports students in the thesis and capstone writing process and addresses the actual needs of student researchers undertaking these requirements.

All information collected will be treated with utmost confidentiality and will be used solely for academic and research purposes. We sincerely hope for your favorable consideration and support. Thank you for your time and valuable contribution to our capstone project.

Respectfully yours

Vince Bella B. Opo
Project Manager

Approved by:

Mr. Joemarie Amparo
Adviser

Mr. Eric P. Ortega
Chairperson, College of Computer Studies
University of Cebu - Banilad Campus

APPENDIX C

Transmittal Letter - University of Cebu Banilad Campus

TRANSMITTAL LETTER

June 15, 2026

Research Coordinator of University of Cebu
Banilad Campus Gov. M. Cuenco Ave.
Banilad, Cebu City, Philippines

Dear Research Coordinator,

We, the fourth-year Bachelor of Science in Information Technology students of the College of Computer Studies, are currently undertaking our Capstone Project 1 entitled **"PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM."** The project aims to develop an intelligent platform that assists students in research writing, formatting, and citation management throughout the thesis and capstone writing process.

As part of the requirements for the completion of our Capstone Project 1, we are conducting interviews and surveys to gather relevant information and validate the need for the proposed system. In this regard, we respectfully request your participation in an interview and/or survey. As a Research Coordinator, your expertise and experience in guiding student researchers will provide valuable insights regarding common challenges in thesis and capstone writing, as well as the effectiveness of existing research support mechanisms. The information gathered will be used for the analysis, design, and evaluation of PAPERPILOT, ensuring that the system effectively supports students throughout the thesis and capstone writing process. All information collected will be treated with strict confidentiality and will be used solely for academic and research purposes. Thank you for your time, support, and valuable contribution to our capstone project.

Respectfully yours

Vince Belle B. Opo
Project Manager

Approved by:

Mr. Joemarie Amparo
Adviser

Mr. Eric P. Ortega
Chairperson, College of Computer Studies
University of Cebu - Banilad Campus

APPENDIX D

Transmittal Letter - University of Cebu Banilad Campus

TRANSMITTAL LETTER

June 15, 2026

Research Office Personnel of
University of Cebu - Banilad
Campus Gov. M. Cuenco Ave.
Banilad, Cebu City, 6000,
Philippines

Dear Research Office Personnel,

We, the fourth-year Bachelor of Science in Information Technology students of the College of Computer Studies, are currently undertaking our Capstone Project 1 entitled **"PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM."** This study aims to develop a platform designed to assist students in producing quality thesis and capstone papers by supporting them throughout the research writing process.

We are gathering data for the study and validation of our proposed system through surveys and interviews as part of the requirements for finishing our Capstone Project 1. We kindly ask that your office participate in this. We want to learn more about the typical difficulties students have when writing their capstone projects or theses, as well as the resources offered to student researchers during this process. The information gathered will greatly contribute to the design, development, and evaluation of PAPERPILOT. Specifically, it will help ensure that the system effectively supports students in the thesis and capstone writing process and addresses the actual needs of student researchers undertaking these requirements.

All information collected will be treated with utmost confidentiality and will be used solely for academic and research purposes. We sincerely hope for your favorable consideration and support. Thank you for your time and valuable contribution to our capstone project.

Respectfully yours

Vince Belle B. Opo
Project Manager

Approved by:

Mr. Joenarth Amparo
Adviser

Mr. Erje P. Ortega
Chairperson, College of Computer Studies
University of Cebu - Banilad Campus

APPENDIX E

Transmittal Letter - CITU Students

TRANSMITTAL LETTER

June 15, 2026

Students of Cebu Institute of Technology
Natalio B. Bacalso Avenue, Cebu City,
Philippines, 6000, Philippines

Dear Students,

We, the fourth-year Bachelor of Science in Information Technology students of the College of Computer Studies, are currently undertaking our Capstone Project 1 entitled **"PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM."** This study focuses on the development of a platform that assists students in research writing by providing intelligent writing support, formatting guidance, and citation assistance throughout the thesis and capstone writing process.

As part of the requirements for the completion of our Capstone 1, we are conducting interviews and surveys to gather insights from students who are currently undertaking their thesis or capstone projects. In this regard, we respectfully request your participation. Your experiences, challenges, and recommendation regarding research writing will greatly contribute to the development and evaluation of our proposed system. The information gathered will be used to assess the usability, effectiveness, and relevance of PAPERPILOT in supporting students throughout the thesis and capstone writing process. All data collected will be treated with strict confidentiality and will be used solely for academic and research purposes. We sincerely hope for your favorable consideration and participation in this study.

Respectfully yours

Vince Bella B. Opo
Project Manager

Approved by:

Mr. Joemarie Amparo
Adviser

Mr. Eric P. Ortega
Chairperson, College of Computer Studies
University of Cebu - Banilad Campus

APPENDIX F

Transmittal Letter - University of Cebu Banilad Campus Students

June 15, 2026

TRANSMITTAL LETTER

Students of University of Cebu -
Banilad Campus Gov. M. Cuenco Ave.
Banilad, Cebu City, 6000, Philippines

Dear Students,

We, the fourth-year Bachelor of Science in Information Technology students of the College of Computer Studies, are currently undertaking our Capstone Project 1 entitled **"PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM."** This study focuses on the development of a platform that helps students with research writing, formatting, and citation management throughout the thesis and capstone writing process.

As part of the requirements for the completion of our Capstone 1, we are conducting interviews and surveys to gather insights from students who are currently undertaking their thesis or capstone projects. In this regard, we respectfully request your participation. Your experiences, challenges, and recommendations regarding research writing will greatly contribute to the development and evaluation of our proposed system. The information gathered will be used to assess the usability, effectiveness, and relevance of PAPERPILOT in supporting students throughout the thesis and capstone writing process. All data collected will be treated with strict confidentiality and will be used solely for academic and research purposes. We sincerely hope for your favorable consideration and participation in this study.

Respectfully yours

Vince Bale S. Opo
Project Manager

Approved by:

Mr. Joemarie Amparo
Adviser

Mr. Eric P. Ortega
Chairperson, College of Computer Studies
University of Cebu - Banilad Campus

APPENDIX G - Research

Instrument Survey Questionnaire

- Students

Dear Respondents,

We, the fourth-year students of the Bachelor of Science in Information Technology program at the University of Cebu - Banilad Campus, are conducting a research study entitled "PaperPilot: An AI-Powered Academic Document Analysis and Compliance Checking System."

This study aims to develop an intelligent academic document analysis system that assists students in evaluating their research papers based on institutional formatting standards and academic writing requirements. The proposed system is designed to analyze the formatting of academic documents, identify compliance issues, and provide explainable feedback and recommendations to help students improve their manuscripts before adviser submission.

The purpose of this study is to determine the common challenges encountered by students during the preparation of thesis, capstone, and other academic documents. Specifically, the study seeks to identify difficulties related to manual formatting verification, document structure validation, reference checking, repeated revisions, and the time-consuming process of ensuring compliance with university manuscript standards. The findings of this study will serve as the basis for developing PaperPilot as an intelligent compliance-checking system that supports students throughout the academic writing process.

To help us develop a solution that effectively addresses these challenges, we respectfully invite you to participate in this survey. Your responses will provide valuable insights into your experiences in preparing academic documents, the difficulties you encounter during manuscript checking and revision, and your perception of the usefulness of an AI-powered compliance-checking system.

Data Privacy Consent:

I consent to having my responses used for the purposes of this research by submitting this form. I understand that the information I provide will be kept strictly confidential and used solely for academic purposes. The researchers are fully responsible for handling my data with care and integrity.

o I Agree

o I Disagree

1. I am familiar with the process of writing a thesis, capstone, or research paper.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

2. I understand the manuscript formatting requirements required by my institution.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

3. I am confident that I can prepare an academic document that complies with my institution's formatting standards without assistance.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

4. I regularly review my manuscript before submitting it to my adviser.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

5. Despite reviewing my manuscript, I am still uncertain whether it fully complies with my institution's formatting requirements.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

6. I Experience delays because i need to wait for feedback from my adviser before proceeding with revisions

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

7. I often overlook formatting mistakes even after reviewing my document several times.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

8. I do not frequently encounter formatting issues involving margins, spacing, headings, fonts.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

9. I worry that I may unintentionally overlook important formatting errors before submission.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

10. Checking every formatting requirement consumes a significant amount of time.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

11. I spend more time checking formatting than improving the quality of my research.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

12. Repeated revisions increase my overall printing expenses.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

13. Waiting for adviser feedback delays the completion of my research paper.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

14. Repeated revisions increase the overall time needed to complete my thesis or capstone project.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

15. The time spent on formatting revisions affects my ability to meet academic deadlines.

○ Strongly Agree

○ Agree

○ Neutral

○ Disagree

○ Strongly Disagree

16. A system that checks formatting based on my institution's standards would be beneficial.

○ Strongly Agree

○ Agree

○ Neutral

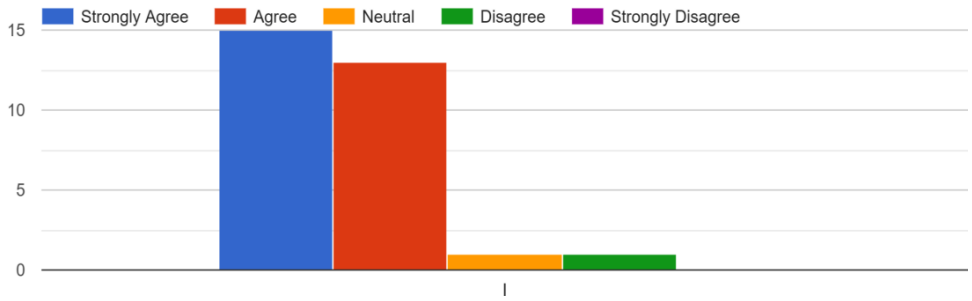
○ Disagree

○ Strongly Disagree

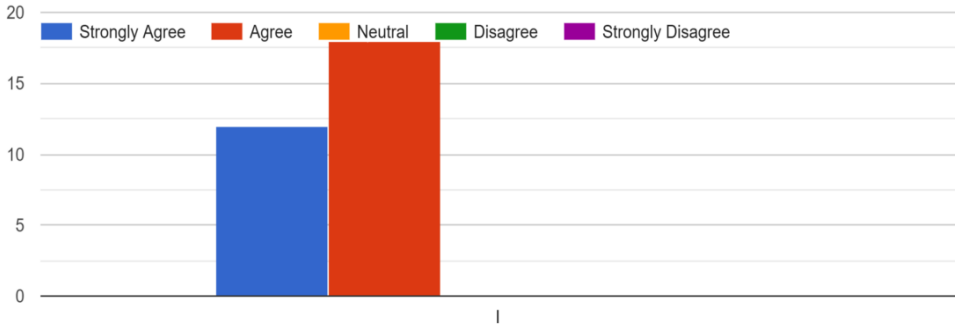
APPENDIX H

Results and Findings

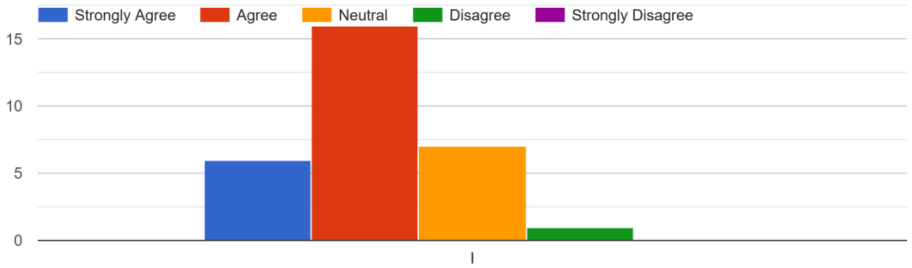
I am familiar with the process of writing a thesis, capstone, or research paper.



I understand the manuscript formatting requirements required by my institution.

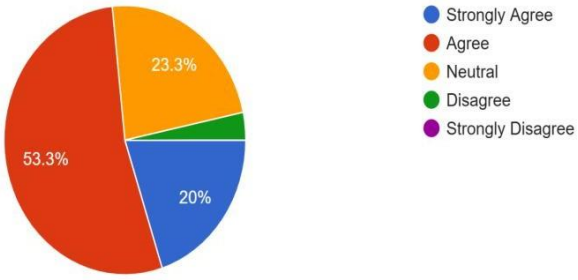


I am confident that I can prepare an academic document that complies with my institution's formatting standards without assistance.



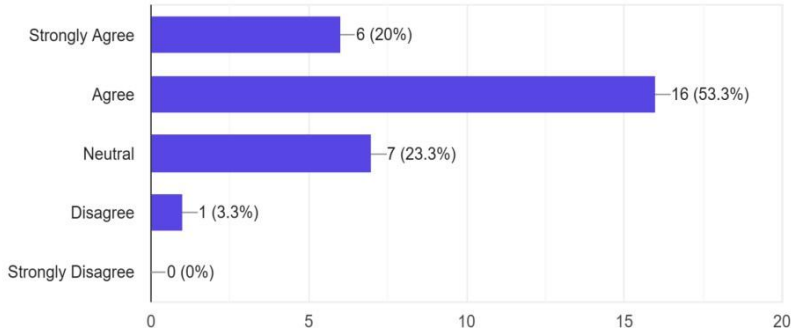
I regularly review my manuscript before submitting it to my adviser.

30 responses

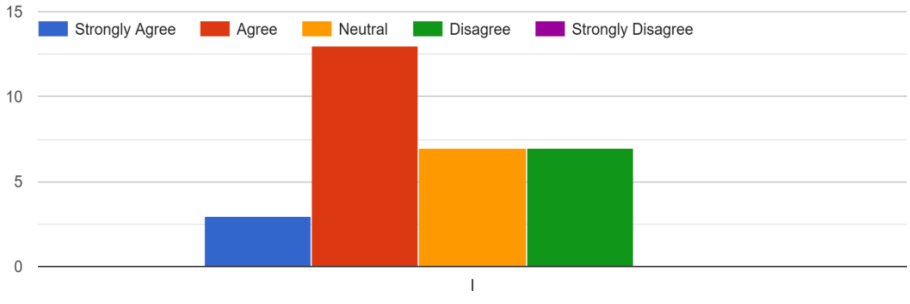


Despite reviewing my manuscript, I am still uncertain whether it fully complies with my institution's formatting requirements.

30 responses

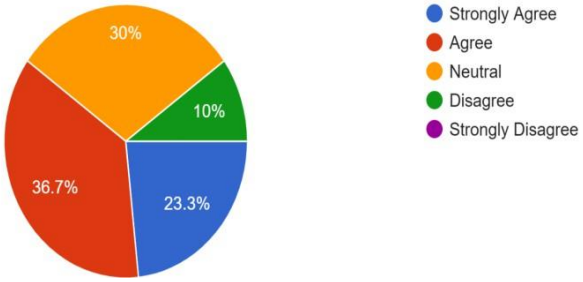


I often overlook formatting mistakes even after reviewing my document several times.

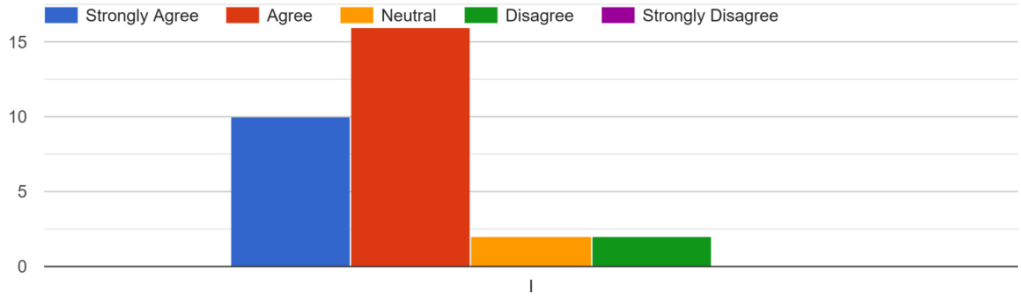


I do not frequently encounter formatting issues involving margins, spacing, headings, fonts, page numbering.

30 responses

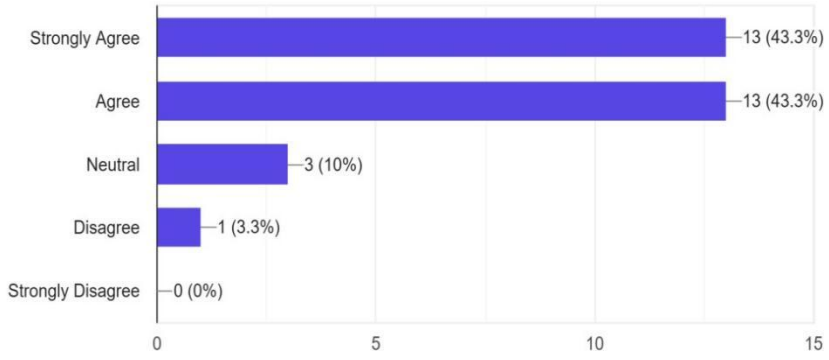


I experience delays because I need to wait for feedback from my adviser before proceeding with revisions.



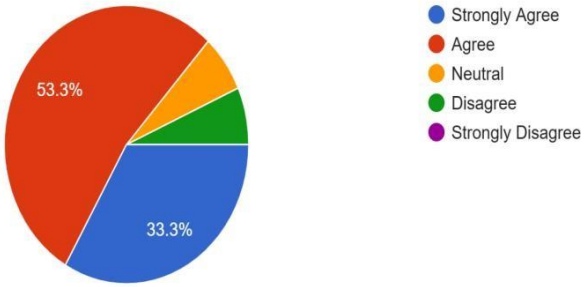
I worry that I may unintentionally overlook important formatting errors before submission.

30 responses



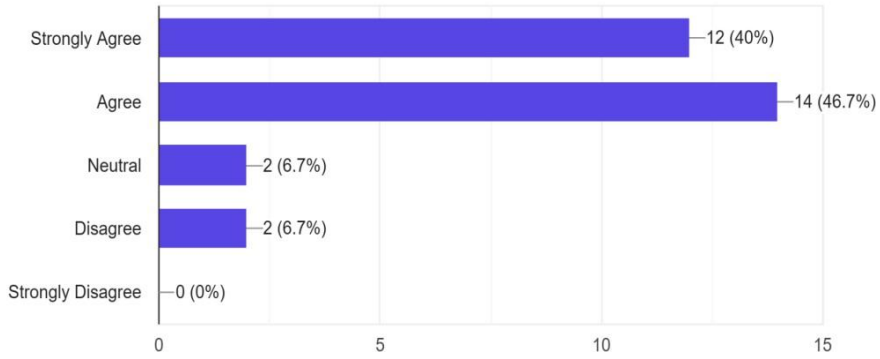
Checking every formatting requirement consumes a significant amount of time.

30 responses

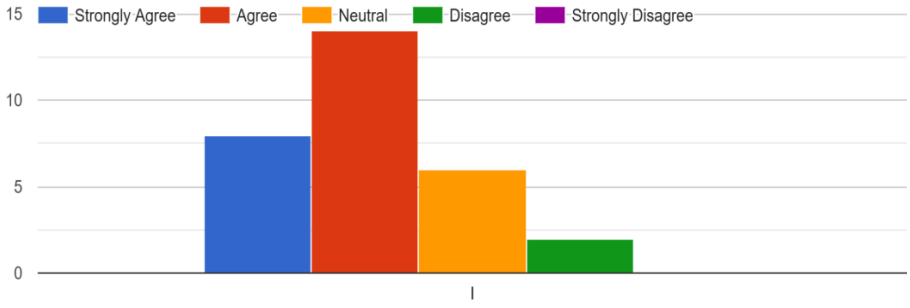


I spend more time checking formatting than improving the quality of my research.

30 responses

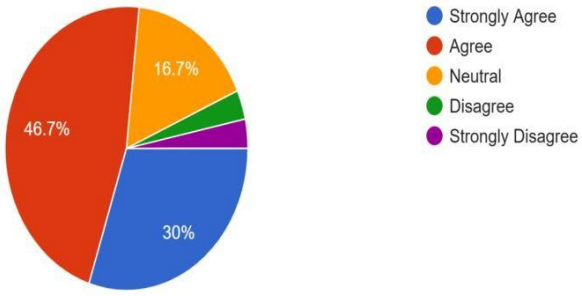


Waiting for adviser feedback delays the completion of my research paper.



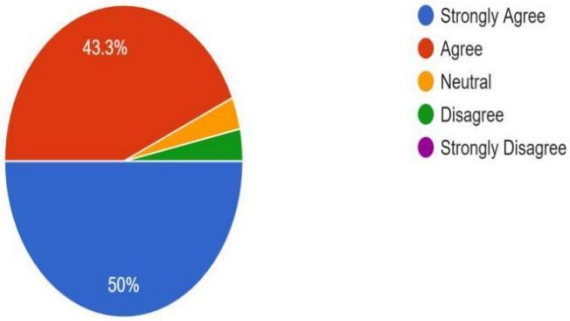
Repeated revisions increase the overall time needed to complete my thesis or capstone project.

30 responses



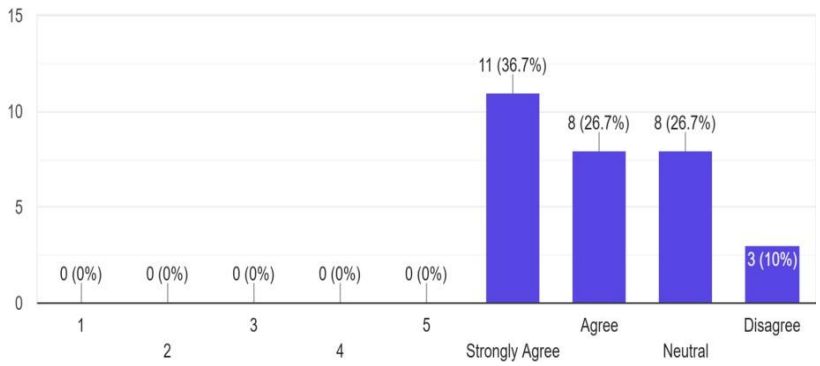
Repeated revisions increase my overall research expenses.

30 responses



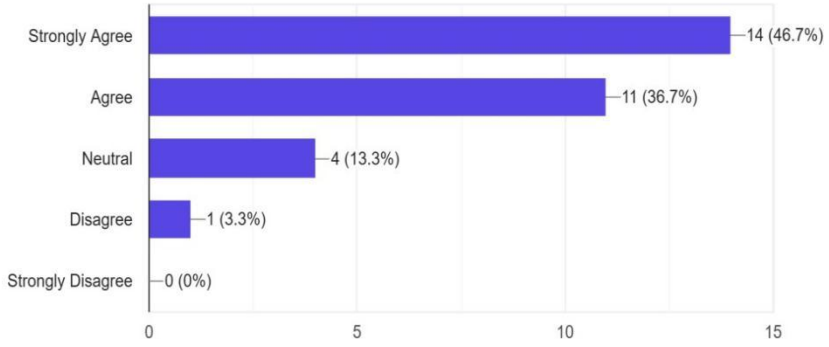
The time spent on formatting revisions affects my ability to meet academic deadlines.

30 responses



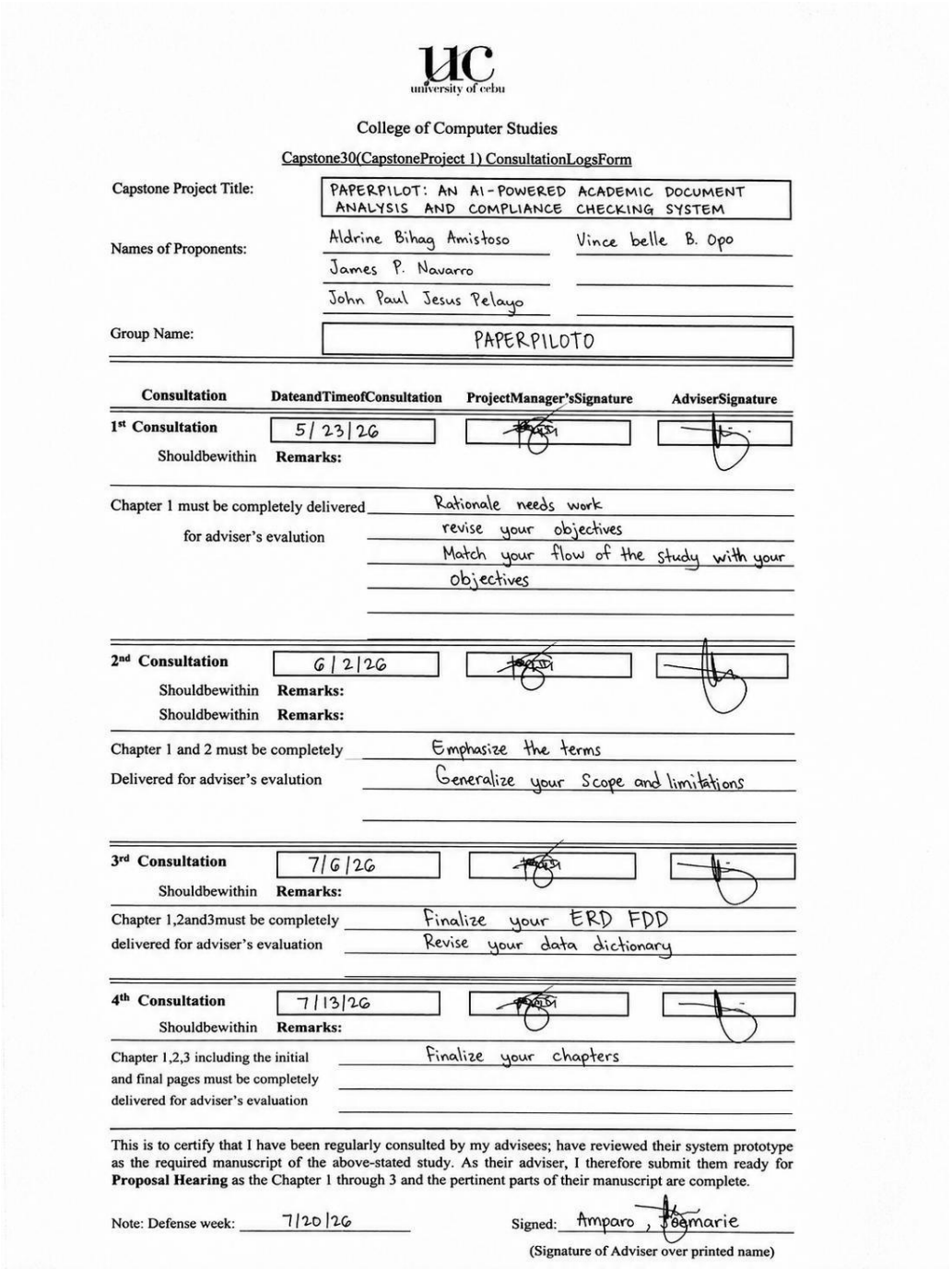
A system that checks formatting based on my institution's standards would be beneficial.

30 responses



APPENDIX I

Consultation Logs - Adviser



university of cebu

College of Computer Studies

Capstone30(CapstoneProject 1) ConsultationLogsForm

Capstone Project Title: PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM

Names of Proponents: Aldrine Bihag Amistoso, Vince belle B. Opo, James P. Navarro, John Paul Jesus Pelayo

Group Name: PAPERPILOTO

Consultation	DateandTimeofConsultation	ProjectManager'sSignature	AdviserSignature
1st Consultation	5/23/26		

Shouldbewithin Remarks:

Chapter 1 must be completely delivered for adviser's evalution

Rationale needs work
revise your objectives
Match your flow of the study with your objectives

2nd Consultation	6/2/26		

Shouldbewithin Remarks:
Shouldbewithin Remarks:

Chapter 1 and 2 must be completely Delivered for adviser's evalution

Emphasize the terms
Generalize your Scope and limitations

3rd Consultation	7/6/26		

Shouldbewithin Remarks:

Chapter 1,2and3must be completely delivered for adviser's evaluation

Finalize your ERD FDD
Revise your data dictionary

4th Consultation	7/13/26		

Shouldbewithin Remarks:

Chapter 1,2,3 including the initial and final pages must be completely delivered for adviser's evaluation

Finalize your chapters

This is to certify that I have been regularly consulted by my advisees; have reviewed their system prototype as the required manuscript of the above-stated study. As their adviser, I therefore submit them ready for **Proposal Hearing** as the Chapter 1 through 3 and the pertinent parts of their manuscript are complete.

Note: Defense week: 7/20/26

Signed: Amparo , Joemarie

(Signature of Adviser over printed name)

APPENDIX J

Consultation Logs - Censor

University of Cebu
College of Computer Studies

Date: 07/14/26

CENSOR'S CERTIFICATE

This is to certify that the undersigned has reviewed and went through all the pages of the proposed project study/research manuscript titled:

Paper Pilot: An AI Powered Academic Document Analysis and Compliance Checking System

as against the set of structural rules that govern the composition of *sentences*, *phrases*, and *words* in the English language as well as the technical terms, syntax (format, etc.) and semantics appropriate for the Information Technology and Computing fields.

Signed:

Grammarian

Conforme:

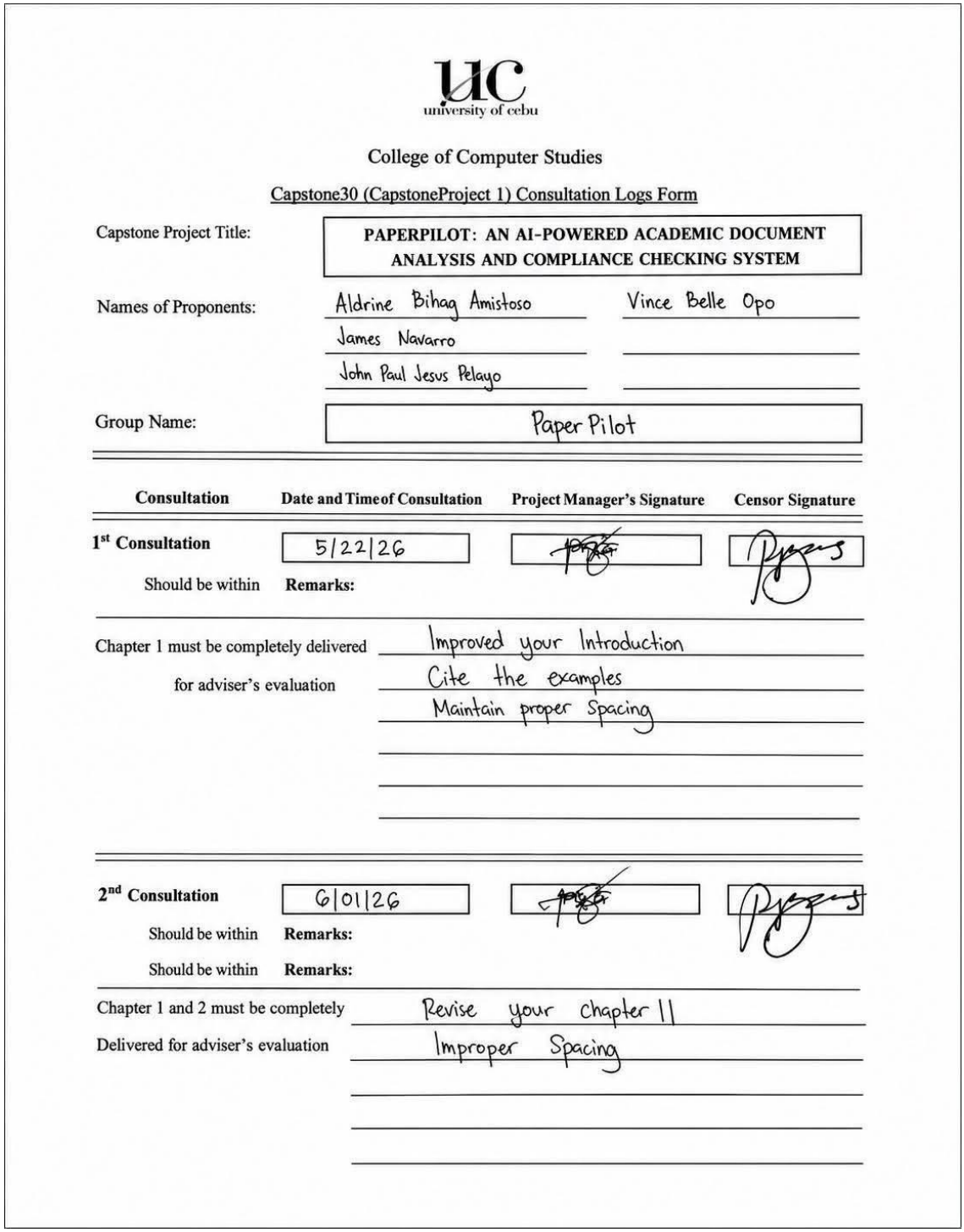
Project Manager

Noted:

Adviser

APPENDIX K

Censor's Certificate



UC
university of cebu

College of Computer Studies

Capstone30 (CapstoneProject 1) Consultation Logs Form

Capstone Project Title: **PAPERPILOT: AN AI-POWERED ACADEMIC DOCUMENT ANALYSIS AND COMPLIANCE CHECKING SYSTEM**

Names of Proponents: Aldrine Bihag Amistoso, Vince Belle Opo, James Navarro, John Paul Jesus Pelayo

Group Name: Paper Pilot

Consultation	Date and Time of Consultation	Project Manager's Signature	Censor Signature
1st Consultation	5/22/26		

Should be within **Remarks:**

Chapter 1 must be completely delivered for adviser's evaluation

Improved your Introduction
Cite the examples
Maintain proper Spacing

2nd Consultation	6/01/26		

Should be within **Remarks:**

Should be within **Remarks:**

Chapter 1 and 2 must be completely Delivered for adviser's evaluation

Revise your chapter II
Improper Spacing

3rd Consultation Shouldbewithin	6/20/26	~~[illegible]~~	Rpzeras
Remarks:	• Do proper Spacing • revise your RRL And Related studies		
Chapter 1,2and3must be completely delivered for adviser's eveuation			

4th Consultation Shouldbewithin	7/13/26	~~[illegible]~~	Rpzeras
Remarks:	• finalize your chapter 1 2 and 3 • make your diagrams clear and readable		
Chapter 1,2,3 including the initial and final pages must be completely delivered for adviser's evualation			

This is to certify that I have been regularly consulted by my advisees; have reviewed their system prototype as the required manuscript of the above-stated study. As their adviser, I therefore submit them ready for **Proposal Hearing** as the Chapter 1 through 3 and the pertinent parts of their manuscript are complete.

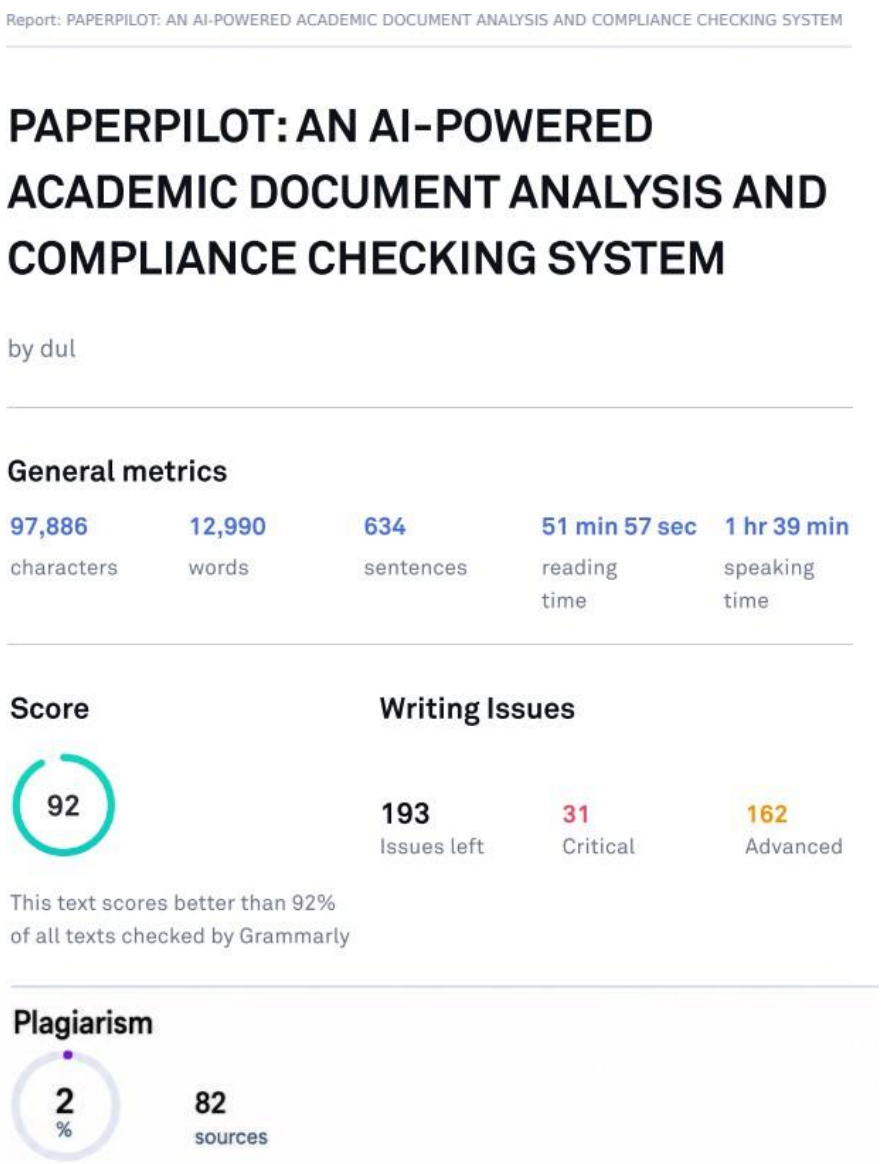
Note: Defense week: 7/20/26

Signed: Rpzeras Rosendo Barzeras

(Signature of Censor over printed name)

APPENDIX L

Grammarly Results



Writing Issues

38	**Correctness**	
5	Text inconsistencies	
1	Citation style options	
3	Improper formatting	
4	Ungrammatical sentence	
2	Determiner use (a/an/the/this, etc.)	
3	Conjunction use	
5	Incorrect phrasing	
3	Incorrect noun number	
3	Incorrect punctuation	
2	Confused words	
1	Pronoun use	
1	Closing punctuation	
3	Mixed dialects of english	
2	Comma misuse within clauses	
151	**Clarity**	
134	Paragraph can be improved	
14	Hard-to-read text	
1	Ineffective or missing emphasis	
1	Wordy sentences	
1	Passive voice misuse	

4	**Delivery**	
1	Potentially sensitive language	
3	Inappropriate colloquialisms	

Unique Words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

13%

unique words

Rare Words

Measures depth of vocabulary by identifying words that are not among the 5,000 most common English words.

49%

rare words

Word Length

Measures average word length

6.2

characters per word

Sentence Length

Measures average sentence length

20.5

words per sentence

TEAM COMPOSITION



Personal Information

Name: Vince Belle B.

Opo Address: Tawason,

Mandaue City Contact

Number: 0907-886-4414

Email Address:

Vincebelleopo@gmail.com Date of

Birth: January 8, 2003

Place of Birth: Cebu City

Civil Status: Single

Religion: Roman Catholic

Educational Background

2022 - Present College Education

University of Cebu - Banilad Campus Bachelor of Science in Information Technology
Gov. M. Cuenco Ave, Cebu City, 6000 Cebu

2020 - 2022 Senior High School

Education Canduman National

High School

H. Abellana Street Housing, Canduman, Mandaue City, Philippines,

6014 2017 - 2020 Junior High School Education

Canduman National High School

H. Abellana Street Housing, Canduman, Mandaue City, Philippines, 6014

Personal Information



Name: Aldrine Bihag

Amistoso Address:

Canduman, Mandaue City

Contact Number: 0968-200-

9489

Email Address:

amistosoaldrine@gmail.com Date

of Birth: August 02, 2003

Place of Birth:

Cebu City Civil

Status: Single

Religion: Roman

Catholic

Educational Background

2022 - Present College Education

University of Cebu - Banilad Campus Bachelor of Science in Information Technology
Gov. M. Cuenco Ave, Cebu City, 6000 Cebu

2016 - 2022 Senior High School

Education Canduman National

High School

H. Abellana Street Housing, Canduman, Mandaue City, Philippines,

6014 2010 - 2016 Junior High School Education

Canduman National High School

H. Abellana Street Housing, Canduman, Mandaue City, Philippines, 6014

Personal Information



Name: John Paul Jesus

Pelayo Address: Mabolo,

Cebu City Contact

Number: 0927-7968-067

Email Address:

pelayo.johnpauljesusrtp@gmail.com Date of

Birth: March 29, 2001

Place of Birth:

Cebu City Civil

Status: Single

Religion: Roman

Catholic

Educational Background

2022 - Present College Education

University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu

2016 - 2022 Senior High School

Education Saint Paul Academy

Suba, Bantayan, Cebu, Bantayan, Philippines,

6052 2010 - 2016 Junior High School

Education

Saint Paul Academy

Suba, Bantayan, Cebu, Bantayan, Philippines, 6052

Personal Information



Name: James P. Navarro

Address:Cubacub

Mandaue City Contact

Number: 09923376134

Email Address:

jemusu2002@gmail.com Date of

Birth:Nov 09, 2002

Place of Birth: Pinamungajan, City

Cebu Civil Status: Single

Religion: Roman Catholic

Educational Background

2022 - Present College Education

University of Cebu - Banilad Campus Bachelor of Science in Information Technology
Gov. M. Cuenco Ave, Cebu City, 6000 Cebu

2016 - 2022 Senior High School

Education Canduman Senior High

School

2010 - 2016 Junior High School

Education Canduman Elementary

School